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IONIZING RADIATION PROTECTION

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This instruction supersedes AFI 48-148, 12 Oct 2001 and implements AFPD 48-1, *Aerospace Medicine Program*, and Department of Defense Instruction (DoDI) 6055.08, *Occupational Ionizing Radiation Protection Program.* It also implements the requirements of NATO Standardization Agreement 2473, *Commanders Guide to Radiation Exposures in Non-Article 5 Crisis Response Operations-ED 2*, for the protection of personnel from low-level radiation exposures. This instruction applies to Department of Defense (DOD) personnel, Air Force Reserve Command (AFRC) units, Air National Guard (ANG) units, Department of Energy (DOE) personnel, DOE prime contractors, and other civilian contractors as specified herein. This instruction is a companion to AFI 40-201, *Managing Radioactive Materials in the US Air Force*, and establishes the requirements outlined in AFMAN 48-125, *Personnel Ionizing Radiation Dosimetry*. It serves as a reference for AFI 91-108, *Air Force Nuclear Weapons Intrinsic Radiation and 91(b) Radioactive Material Safety Program.* This instruction specifies the requirements for the protection of AF personnel and their dependents as well as the public from exposure to ionizing radiation resulting from AF activities. It defines responsibilities for the protection, monitoring and medical follow-up of military personnel for the full spectrum of military operations. This instruction applies to uniformed AF personnel, AF civilians and

individuals living on AF installations who might be exposed to radiation. This AFI has limited applicability to AF contractors and only applies as indentified herein.

This AFI ensures that all records created as a result of processes prescribed in this publication are maintained in accordance with AFMAN 33-363, Management of Records, and disposed of in accordance with the Air Force Records Disposition Schedule (RDS) located at **https://www.my.af.mil/afrims/afrims/afrims/rims.cfm**. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the AF Form 847, *Recommendation for Change of Publication;* route AF Form 847s from the field through the appropriate chain of command. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the AF. See Attachment 1 for a Glossary of References and Supporting Information. This publication requires the collection and or maintenance of information protected by the Privacy Act (PA) of 1974. The authority to collect and or maintain the records prescribed in this publication is DODI 6055.08 "Occupational Ionizing Radiation Protection Program."

(MOODYAFB) AFI 48-148, 21 Sep 2011 is supplemented as follows: This supplement applies to Moody Air Force Base. It applies to all tenant organizations, Air National Guard, Air Force Reserve Components, and contractor operations on Moody AFB. The objective of the radiation protection program is to ensure all occupational exposures to ionizing radiation are kept ALARA. This instruction applies to all military and civilian personnel with a potential exposure to ionizing radiation. Refer recommended changes and questions about this publication of the Office of Primary Responsibility (OPR), using Air Force Form 847, *Recommendation for Change of Publication;* route AF Forms 847 from the field through the appropriate functional change of command.

Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with Air Force Manual 33-363, *Management of Records*, and disposed of in accordance with Air Force Records Information Management System Records Disposition Schedule located at **https://www.my.af.mil/afrims/afrims/afrims/rims.cfm** Contact supporting records managers as required.

SUMMARY OF CHANGES

This instruction has been revised in its entirety. Major changes include the creation of the AF Radiation Safety Committee which will provide direction for radiation protection, not otherwise covered in AFIs 40-201 and 91-108. Roles and responsibilities of the Installation Radiation Safety Officer have been modified and expanded. New functional training requirements are outlined. Finally, new, detailed radiation protection requirements for the medical use of ionizing radiation have been developed.

(MOODYAFB) Established goals of the Radiation Protection Program for Moody Air Force Base. Established additional requirements for Installation, Unit, and Permit Radiation Safety Officers; and shop supervisors. Added items to address commanders' responsibilities and requirements IAW federal and Air Force regulations for ionizing radiation protection. Deletes portions not applicable to Air Force Instruction 48-148, *Ionizing Radiation Protection*, which are

covered in other publications. Expanded overexposure investigation procedures. Added health risk assessment and communication procedures in the context of all-hazard response.

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Chapter 1

INTRODUCTION

1.1. Purpose. This instruction addresses the concerns associated with exposure to ionizing radiation, regardless of source, and seeks to keep exposures as low as reasonably achievable (ALARA). Specific USAF requirements for USAF Radioisotope Committee regulated and 91(b) radioactive materials are found in AFIs 40-201 and 91-108, respectively.

1.2. Applicability. The requirements outlined in this instruction apply to all AF, or AF-led, installations with operations that involve:

1.2.1. Possession or use of radioactive material (RAM) as specified in AFIs 40-201 and 91-108;

1.2.2. Possession or use of radiation producing devices (hereafter referred to as radiation sources), including situations where radiation emissions are incidental to use;

1.2.3. Use of RAM or radiation devices by contractors, as specified by contract. **Note**: Contractors are solely responsible for the health and safety of their personnel as specified in their contract; or

1.2.4. Avertable general public and occupational doses.

1.3. Objectives of the AF Radiation Protection Program.

1.3.1. Prevent the occurrence of clinically significant radiation induced deterministic effects (i.e., effects for which a threshold dose exists and whose severity is proportional to dose; examples include cataracts and acute skin effects). Such shall be accomplished by adherence to dose limits.

1.3.2. Limit the risk of stochastic effects (i.e., effects whose probability of occurrence is proportional to dose; examples include cancer and hereditary effects). Such shall be kept to a reasonable level in relation to the requirements of the military mission, other societal or military risks, benefits gained and economic factors.

1.3.3. Ensure workers are properly trained on the risks of radiation, methods to reduce exposure, and the ALARA principle.

1.3.4. Performance of health risk assessments, adoption of controls, and documentation of radiation exposure.

1.3.5. **(Added-MOODYAFB)** The goals of the Moody Air Force Base Radiation Protection Program include:

1.3.5.1. **(Added-MOODYAFB)** Maintaining exposures ALARA within the operational context,

1.3.5.2. **(Added-MOODYAFB)** Compliance with federal regulations,

1.3.5.3. **(Added-MOODYAFB)** Maintain control of radioactive material and radiation-producing devices,

1.3.5.4. **(Added-MOODYAFB)** And limit subsequent exposures to occupational workers and members of the general public ALARA.

1.4. Types of Radiation Exposure. For the purposes of this instruction, there are four categories of radiation exposure:

1.4.1. Occupational exposures are routine exposures incurred as a necessary part of supporting the military mission. Common practices that may incur occupational exposures include, but are not limited to: industrial radiography, depleted uranium munitions handling, research involving nuclear materials or radiation sources, well-logging, diagnostic radiology, nuclear medicine, radiation therapy, cosmic or solar radiation during flight, and activities involving maintenance of nuclear weapon systems. Furthermore, occupational exposures include those due to radon in workplaces;

1.4.2. Medical exposures are exposures incurred by individuals as part of their own medical diagnosis and treatment. **Note:** Occupational exposure to medical staff or incidental radiation exposure to the general public from medical sources are not medical exposures;

1.4.3. Public exposure encompasses all exposures other than occupational and medical exposures. Public exposures of concern, for the purposes of this instruction, include those resulting indirectly from AF occupational and medical practices, radon, and from accidents or incidents; and

1.4.4. Incident or contingency type exposures. Such pertains to specific emergency response activities to save life or property, humanitarian assistance operations, and military operations conducted where national interests may force personnel to incur radiation doses beyond occupational limits.

1.5. Practices and Interventions. Activities that may result in radiation exposure can be broadly divided into two categories: practices and interventions (reference ICRP Publication 60, *1990 Recommendations of the International Commission on Radiological Protection*).

1.5.1. Radiation Protection Policy for Practices. Routine and controlled operations that incur radiation exposure as an unavoidable and unintentional aspect of the activity are considered practices. This instruction considers all AF activities that involve the routine use of radiation sources or RAM in medicine, research, industry and training to be practices.

1.5.1.1. Justification: Any proposed activity that may cause exposure to persons should yield a sufficient benefit to society or the military to justify the risks incurred by the radiation exposure.

1.5.1.2. Optimization: The magnitude of individual doses and the number of people exposed shall be kept ALARA. Economic, military, and social factors shall be taken into account.

1.5.1.3. Dose Limits: The dose to an individual resulting from a combination of all relevant practices shall not exceed the limits specified in this instruction.

1.5.2. Radiation Protection Policy for Interventions. Interventions encompass two broad types of activity, operations that seek to reduce existing exposures that are not part of a controlled practice and activities conducted to mitigate threats greater than that posed by radiation or otherwise conduct operations necessary to achieve higher objectives, including those of national security.

1.5.2.1. Justification: A proposed intervention should do more good than harm. The action should take into account the goals of military objectives or humanitarian assistance in context with risk management.

1.5.2.2. Optimization: The form, scale, and duration of the intervention should be optimized so the net benefit is maximized and the net detriment is minimized. Again, individual doses shall be maintained ALARA.

1.5.2.3. Dose Guidance: Dose limits do not apply for interventions. Instead, dose guidance is utilized in context with surveillance and protection of forces.

Chapter 2

ROLES AND RESPONSIBILITIES

2.1. SAF/IE (Assistant Secretary of the Air Force (Installations, Environment and Logistics)).

2.1.1. Appoints a voting representative and alternate to the Air Force Radiation Safety Committee (AF-RSC).

2.1.2. Provides guidance, direction, and oversight on all matters pertaining to the formulation, review, and execution of Environment, Safety, and Occupational Health (ESOH) policies, plans, programs, and budgets.

2.1.3. Provides oversight to USAF/SG whose executive is located in AFMSA/SG3.

2.2. SAF/AQ (Assistant Secretary of the Air Force (Acquisition)).

2.2.1. Appoints a voting representative and alternate to the AF-RSC.

2.2.2. Ensures adequate acquisition procedures exist governing the life-cycle management of sources of radiation not covered by AFI 40-201 or AFI 91-108.

2.2.3. Ensures the AF-RSC is informed regarding the acquisition of new systems and upgrades that utilize sources of radiation not covered by AFI 40-201 or 91-108 prior to fielding.

2.3. USAF/A4/7 (Deputy Chief of Staff of the Air Force (Logistics, Installations, and Mission Support)).

2.3.1. Appoints a voting representative and alternate to represent Civil Engineering and Security Forces to the AF-RSC.

2.3.2. Ensures adequate procedures exist governing radiation safety as they pertain to logistics, maintenance, civil engineering, and security.

2.3.3. Provides consultation on ionizing radiation matters associated with accidents/incidents/attacks involving RAM, radiation sources or weapons of mass destruction.

2.4. USAF/SG (The Surgeon General).

2.4.1. Establishes AF policy for controlling radiation hazards and sets limits for exposure to radiation. Ensures compliance with relevant Federal policies and accepted scientific practice.

2.4.2. Appoints a Health Physics Consultant and a Medical Physics Consultant, who can be one individual if qualified for both positions. Receives reports from the Consultants, through the Associate Corps Chief for Bioenvironmental Engineering on matters that require SAF visibility.

2.5. USAF/SG3 (Health Care Operations, Office of the Surgeon General).

2.5.1. Oversees the use of sources of radiation that are not covered by AFIs 40-201 or 91-108 to ensure that they are being used in accordance with all Federal, DOD and AF requirements.

2.5.2. Establishes the AF-RSC that will serve as the focal point for ionizing radiation issues and specific, medical non-ionizing radiation issues (magnetic resonance imaging, medical lasers). Chairs, or delegates someone to chair, the AF-RSC (preferably from AFMSA/SG3P). Authorizes the Chief, Radiation Health to act in the Chairperson's absence.

2.5.3. Appoints the AF/SG Consultant on Health Physics to be the Executive Secretary of the AF-RSC. Provides the Executive Secretary authority to conduct business on behalf of the AF-RSC.

2.5.4. Appoints a voting representative and alternate from the Radioisotope Committee Secretariat (RICS) to the AF-RSC. The voting representative is the Executive Secretary of the AF-RSC.

2.5.5. Appoints a voting representative and alternate from Bioenvironmental Engineering (AFMSA/SG3PB) to the AF-RSC.

2.5.6. Appoints the AF/SG Consultant on Medical Physics as a voting representative to the AF-RSC. The Consultant on Medical Physics serves as the liaison for the Air Force Medical Physics Working Group (AF-MPWG) to the AF-RSC.

2.5.7. Appoints a voting representative and alternate from Medical Readiness (AFMSA/SGX) to the AF-RSC.

2.6. USAF/A7C (The Civil Engineer). Appoints a voting representative and an alternate to the AF-RSC to provide consultation on ionizing radiation matters associated with accidents/incidents/attacks involving RAM, radiation sources or weapons of mass destruction.

2.7. All MAJCOM Surgeons. Appoint a voting representative and an alternate to the AF-RSC to provide consultation on research/development, training, health risk surveillance, and medical readiness issues.

2.8. AFIA/SGO.

2.8.1. Appoints a voting representative to the AF-RSC. Resources and prioritizes inspections according to AFI 90-201, *Inspector General Activities*.

2.8.2. Maintains a Bioenvironmental Engineer (BE) qualified in health physics (43E3G) who is trained to conduct inspections of organizations possessing radiation sources covered by this AFI.

2.8.3. Conducts special emphasis inspections (SEIs) as requested by the AF-RSC. Reports on trends associated with findings from routine inspections on a quarterly basis.

2.8.4. Conducts inspections to assess AF compliance with the conditions of this instruction and AFI 91-108.

2.9. 711 HPW/CC.

2.9.1. Appoints a voting representative and an alternate to the AF-RSC to provide consultation on research/development, training, and technical matters.

2.9.2. Through USAFSAM, resources and maintains a consultative service capable of addressing the full spectrum of radiation issues facing the AF. Such shall be capable of publishing technical reports as requested by the AF-RSC.

2.10. Director, USAFSAM/OEH. Provides and manages the following services.

2.10.1. Radiation Dosimetry Laboratory. Serves as the office of primary responsibility for AFMAN 48-125, *The US Air Force Personnel Dosimetry Program.*

2.10.1.1. Maintains the AF Master Radiation Exposure Registry (MRER). The MRER will archive comprehensive dosimetry records for all military personnel and for other personnel who use AF dosimetry services. Records will include negative and positive results of bioassays, administrative dose assignments (including copies of documents supporting dose assignments), and supplementary occupational dose equivalent information (e.g., dosimetry information resulting from off-duty employment).

2.10.1.2. Data from classified operations will be included to the fullest extent possible. An individual's assessed dose shall not be, in most circumstances, classified. Where classification is deemed necessary, see **paragraph 6.5.4.2** for further guidance.

2.10.1.3. Immediately notifies the IRSO, AFMSA/SG3PB (Radiation Health), and MAJCOM BE with dosimetry results indicating personnel received a dose exceeding the values in **Table A5.1**.

2.10.2. Radioanalytical Services.

2.10.2.1. Maintains a capability suitable for the collection of environmental and bioassay samples, conducts in-vivo monitoring, and performs radioanalyses of samples. The capability shall support both operational requirements and medical-legal documentation of individual internal exposures during peacetime, combat and non-combat military operations, and war.

2.10.2.2. Processes, analyzes and interprets bioassay and environmental samples IAW the most current and scientifically effective analytical procedures.

2.10.2.3. Maintains complete records of all bioassay samples, sample analysis results, and estimation of internal dose in perpetuity, including estimates below detection limits of the analytical method.

2.10.2.4. Provides bioassay results to the supported commander's medical staff. Bioassay results will be provided with necessary interpretation for clear understanding of their meaning and significance.

2.10.2.5. Reports negative or positive occupational bioassay sample results to the Radiation Dosimetry Laboratory for inclusion in the individual's occupational dosimetry record (MRER).

2.10.3. Health Physics Consulting.

2.10.3.1. Appoints a voting representative and an alternate to the AF-RSC to provide consultation on the areas pertaining to radiation programs under USAFSAM.

2.10.3.2. Provides technical advisory services on all radiation protection issues including health risk assessments, exposure reconstructions, radiation safety program reviews, radiation safety quality assurance program development, radiation safety training, shielding assessments, medical and industrial scatter surveys, public dose assessments, and decommissioning surveys.

2.10.3.3. Provides advice to unit commanders with missions that require flight operations with average flight profiles in excess of 40,000 feet to establish radiation protection protocols.

2.10.3.4. Publishes technical reports as requested by the AF-RSC.

2.10.4. Air Force Radiation Assessment Team (AFRAT).

2.10.4.1. Provides operational health physics support during nuclear or radiological contingencies and is supported by the Health Physics Section, Radioanalytical Laboratory and Radiation Dosimetry Laboratory.

2.11. Air Force Radiation Safety Committee (AF-RSC).

2.11.1. Provides direction on uses of radiation, not otherwise covered in AFIs 40-201 and 91-108, and grants authority to the AF-RSCES to conduct all business on its behalf (reference **Attachment 9**). **Note:** Issues related to the AF Master Materials License, or directed energy, to include non-medical lasers, are addressed by other groups.

2.11.2. Serves as the primary AF point of contact for communications with Federal, state, and Host Nation regulatory authorities regarding radiation issues, not otherwise covered in AFIs 40-201 and 91-108. **Note**: HQ AFSC is the single point of contact for 91(b) material.

2.11.3. Prior to procurement and fielding of new sources of ionizing radiation covered by this AFI, ensure that safe operating procedures exist, that the ionizing radiation hazard has been fully characterized, and that the exposures to operators, maintainers, and personnel are kept ALARA using administrative controls, engineering controls, or personal protective equipment.

2.11.3.1. Prior to procurement and fielding of new ionizing radiation detectors or dosimeters, ensures the devices are appropriate for the desired use, that proper calibration requirements exist, and that adequate training requirements are identified.

2.11.4. Recommends policies to AF/SG for keeping radiation exposure ALARA.

2.11.5. Identifies new or special inspection needs and reports them to AFIA/SG.

2.11.6. Establishes the Air Force Medical Physics Working Group (AF-MPWG) via the Associate Corps Chief for Bioenvironmental Engineering and the AF/SG Consultant for Medical Physics.

2.11.6.1. The AF-MPWG reports on medical physics issues to the AF-RSC and to the Associate Corps Chief for Bioenvironmental Engineering for planning purposes of the Bioenvironmental Engineering Corporate Board (BCB).

2.11.7. Meets as often as necessary but not less than twice a year and delegates action items to voting representatives or other technical representatives. A quorum of at least 50% of the appointed representatives is required for each meeting.

2.11.8. Publishes and makes available minutes to all members.

2.11.9. Provides final resolution for allegations concerning the safe and regulatory compliant use of radiation, not otherwise covered in AFIs 40-201 and 91-108.

2.11.10. Is comprised of the members identified above, but may add members as deemed necessary.

2.12. AF-RSC Executive Secretariat (AF-RSCES).

2.12.1. The AF-RSCES is led by the Chief, Radiation Health (AFMSA/SG3PB). The AF-RSCES manages the affairs and executes the decisions of the AF-RSC and maintains AF policy pertinent to ionizing and medical non-ionizing radiation safety.

2.12.2. In collaboration with relevant AF/SG Consultants, determines whether individuals are qualified by training, education, and experience to use sources of radiation covered by this AFI.

2.12.3. Prior to procuring and fielding new sources of ionizing radiation covered by this AFI, ensure that safe operating procedures exist, that the ionizing radiation hazard has been fully characterized, and that the exposures to operators, maintainers, and personnel are kept ALARA using administrative controls, engineering controls, or personal protective equipment. Equipment procured through the Air Force Diagnostic Imaging and Radiotherapy Board (AFDIRB) is outside the scope of the AF-RSC.

2.12.3.1. Prior to procuring and fielding new ionizing radiation detectors or dosimeters, ensures the devices are appropriate for the desired use, that proper calibration requirements exist, and that adequate training requirements are identified.

2.12.4. In collaboration with AFLOA/JACE, provides provisional interpretation of Federal regulations, DODIs, and AFIs until a final ruling can be obtained from the AF-RSC or applicable Federal or State agency.

2.12.5. Conducts visits and responds to incidents and mishaps within the scope of this AFI. As necessary, accompanies AFIA/SGO during inspections where SEIs are being addressed.

2.13. AFLOA/JACE. Provide legal counsel to both the AF-RSC and to the AF-RSCES, as appropriate, on all legal and policy requirements affecting or otherwise applying to AF implementation of the radiation safety program. Specific tasks include:

2.13.1. The Chief, AFLOA/JACE serves as chief counsel to both the AF-RSC and to the AF-RSCES. They may appoint a qualified attorney from within JACE, military or civilian, to provide legal counsel to the AF-RSC and the AF-RSCES on an as needed basis.

2.13.2. Attending all scheduled meetings of the AF-RSC.

2.13.3. Collaborating with AF-RSCES in obtaining and providing accurate interpretations of all statutes, regulations, instructions and guidance documents that affect implementation of the AF radiation protection program for all AF components covered by this AFI.

2.14. Wing or Installation Commanders, as appropriate. Ensure the Wing or Installation radiation safety program is comprehensive, compliant with current requirements, and fully integrates the radiation safety programs of units, tenant units, and geographically separated units (GSUs). Specific tasks include:

2.14.1. Appointing, in writing, a qualified individual to be the IRSO. Qualifications for IRSOs are listed in **Attachment 3**.

2.14.2. Ensuring that tenant organization and unit radiation safety programs (including nuclear capable units, units supporting nuclear capable units and units with 91(b) material *not* in current nuclear weapons or components) are fully integrated into the Wing or Installation radiation safety program.

2.14.3. Affording the AFIA the opportunity to inspect radiation use activities, and the premises and facilities wherein such is used or stored. Upon receiving notice from AFIA, Commanders shall make records kept by the unit, pursuant to Federal, DOD, and AFIs, available for inspection.

2.15. Organization or Unit Commanders, as appropriate.

2.15.1. Designate, in writing, a Unit RSO (URSO) when in possession of RAM or radiation sources. Units that possess only equipment with magnesium-thorium components or depleted uranium (DU) munitions, not covered by permits, generally do not require URSOs. Additionally, designate, in writing, a Permit Radiation Safety Officer (PRSO) when required by AFI 40-201 or AFI 91-108. Consult with the IRSO for guidance on PRSO requirements.

2.15.1. **(MOODYAFB)** In the event of a change in URSO and/or PRSO, notify the IRSO within 30 days of the change.

2.15.2. Provide facilities, equipment, and resources for radiation protection and safety. The nature and extent of which must be commensurate with the ALARA concept and the radiation hazards of the workplace.

2.15.3. Implement policies, procedures, and a radiation protection program to ensure the requirements of this instruction are met.

2.15.3. **(MOODYAFB)** A written radiation protection program must be reviewed by the IRSO annually. The program will contain the elements listed in paragraph 2.20.3 and Chapter 3 of this instruction. The IRSO is available to assist with the development of the written radiation protection program.

2.15.4. Ensure implementation of AF radiation dosimetry and/or bioassay program, as necessary.

2.15.5. Ensure personnel receive education and training IAW this instruction.

2.15.6. Ensure reports are made and records are maintained IAW this instruction.

2.15.6.1. **(Added-MOODYAFB)** All reports regarding the radiation safety program, to include compliance with federal, state, and Air Force regulations, will be sent to the IRSO within 30 days.

2.15.7. Ensure workers incidentally exposed to ionizing radiation in their workplace (not otherwise directly related to or required by their work), receive the same level of protection as if they were members of the public.

2.15.8. Ensure patient dose records, either measured or estimated, are maintained in the patient's medical record.

2.15.9. Establish, review, and approve procedures for conducting Operational Risk Management (ORM) with respect to radiation exposure IAW established policy (DODI

6055.1, *Safety and Occupational Health (SOH) Program*) and procedure (AF Pamphlet 91-215, *Operational Risk Management Guidelines and Tools*).

2.15.10. Notify the IRSO before making changes regarding RAM or radiation sources (i.e., the amount or types of RAM; new or altered radiation sources; special operations or construction of new facilities). The IRSO shall also be notified prior to any change in facilities that affect source or device security requirements, increased potential for personnel exposures, the location of RAM or radiation sources, or the potential for release of RAM.

2.15.10.1. **(Added-MOODYAFB)** Request authorization from the IRSO if radioactive material or radiation-producing devices are brought on the base. The request must include the following:

2.15.10.1.1. **(Added-MOODYAFB)** Description of the RAM or device,

2.15.10.1.2. **(Added-MOODYAFB)** Use of the RAM or radiation-producing device,

2.15.10.1.3. **(Added-MOODYAFB)** List of authorized users,

2.15.10.1.4. **(Added-MOODYAFB)** Copy of the RAM permit, if applicable,

2.15.10.1.5. **(Added-MOODYAFB)** Point of contact,

2.15.10.1.6. **(Added-MOODYAFB)** Security procedures for the RAM or device,

2.15.10.1.7. **(Added-MOODYAFB)** Date of the RAM or device coming on to the base, and

2.15.10.1.8. **(Added-MOODYAFB)** Duration of time the RAM or device will be used on the base.

2.15.10.2. **(Added-MOODYAFB)** If the duration of time the RAM or device will be used on the base exceeds one (1) year, a separate request must be submitted to the IRSO each year.

2.15.11. Execute the specific requirements for 91(b) material as outlined in AFI 91-108, as required.

2.15.12. Execute the specific requirements for control of permitted RAM as outlined in AFI 40-201, as required.

2.15.13. Consult with the IRSO and USAFSAM to determine radiation protection requirements for missions that require flight operations with flight profiles above 40,000 feet.

2.15.14. **(Added-MOODYAFB)** Allow IRSOs to have access to radiation protection program documents, as they have oversight of the base's radiation safety program.

2.15.15. **(Added-MOODYAFB)** Allow IRSOs to annual reviews on the unit's radiation safety program. If findings are recorded, correct all findings in writing within 60 days of the inspection.

2.15.16. **(Added-MOODYAFB)** Ensure compliance with all applicable federal, state, and Air Force regulations.

2.16. Organization , Unit, and/or Wing Commanders During Contingencies, and Deployed AF AOR Commanders.

2.16.1. Appoint, in writing, a qualified individual to be the deployed IRSO, as appropriate. Qualifications for IRSOs are listed in **Attachment 3**. Ensure all practices meet the requirements of this AFI.

2.16.2. Manage interventions, as defined in this instruction, using ORM and the guidance in this AFI.

2.16.3. Include the Deployed Medical Commander (DMC) or deployed Senior Medical Officer (SMO) in the operations and planning staff for all tasks with radiation safety implications. Convey intelligence regarding the nature and extent of actual or potential radiological hazards to the DMC or deployed SMO.

2.16.4. Request additional expertise and support, as necessary, from higher headquarters and from host nation radiation protection experts, as necessary.

2.16.5. Establish or confirm a dose IAW the guidance in **Table A7.2**, for public interventions, and **Table A8.1**, for operations involving interventions.

2.16.6. Implement control measures necessary to contain the radiological hazard as indicated in AFI 10-2501, *Air Force Emergency Management (EM) Program Planning and Operations.*

2.16.7. Ensure facilities, equipment and protective clothing are decontaminated to the levels designated in **Table A5.2**; **Table A8.2** and **Table A8.3**, as applicable; or IAW host nation agreement.

2.17. MTF Commanders.

2.17.1. Ensure complete records are maintained in the member's medical record, of either measured or estimated radiation dose received by personnel during occupational practices and/or contingency operations and that all releasable data is available to the monitored individual.

2.17.1.1. Forward dose determinations records to USAFSAM/OEH for incorporation into the MRER (to include locally performed bioassays).

2.17.1.2. Maintain and be able to access all radiation exposure data for organizations or units conducting classified operations.

2.17.2. Ensure collection of bioassay and laboratory specimens as necessary to assess internal exposures from ingested or inhaled RAM or contaminated wounds IAW NATO Allied Engineering Publication-49, *NATO Handbook for Sampling and Identification of Radiological Agents (SIRA)*. Samples shall be forwarded to USAFSAM/OEH for analysis and interpretation.

2.17.3. Ensure compliance through designation of appropriate staff and resources IAW the responsibilities and requirements, to include staffing qualifications, as specified in **Chapters 4 and 6.**

2.17.4. Ensure medical follow-up of personnel receiving significant exposures IAW **Chapter 6**.

2.18. Deployed Medical Commander (DMC) or Deployed Senior Medical Officer (SMO).

2.18.1. Consults with personnel possessing specialized expertise. These may include: nuclear medicine physicians, diagnostic radiologists or radiation oncologists, BEs, health physicists, medical physicists, occupational health physicians, and preventive medicine physicians. If local expertise is not available, contact USAFSAM and/or the Health or Medical Physics Consultant to the AF/SG.

2.18.2. Applies the framework for radiation protection presented in **Chapter 6**, including justification and optimization of exposures.

2.18.3. Applies the principles of ORM to manage all hazards faced by personnel, including the short and long-term health risks from radiation exposure.

2.18.4. Recommends to the AF component of a Joint Task Force (JTF) a dose IAW the guidance in **Table A8.1**, for all operations considered interventions.

2.18.5. Ensures exposure control measures are implemented for adherence to the applicable dose limits or dose guidance.

2.18.6. Ensures a personnel dosimetry and bioassay program IAW **Chapters 3 and 6** is established, as necessary. Ensures dose records and local bioassay determinations are entered into deployment medical record DD Form 2766, *Adult Preventive and Chronic Care Flowsheet.*

2.18.7. Ensures there is an active environmental surveillance and radiation survey program. The program must provide for evaluation of radiological hazards, assessment of individual doses and exposures, and implementation of protective measures.

2.18.8. Provides diagnostic and treatment services for radiation injury, to include psychiatric casualties and internal radioactive contamination, IAW AF Manual 44-161 (l), *Treatment of Nuclear and Radiological Casualties*, NCRP Report 161, *Management of Persons Contaminated with Radionuclides*; and NATO Manual AMedP-6(B), *NATO Handbook on the Medical Aspects of NBC Defensive Operations, Part I - Nuclear*, or as otherwise consistent with accepted practice.

2.19. Installation Radiation Safety Officer (IRSO). Qualifications for the IRSO are covered in **Attachment 3**.

2.19.1. Executes the responsibilities outlined in AFI 40-201, AFI 91-108, AFMAN 48-125 and this AFI as they apply to the installation.

2.19.2. Ensures the overall coordination of installation radiation safety activities. Provides direct support and information to the installation commander on radiation health and safety issues and effectiveness of measures to control radiation hazards to comply with Federal, DOD and AF requirements. Note: USAF activities overseas shall also follow applicable laws and regulations of either the United States or the host nation. Host nation laws and regulations shall take precedence according to the terms of an applicable Treaty or SOFA (or similar document) with the host nation. Radiation safety standards and requirements followed by USAF organizations overseas will always be at least as stringent as those followed by USAF organizations within the United States.

2.19.2. **(MOODYAFB)** The URSO evaluates requests from unit commanders to bring RAM or radiation-producing devices onto the base. If the requests meet all of the federal,

state, and Air Force requirements, grant approval. Continue to assist the commanders, if requested.

2.19.3. Establishes and manages the overall installation radiation safety program IAW **Chapter 3**. The program must include periodic, but at least annual, reviews of procedures and practices, facility design and classification, training, exposure control, monitoring and surveillance activities. IRSOs shall provide oversight of permitted activities. This includes, but is not limited to, ensuring primary and alternate PRSOs are correctly identified on existing USAF radioactive Material Permits, and verifying the accuracy of RAM inventories. Reports deviations from this AFI and the base instruction or supplement, as applicable, to the Installation Commander and AFMSA/SG3PB or AFSC/SEW, as applicable.

2.19.3.1. **(Added-MOODYAFB)** Provide certification testing, oversight, and guidance to Permit RSOs.

2.19.3.2. **(Added-MOODYAFB)** Annually review exposure surveys for all workplaces which own, store, maintain, and/ or operate any radioactive materials, sources, emitters, or USAF Radioactive Material Permits.

2.19.3.3. **(Added-MOODYAFB)** Conduct annual reviews of all radiation protection surveys for all workplaces which own, store, maintain, and/ or operate any radioactive materials, sources, emitters, or USAF Radioactive Material Permits.

2.19.4. Assists commanders with the development of installation radiation safety operating instructions or radiation safety manuals, as appropriate.

2.19.5. Conducts public dose assessments and radon exposure monitoring described in **Chapter 5**.

2.19.6. Works with Civil Engineering (CE) to ensure adequate design of facilities that will contain radiation sources.

2.19.7. Manages the distribution and recordkeeping requirements of the personnel dosimetry and bioassay program for both occupational exposures and interventions.

2.19.8. Oversees routine radiological decontamination and site remediation activities.

2.19.9. Publishes an installation radiation safety instruction or supplement, as appropriate, detailing local procedures for complying with this instruction. Annually reviews the installation instruction and coordinates changes with affected installation units. Obtains approval for changes to the installation radiation safety instruction from the Installation Commander, or equivalent. At least annually briefs the Installation Commander regarding the results of the review. (**Note**: Providing this briefing to the Environmental, Safety, Occupational Health Council satisfies this requirement.)

2.19.10. IRSOs supporting nuclear capable units shall issue radiation dosimeters to all members of the 2W2 (Nuclear Weapons Specialist) career field assigned to nuclear capable units unless they are assigned to duties that do not have the potential for intrinsic radiation exposure (i.e., administrative positions). 2W2 personnel shall be monitored for gamma and neutron dose IAW AFI 91-108.

2.19.11. Ensures discrepancies in radiation dosimetry reports due to lost or damaged radiation monitoring devices are corrected in the MRER, in coordination with the AF Dosimetry Center. Corrections shall be accomplished within 30 days of identification.

2.19.12. **(Added-MOODYAFB)** Implement and maintain a radiation dosimetry program for monitoring the radiation exposures of all workers whose exposure may exceed 10% of any occupational exposure limit outlined in this instruction.

2.20. Unit Radiation Safety Officer (URSO) and Permit Radiation Safety Officer (PRSO). The qualifications for URSOs and PRSOs are covered in **Attachment 3**. PRSO responsibilities are further prescribed in AFI 40-201.

2.20.1. Execute the responsibilities outlined in AFIs 40-201 and 91-108 and this AFI as they apply to the unit.

2.20.2. Provide technical support to organization or unit commanders on radiation protection issues. Keep organization and unit commanders, and the IRSO, informed about radiation health and safety issues and effectiveness of measures to control radiation hazards.

2.20.2.1. **(Added-MOODYAFB)** Conduct an annual comprehensive radiation safety program review. The IRSO can provide checklists upon request. If findings are recorded, correct all findings in writing within 60 days of the review.

2.20.2.2. **(Added-MOODYAFB)** All reports regarding the radiation safety program, to include compliance with federal, state, and Air Force regulations, will be sent to the IRSO within 30 days.

2.20.2.3. **(Added-MOODYAFB)** Allow IRSOs to have access to radiation protection program documents, as they have oversight of the base's radiation safety program.

2.20.3. Establish and manage the organization or unit radiation safety program IAW **Chapter 3**, as applicable. The program must include review of procedures and practices, facility design review and classification, training, exposure control activities and routine monitoring and surveillance activities.

2.20.3.1. **(Added-MOODYAFB)** Request assistance from IRSOs, to include but not limited to compliance issues, public exposure surveys, shipping RAM, dosimetry, and training.

2.20.3.2. **(Added-MOODYAFB)** A written radiation protection program must be reviewed by the IRSO annually. The program will contain the elements listed in paragraph 2.20.3., and Chapter 3 (for industrial procedures) or Chapter 4 (medical use of ionizing radiation) of this instruction. The IRSO is available to assist with the development of the written radiation protection program.

2.20.3.3. **(Added-MOODYAFB)** Report any suspected or actual overexposures to the IRSO or alternate within 24 hours and assist in the subsequent investigation.

2.20.3.4. **(Added-MOODYAFB)** Inform the Installation RSO of any proposed change in the facilities, any new systems, any modifications or repairs made to existing systems, for all operations that involve radioactive materials or x-ray emitters.

2.20.3.5. **(Added-MOODYAFB)** Ensure shop personnel receive initial and annual refresher site- or workplace-

2.20.3.6. **(Added-MOODYAFB)** Ensure personnel are using appropriate radiation monitoring devices if required by BE (i.e. personnel dosimeter badges) and approved protective equipment for work in areas with potential exposure to ionizing radiation, ensure that proper radiation signs are posted and legible, and all radiation safety devices (lights, interlocks, klaxons, etc.) are operating properly.

2.20.3.7. **(Added-MOODYAFB)** Ensure workers who are on the Thermoluminescent Dosimeter (TLD) program are briefed of their radiation dosimetry results reported on the Armstrong Listing Forms 1499 and ensure the forms are maintained for at least 1 year.

2.20.3.8. **(Added-MOODYAFB)** Additional PRSO duties are listed in the Moody AFB Supplement to AFI 40-201.

2.20.4. Provide commanders assistance in developing organization specific radiation safety operating instructions and radiation safety manuals.

2.20.5. Maintain and manage records as required by this AFI.

2.20.6. **(Added-MOODYAFB)** In the event of a change in URSO and/or PRSO, notify the IRSO within 30 days of the change.

2.20.7. **(Added-MOODYAFB)** Ensure compliance with all applicable federal, state, and Air Force regulations.

2.21. Base Civil Engineering (CE).

2.21.1. Design facilities IAW **paragraph 3.2** of this instruction.

2.21.2. In response to major accidents/incidents, enemy attack and terrorist use of weapons of mass destruction involving nuclear or radiological materials, EM, with guidance from the IRSO shall:

2.21.2.1. Mitigate and remediate radiological hazards, as necessary, to keep exposures ALARA and less than guidance presented in **Attachments 7 and 8**, respectively.

2.21.2.2. Assist in conducting radiation surveys to evaluate or confirm the extent and nature of the radiological hazards.

2.21.3. Mitigate structures where exposures to radon or radon progeny exceed the remedial action level specified in **paragraph 5.3.1**. Incorporate radon reduction measures in the construction of new facilities at medium and high risk installations as identified in the 1987 Radon Assessment and Mitigation Program (RAMP).

2.21.4. Manage and control radioactive wastes generated during remedial actions or interventions. Ensure wastes are disposed of via the Air Force Radioactive Recycling and Disposal (AFRRAD) Office IAW AFIs 40-201, 91-108 and 32-7086.

2.21.5. Conduct training IAW AFI 10-2501, *Air Force Emergency Management (EM) Program Planning and Operations,* and include all subjects identified in **Chapter 6**.

2.22. Workplace Supervisors.

2.22.1. Ensure workplace adherence to the requirements of this AFI.

2.22.2. Ensure protection of Airmen and AF civilians from occupational exposures. Contractors shall comply with this instruction regarding the use and control of radiation

devices and are solely responsible for the health and safety of their personnel as specified in their contract.

2.22.3. Ensure protection of the public from non-occupational exposures from AF practices.

2.22.4. Ensure personnel are trained on radiation hazards in the workplace and appropriate protection requirements.

2.22.5. Ensure radiation safety procedures are current and adhered to by workers.

2.22.6. Ensure declared pregnant workers notify Public Health of their pregnancy status.

2.22.7. Notify the responsible RSO of changes in practices or procedures involving radiation sources potential violations of this instruction, unsafe work practices involving radiation sources, or accidents or incidents involving radiation.

2.22.8. **(Added-MOODYAFB)** Ensure Public Health and BE possess current shop personnel rosters.

2.22.9. **(Added-MOODYAFB)** Ensure shop personnel complete initial and annual ALARA training (if on the dosimetry program) or radiation safety awareness training (if not on the dosimetry program). Document training attendance, and maintain documentation of training attendance and training materials for at least five (5) years.

2.23. Radiation Workers.

2.23.1. Follow applicable rules and procedures for radiation protection and safety specified by organizational management and this AFI.

2.23.2. Use issued dosimeters and personal protective equipment correctly and IAW AFMAN 48-125.

2.23.3. Comply with commander directed radiation protection and dose assessment programs as well as radiological health surveillance.

2.23.4. Provide the URSO and/or IRSO information on past and current work relevant to ensure comprehensive effective protection and safety for themselves and others. This includes work outside the AF where radiation exposures may also occur (IAW 10 CFR 20.2104).

2.23.5. Perform operations in a manner that maintains doses ALARA.

2.23.6. Accept information, instruction and training concerning protection and safety to conduct work IAW this AFI.

2.23.7. Notify workplace supervisors of changes to procedures or operations that could affect exposure, potential violations of this instruction, unsafe work practices involving radiation sources, or accidents or incidents involving radiation.

2.23.8. An active duty pregnant female, shall, on becoming aware she is pregnant, notify her Commander, workplace supervisor, and the Public Health office. A non-military member should notify the Commander, workplace supervisor, and Public Health office. **Note**: It is important to remember that it is the decision of a civilian woman whether or not she declares her pregnancy.

Chapter 3

RADIATION PROTECTION FOR OCCUPATIONAL PRACTICES

3.1. Organization and Administration. Every organization or installation that uses non-exempt quantities of RAM (defined in AFI 40-201), radiation sources, or has the potential for exposures to residual RAM associated with AF operations shall implement a radiation safety program commensurate with the scope of the program and its potential health hazards. Critical program elements are found in this chapter, and are generic to all AF practices involving potential exposure to ionizing radiation. Specific Technical Orders (TOs) or instructions should be referred to for additional detailed information on radiation protection for unique practices (e.g., T.O. 33B-1-1, *Non-destructive Inspection*).

3.1.1. Installation Radiation Safety.

3.1.1.1. IRSO: The wing or installation commander shall appoint an IRSO in writing. IRSO responsibilities are covered in **paragraph 2.18**. The IRSO has overall responsibility and authority over the installation radiation safety program.

3.1.1.2. URSO or PRSO. The unit or organization commander shall appoint the URSO or PRSO in writing when one is required by **paragraph 2.14.1**. URSO and PRSO responsibilities are covered in **paragraph 2.19**. In addition, nuclear capable units and units with 91(b) material will appoint URSOs in accordance with AFI 91-108. This individual shall be directly responsible to senior management of the organization, properly resourced to execute the requirements of the radiation safety program and have ready access to all levels of the organization that may use radiation sources.

3.1.1.3. Radiation Safety Commitee (RSC). Facilities, organizations or installations with extensive radiation protection program requirements shall establish a RSC composed of senior management, the IRSO, organization or URSO and other individuals knowledgeable and responsible for RAM and radiation sources. Examples of extensive programs would include large medical facilities and research organizations to which a health or medical physicist (43EXG) is assigned. The RSC shall meet at a frequency appropriate to evaluate the purpose, safety and compliance of the radiation safety program and regulatory requirements, except RSCs established as a condition of an AF RAM permit which shall meet quarterly.

3.1.2. Radiation Safety Program Policy, Instructions, Procedures and Manuals.

3.1.2.1. Commanders, organization and URSOs, and workplace supervisors shall institute a radiation safety policy that defines the goals of the radiation safety program, the organization and administrative controls required for use of RAM and radiation sources, and state a commitment to the radiation protection policy for practices.

3.1.2.2. Commanders, organization and URSOs, and workplace supervisors shall ensure radiation safety procedures are incorporated into appropriate procedures or instructions. These procedures and instructions should describe the actions or steps necessary to safely conduct a particular task involving a radiation source and document performance of the task. Radiation safety procedures and instructions shall be clearly written, readily available to all users of radiation sources, annually reviewed and updated, as necessary.

They shall describe the safety controls and procedural safeguards necessary to limit exposure, and actions to be followed in the event of a mishap or emergency.

3.1.2.3. Commanders of organizations requiring a RSC shall, ensure the committee meets the requirements of this AFI and, where applicable the requirements of AFI 40-201, 10 CFR 33.13, and 10 CFR 35.24.

3.1.3. Quality Assurance. Commanders, RSOs, and workplace supervisors shall develop and maintain a quality assurance program (QAP) to oversee radiation safety.

3.2. Facility Design, Layout and Area Classification.

3.2.1. Facility Design.

3.2.1.1. Facilities in which non-exempt quantities of RAM, nuclear weapons or components, 91(b) material, or radiation sources are used shall be designed so that exposures from normal operation of the facility are ALARA, and do not result in exposures that exceed applicable limits as provided in **Attachment 4**. Where appropriate, facilities shall also be designed so as to prevent or mitigate radiation mishaps.

3.2.1.2. Consideration shall be made, but not limited to: site selection, facility layout, equipment and system design, shielding, ventilation (including fume hoods and glove boxes), RAM waste management, monitoring and surveillance requirements, and access controls (References: NCRP Report No 127, *Chapter 4, Facility Design*, NCRP Report No 88, *Radiation Alarms and Access Control Systems*, NCRP Report No. 147, *Structural Shielding Design for Medical X-Ray Imaging Facilities, and* NCRP Report No. 151, *Structural Shielding and Evaluation Megavoltage X- and Gamma-Ray Radiotherapy Facilities*).

3.2.1.3. New AF facilities designed for use of RAM or radiation sources shall be constructed so that a member of the public will not likely receive more than 0.02 mSv (2 mrem) deep-dose equivalent in any one hour and 1 mSv (100 mrem) total effective dose equivalent (TEDE) in a calendar year from the normal operation of the facility.

3.2.1.4. The planning and design of new or significantly modified facilities shall include a review by a qualified expert (IRSO or medical physicist, as defined in the Glossary) to ensure appropriate radiation safety features are incorporated. Whereas local expertise is unavailable, contact USAFSAM for assistance.

3.2.1.5. To the greatest extent possible, administrative controls and personal protective equipment shall not be used as a substitute for engineering controls and appropriate facility design.

3.2.1.6. Consideration shall be made for decommissioning requirements during the design phase if facilities are to be used for unsealed RAM, accelerators producing photons with energies greater than 13 MeV, or neutron sources.

3.2.2. Classification of Areas.

3.2.2.1. Restricted Areas. Restricted areas shall be established where specific protective measures are or could be required for controlling routine radiation exposures, preventing

access to radiation sources or preventing the spread of contamination during normal work practices. Restricted areas shall:

3.2.2.1.1. Be delineated appropriately through engineered and physical controls, signage and/or administrative controls, as appropriate.

3.2.2.1.2. Have access controlled so that only the IRSO or personnel having approval of the unit or organization are permitted unescorted access. Visitors and other personnel (e.g., patients) entering restricted areas shall be escorted.

3.2.2.2. Radiation Areas. Any area, accessible to individuals, where radiation levels could result in an individual receiving a dose equivalent in excess of 0.05 mSv (5 mrem) in 1 hour at 30 centimeters from the radiation source or from any surface that the radiation penetrates, shall be designated a radiation area. All such areas shall:

3.2.2.2.1. Be considered restricted areas.

3.2.2.2.2. Be posted with a sign that includes a magenta or black radiation symbol on a yellow background and the words "Caution, Radiation Area", with the exception of temporary areas, medical x-ray facilities and rooms otherwise exempted IAW 10 CFR 20.1903.

3.2.2.3. High Radiation Areas. Any area, accessible to individuals, where radiation levels could result in an individual receiving a dose equivalent in excess of 1 mSv (100 mrem) in 1 hour at 30 centimeters from the radiation source or for any surface that the radiation penetrates, shall be designated a high radiation area. All such areas shall:

3.2.2.3.1. Be considered restricted areas.

3.2.2.3.2. Be posted with a sign that includes a magenta or black radiation symbol on a yellow background and the words "Caution, High Radiation Area", with the exception of temporary areas, medical x-ray facilities, and rooms otherwise exempted IAW 10 CFR 20.1903.

3.2.2.3.3. Adhere to the requirements of 10 CFR 20.1601.

3.2.2.4. Very High Radiation Area. Any area, accessible to individuals, where radiation levels could result in an individual receiving an absorbed dose in excess of 5 Gy (500 rad) in 1 hour at 1 meter from a radiation source, shall be designated a very high radiation area. All such areas shall:

3.2.2.4.1. Be considered restricted areas.

3.2.2.4.2. Be posted with a sign that includes a magenta or black radiation symbol on a yellow background and the words "Grave Danger, Very High Radiation Area", with the exception of temporary areas, medical therapy facilities, and facilities otherwise exempted IAW 10 CFR 20.1903.

3.2.2.4.3. Adhere to the requirements of 10 CFR 20.1602.

3.3. Training.

3.3.1. General: All personnel (military, civilians and in-house contractors) who have the potential to be occupationally exposed to 1 mSv (100 mrem) in a year shall receive initial and annual training that is appropriate in breadth and depth to the radiation hazards present in the

workplace. Traning may include other populations based on the judgment of the IRSO. Training should address the following topics, as applicable:

3.3.1.1. Types and characteristics of radiation of concern;

3.3.1.2. Radioactivity, radioactive decay or x-ray production, as appropriate;

3.3.1.3. Modes of exposure (i.e., internal versus external);

3.3.1.4. Health risks posed by this exposure including: deterministic and stochastic effects, somatic and genetic effects, as well as effects on the unborn fetus, as appropriate; and

3.3.1.5. General radiation protection principles including:

3.3.1.5.1. ALARA and dose limits;

3.3.1.5.2. External protection through time, distance, and shielding;

3.3.1.5.3. Internal protection through respiratory protection, protective clothing and hygiene, as appropriate; and

3.3.1.6. Use of instruments, equipment, and personal dosimetry, as appropriate to:

3.3.1.6.1. Identify sources of radiation emission and radioactive contamination;

3.3.1.6.2. Measure radiation exposure rates or dose rates;

3.3.1.6.3. Monitor individual radiation doses; and

3.3.1.7. Emergency procedures;

3.3.1.8. Reporting requirements;

3.3.1.9. RAM permit requirements, as appropriate; and

3.3.1.10. Other occupation specific hazards and the related skills and procedures that are required for working with the RAM or radiation-producing devices of concern (e.g., depleted uranium awareness training).

3.3.1.11. Intrinsic radiation safety training shall be given to all nuclear weapons specialists (2W2) assigned to nuclear weapons capable units or to units with 91(b) materials. Such shall occur within 90 days of assignment with a refresher given every 15 months thereafter. The IRSO and URSO can expand the scope of this training as appropriate (e.g. handlers, loaders, security forces). The content of training is provided at the AFMSA Radiation Programs website located on Knowledge Exchange.

3.3.2. Training Plan: A written training plan shall be developed by the unit, organization, or IRSO. The training program shall be reviewed and revised as necessary to reflect changes in practices in the workplace.

3.3.2. **(MOODYAFB)** Written training plans must be reviewed and approved by the IRSO. It is the URSO's or PRSO's responsibility to submit training plans to the IRSO. If any modifications are made to the training plan, the updated plan must be reviewed and approved by the IRSO.

3.3.3. Record keeping. Training programs presented, course curricula, and attendance shall be maintained for a period of five (5) years unless otherwise specified. Training shall be

documented on personnel's AF Form 55, *Employee Safety and Health Record,* or other AF sanctioned training database.

3.4. Radiation Exposure Control.

3.4.1. Dose Limits. Occupationally exposed personnel shall not exceed dose limits specified in **Table A5.1**.

3.4.2. Reference Levels. Personnel should not receive a dose in excess of 25% of the applicable annual dose limits in a quarter, or 10% of the applicable annual dose limits in a month without proper justification and optimization of the procedure.

3.4.2. **(MOODYAFB)** The IRSO must perform and document a health risk assessment on the aforementioned doses and procedures prior to exposure. The IRSO must also approve the aforementioned doses and procedures prior to exposure.

3.4.3. Investigation Levels. IRSOs, or the RSC where applicable, shall establish a program that incorporates acceptable threshold levels for dose. Based upon dosimetry, bioassay, or surveillance, IRSOs shall conduct an investigation to determine the causative factors, and identify corrective measures, as appropriate. Refer to AFMAN 48-125 for additional information.

3.4.3.1. **(Added-MOODYAFB)** The investigation action levels for Moody Air Force Base are:

3.4.3.1.1. **(Added-MOODYAFB)** 62.5 mrem per quarter,

3.4.3.1.2. **(Added-MOODYAFB)** 250 mrem per year,

3.4.3.1.3. **(Added-MOODYAFB)** 28 mrem per month for pregnant occupational radiation workers for the duration of her pregnancy.

3.4.3.2. **(Added-MOODYAFB)** The URSO or PRSO will assist in the investigation, if necessary.

3.4.4. Exposure Control**.** Workers shall use the following techniques, as appropriate, and under the judgment and discretion of the IRSO, to ensure dose limits are not exceeded and exposures are ALARA (reference NCRP Report No. 127).

3.4.4.1. Time, Distance, and Shielding: As appropriate, minimize the time around sources of external radiation, maximize the distance to radiation sources, and utilize radiation shielding between radiation sources and potential exposed personnel to control external radiation doses.

3.4.4.1.1. **(Added-MOODYAFB)** Personnel may be used to secure the cordon around a controlled area (i.e. NDI x-ray procedure at an unshielded facility). It is recommended that personnel use radiation detection instruments to monitor exposures at the borders of the controlled area.

3.4.4.1.2. **(Added-MOODYAFB)** If multiple sources of radiation placed in close proximity to one another result in significant exposures, the sources may be placed further away from each other in order to decrease the exposure rate. Proper precautions IAW the unit radiation protection program will be taken while moving the sources apart.

3.4.4.2. Personal Protective Clothing:

3.4.4.2.1. Personal protective clothing including lead aprons, glasses and thyroid shields to protect from x-rays, plastic face shields and glasses to protect from beta particles, and clothing and gloves to protect from contamination shall be used to the greatest extent possible.

3.4.4.2.2. Personal protective clothing is not warranted where its use may result in overall more significant internal or external exposure to radiation or other health risks more severe than that posed by the potential radiation exposure.

3.4.4.2.3. Respiratory protection use shall be IAW AFOSH STD 48-137, *Respiratory Protection Program*.

3.4.4.3. Contamination Control: In the absence of superseding regulatory or advisory guidance, a surface is contaminated if either the removable or total radioactivity is above the levels in **Table A5.2**.

3.4.4.3.1. If a surface cannot be decontaminated promptly to levels below those in **Table A5.2**, the area should be controlled and labeled as contaminated. An exemption is granted to routine low-level contamination by short-lived radionuclides (half-life less than 120 days)) used in nuclear medicine departments. Contact USAFSAM/OEH for additional guidance.

3.4.4.3.2. Always prevent or keep radioactive contamination ALARA.

3.4.4.4. Access Control and Alarm Systems: Provide access control and/or alarm systems to prevent access to or warn of a radiation hazard, as appropriate for areas that can be classified as high or very high radiation areas under **paragraph 3.2.2**. For additional information, consult USAFSAM/OEH and refer to NCRP Report No. 88.

3.4.4.5. Radiation Safety Procedures and Work Permits: Implement, use and periodically review radiation safety procedures and radiation work permits, as appropriate.

3.4.4.6. Change of Duty or Curtailment: Individuals who are likely to exceed the dose limits of **Table A5.1**, even with application of the above measures, shall have duties modified or curtailed so that limits are not exceeded. Such changes in duty shall remain in effect until the individual's projected dose will be less than the prescribed limits. This situation is most frequently encountered for the pregnant worker who has declared her pregnancy. Based on the exposure information and workplace assessment of the IRSO, the female worker's PCM (if military) or health care provider (if civilian) will determine any restrictions to the worker's duties involving occupational radiation exposure. The following guidelines are prudent for declared pregnant workers:

3.4.4.6.1. Declared pregnant technologists or providers should be restricted from performing fluoroscopic and interventional procedures.

3.4.4.6.2. Declared pregnant technologists can perform routine radiographic studies, portable examinations to include the operating room (except fluoroscopy or C-arm), computed tomography, mammography, and radiation therapy.

3.4.4.6.3. Declared Pregnant Nuclear Medicine Technologists may work in imaging rooms, but should be restricted from compounding radiopharmaceuticals, dosing patients and working in the hot lab.

3.4.4.6.4. Female nuclear weapons specialists (2W2) who declare pregnancy should be restricted from duties requiring contact with or occupancy in rooms where Hydrogen-3 (tritium) vapors/oxides are present. If operational requirements of the unit make it necessary for pregnant females to work in those areas, then the unit commander shall review the exposure potential. The unit commander should consult AFSC/SEW, the IRSO, and/or the worker's PCM to discuss the potential for exposure and risk.

3.5. Radiation Dosimetry, Reporting and Record Keeping.

3.5.1. Personnel Monitoring Criteria. Individuals who are occupationally exposed to ionizing radiation as part of their duties must be provided dosimetry and bioassays as described in AFMAN 48-125 when any of the following apply:

3.5.1.1. Exposures are measured or calculated to exceed 1 mSv (100 mrem) TEDE in a year, 2% of a CEDE based ALI or 2% of the occupational limits listed in **Table A5.1**.

3.5.1.2. Dosimetry services may be provided to individuals not meeting the above criteria. If the type of radiation to which the individual could be exposed is detectable by the AF personnel monitoring program and at the discretion of the IRSO, dosimetry services may be provided in the following cases:

3.5.1.2.1. Monitoring would be helpful in demonstrating compliance with ALARA, or

3.5.1.2.2. Monitoring is desirable to evaluate potential exposure conditions to allay worker concern.

3.5.1.2.3. Requested by the individual.

3.5.1.3. Individuals entering a high or very high radiation area.

3.5.1.4. Members of the 2W2 career field assigned to a nuclear capable unit except those assigned to duties that do not have the potential for INRAD exposure (i.e., administrative positions). These individuals shall be monitored for both gamma and neutron dose IAW AFI 91-108.

3.5.1.5. Pregnant occupational radiation workers must be monitored throughout their gestational period. **Note**: Contractors should not be provided dosimetry. The contract, IAW 29 CFR 1910.1096, should state the contractor's employer will provide dosimetry. The AF always has the responsibility to control exposures due to licensed radioactive material IAW 10 CFR 20.1201. The IRSO may determine that a contractor can receive personal dosimetry if any of the above criterion is met and their employer is not able to provide the service.

3.5.2. Reference Levels. Any dosimeter and/or bioassay result which indicates a dose in excess of 25% of the applicable annual dose limit, received for a quarterly monitoring period, or 10% of the applicable annual dose limits if monitored monthly, shall be investigated within 10 days following discovery. An investigation, which validates excessive exposure

occurred, shall require a report be submitted to AFMSA/SG3PB and USAFSAM/OEH through the MAJCOM BE within 30 days. The report shall include:

3.5.2.1. Name, SSAN, occupational dosimetry code, and AFSC of the individual(s) involved;

3.5.2.2. Description of circumstances surrounding the exposure;

3.5.2.3. Estimates of each individual(s) dose equivalent to include a detailed discussion of how this value was determined;

3.5.2.4. Cause of the exposure;

3.5.2.5. Corrective actions taken to prevent recurrence;

3.5.2.6. Statement signed by the individual(s) involved either supporting or contesting the investigation report;

3.5.2.7. Results of any medical examinations;

3.5.3. Overexposures. Any dosimeter and/or bioassay result exceeding the applicable limit in **Table A5.1** shall be considered an overexposure and immediately investigated.

3.5.3.1. Notification:

3.5.3.1.1. When a dosimeter and (or) bioassay indicates an overexposure may have occurred, USAFSAM/OEH will immediately notify the IRSO by telephone and follow up with an email or faxed letter within three (3) hours.

3.5.3.1.2. The individual shall be removed, when warranted, from all duties involving potential radiation exposure until an investigation of the incident can be completed.

3.5.3.2. Reporting:

3.5.3.2.1. The IRSO shall investigate suspected overexposures, with a written report of the investigation submitted through the MAJCOM BE to USAFSAM/OEH and AFMSA/SG3PB within 10 duty days after notification. The written report must include those elements required in **paragraph 3.5.2**.

3.5.3.2.1.1. **(Added-MOODYAFB)** The investigation may include a review of the current radiation exposure survey, reenactment of the incident, dose estimation, additional measurements IAW the ALARA principle, and interviews.

3.5.3.2.1.2. **(Added-MOODYAFB)** Although the IRSO is responsible for writing and submitting the report, the URSO or PRSO will assist the IRSO with the investigation. This includes coordinating investigation actions (i.e. interviews, providing radiation protection program information).

3.5.3.2.2. The reporting requirements established here do not replace or supersede the reporting requirements associated with a nuclear reactor or radiological mishap as established in AFI 91-204, *Safety Investigation and Reports*.

3.5.4. Record Keeping.

3.5.4.1. Annual Report of Occupational Exposure. The IRSO shall manage AF Form 1527-1, *Annual Report of Individual Occupational Exposure to Ionizing Radiation*, IAW AFMAN 48-125.

3.5.4.2. Cumulative History of Occupational Exposure. Upon written request by the individual, IRSO, or other authorized organizations and individuals, USAFSAM/OEH shall provide a copy of all AF Forms 1527-2, *Cumulative History of Individual Occupational Exposure to Ionizing Radiation*. All requests other than those made for official AF use must have a release signed by the individual for whom the report pertains.

3.5.4.3. Previous or Concurrent Occupational Dose. The IRSO shall make a reasonable effort to collect dosimetry records for individuals having either past or present non-AF employment that involve radiation exposure. AF personnel moonlighting in jobs where they are monitored for radiation exposure shall make arrangements to routinely (e.g., annually or based on monitoring period) provide these results to the IRSO or URSO. The IRSO shall ensure these results are forwarded to USAFSAM/OEH for incorporation in the MRER. The individual bears ultimate responsibility for ensuring any non-AF dosimetry results become part of the MRER.

3.5.5. Administrative Doses. Assignment of administrative doses are required for lost or damaged dosimeters or for those individuals with exposures other than the purpose for which it was intended. Exposure information from recovered dosimeters shall replace administrative dose estimates. Assigned doses must be estimated by the IRSO using the procedures in AFMAN 48-125. The IRSO shall ensure the assigned dose replaces the administrative dose in MRER within 30 days.

3.6. Monitoring and Surveillance Programs, and Instrumentation.

3.6.1. Types of Surveys: Surveys shall be conducted in areas where the potential exists for exposure to external radiation fields, airborne contamination or surface contamination. Two types of surveys are common: routine surveys conducted on a fixed, periodic basis at common locations to determine changes or trends in the radiation environment and non-routine surveys performed to evaluate new or expected changes in a radiation field.

3.6.1.1. Routine surveys shall be performed as specified by AFMSA/SG3PB, this instruction or based on AFMAN 48-153, *Health Risk Assessment*.

3.6.1.2. Non-routine surveys shall be performed by a qualified expert (as defined in the Glossary) on any new or substantially modified facility where a radiation source is used, or if the characteristics of the radiation source are significantly changed. This includes nuclear weapons storage and maintenance facilities, as well as facilities housing 91(b) materials. The survey shall determine the efficacy of any installed shielding to protect surrounding areas from primary or scattered radiation, as applicable.

3.6.1.3. In all cases, the survey shall be conducted within 90 calendar days of the facility acceptance date or the change in the radiological characteristics of the source. New survey measurements must be taken after any modification of the facility impacting the shielding or RAM control, replacement or change of the radiation source, and/or a significant change in practice or the uses of the radiation source.

3.6.1.4. **(Added-MOODYAFB)** Surveys will be conducted to support the Occupational and Environmental Health Site Assessment IAW AFMAN 48-154.

3.6.2. Instrumentation: Instrumentation used to perform surveys shall be appropriate to the type of radiation(s) and energy of radiation(s) being detected. IRSOs shall consult with USAFSAM/OEH on instrumentation requirements prior to performing intrinsic radiation, atmospheric radiation, high energy (>13 MeV) accelerator surveys or other mixed radiation environments.

3.6.2.1. Instrumentation shall be calibrated annually, or at a frequency specified in the applicable USAF RAM Permit or TO, using National Institute of Standards and Technology (NIST) traceable radiation sources.

3.6.2.2. Records of instrument calibration shall be maintained by the installation, organization or URSO, as appropriate, for three (3) years.

3.6.2.3. Instrument performance shall be checked before each use, IAW manufacturer's recommendations, as applicable.

3.6.3. Record keeping.

3.6.3.1. Survey results should be maintained for a period of no less than three years, or as specified in the applicable AF RAM permit, TO or AF Files Disposition Instruction, whichever is most stringent. Specific record keeping requirements for radiation protection programs can be found in AFI 40-201, Attachment 7

3.6.3.2. Survey results shall include: a description or drawing of each measurement location; measured dose or contamination levels at each location; the type, model number, serial number, and calibration date of the instrument; name of individual performing the survey, date and time of the survey and applicable comments.

3.7. Radioactive Waste Management.

3.7.1. Radioactive Waste Minimization: Waste generation shall be minimized, or preferably prevented at the source. Unavoidable wastes should be recycled, when feasible, or be reduced in volume and treated, when feasible, to render it less hazardous.

3.7.2. Radioactive Waste Storage: All waste pending disposition shall be stored in a restricted area, in clearly marked containers.

3.7.2.1. Rooms where the amount of RAM exceeds ten (10) times the quantity of material specified in Appendix C of 10 CFR 20 shall be posted with a conspicuous sign or signs bearing a radiation symbol, and the words, “CAUTION” or “DANGER”, “RADIOACTIVE MATERIAL(S).”

3.7.3. Radioactive Waste Disposal and Material Recycling, including 91(b) material, shall be requested through the AFRRAD Office (88 ABW/CEAR) IAW AFI 40-201, AFI 91-108 and AFI 32-7086.

3.8. Occupational Medical Surveillance and Follow-Up. For personnel who are occupationally exposed and whose doses do not exceed the applicable dose limits in **Table A5.1**, testing and monitoring should be the same as standard clinical preventive services that follow guidelines for the general population (e.g., routine mammography, pap smears, and prostate

specific antigen testing). Medical surveillance of overexposures shall be evaluated on a case-by-case basis by the MAJCOM/SG in coordination with AFMSA/SG3PB.

Chapter 4

RADIATION PROTECTION REQUIREMENTS FOR MEDICAL USES OF RADIATION

4.1. Introduction.

4.1.1. This chapter applies to all USAF Medical Treatment Facilities (MTFs) worldwide (i.e., CONUS, OCONUS, deployed AF locations, and Joint DOD installations where the AF is the lead service) that acquire, possess, install, calibrate, maintain, evaluate, use or dispose of sources of radiation used for diagnosis, therapy or for medical research. Any exceptions to this policy must be approved by the Associate Corp Chief for Bioenvironmental Engineering in consultation with the AF/SG Consultant for Medical Physics.

4.1.1.1. For Joint DOD or Joint AF-Veterans Affairs MTFs, the Memorandum of Agreement (MOA) shall include specific instructions and/or guidance regarding the scope of medical physics support and required services. Coordination of all proposed MOAs must be coordinated through the Associate Corps Chief for Bioenvironmental Engineering in consultation with the AF/SG Consultant for Medical Physics. If the AF is the lead service, coordination through the designated Regional Consulting Medical Physics Office (reference **Attachment 6**) is also required.

4.1.2. This chapter describes radiation protection requirements that are applicable to medical practices to ensure patient, practitioner and public radiation doses are ALARA, while obtaining clinical objectives. Responsibility for complying with the requirements in this chapter rests with the MTF commander or duly appointed individual(s) who possess appropriate training, skills, credentials, and resources to ensure compliance.

4.2. Responsibilities.

4.2.1. MTF Commander.

4.2.1.1. If the MTF maintains qualified medical physicists (reference **paragraph 4.3.3**):

4.2.1.1.1. The MTF shall ensure that the qualified organizational medical physicists are directly responsible to the organization to which they are assigned, and that appropriate staff is maintained to provide the medical physics services referenced in paragraphs 4.7.1 and 4.7.2 to support those MTFs within their host organization's region of responsibility (reference **Attachment 6**).

4.2.1.1.2. The MTF shall provide resources to support the training of active duty medical physics trainees. If qualified active duty or civil service medical physicists are not available for all diagnostic imaging and/or radiation therapy modalities, qualified contracted medical physics services shall include a contract provision to provide clinical training to active duty medical physicists to a fully-qualified status.

4.2.1.2. If the MTF does not maintain qualified medical physicists:

4.2.1.2.1. The MTF shall designate annual consulting (TDY) funds for medical physics services in accordance with **paragraph 4.2.6.3** and referencing **paragraph 4.2.6.4.1**

4.2.1.2.2. An estimate for the clinical physics testing services for the scope of the MTF (i.e., diagnostic imaging, nuclear medicine, and other users of radiation imaging devices) can be provided by the designated regional consulting medical physics office (reference **Attachment 6**) or the AF/SG Consultant for Medical Physics (reference **paragraph 4.2.6.4**).

4.2.1.2.3. The MTF shall maintain a radiation safety program and implement appropriate quality control (QC) programs for the various diagnostic imaging, nuclear medicine, and radiation therapy modalities offered at the MTF. The designated regional consulting medical physics office can provide consultation regarding these programs.

4.2.1.2.4. New diagnostic imaging and/or radiation therapy systems shall not be used clinically until acceptance testing has been performed as specified in **paragraphs 4.7.1, 4.7.2, and 4.7.3**. In addition, commissioning of radiation therapy systems must be performed by a qualified radiation therapy medical physicist prior to clinical use.

4.2.1.2.4.1. Human test subjects or volunteers shall not be used for device/system evaluation. An exception may be granted by AFMSA/SG3PB, but only after a medical physics acceptance test has been performed and appropriate participant consent forms signed and approved by the MTF/ Chief of Hospital Services (SGH), in consultation with the MTF Radiology flight commander and the regional medical physicist.

4.2.2. Physicians. Radiologists, radiation oncologists, nuclear medicine physicians, and other qualified physicians that clinically use radiation devices and/or RAM shall:

4.2.2.1. Control all aspects of the conduct and extent of the examinations;

4.2.2.2. Ensure radiographic examinations, radiation therapies or nuclear medicine procedures are only performed by properly trained personnel with adequate knowledge of the physical properties and harmful effects of radiation.

4.2.2.3. Ensure the procedures used are appropriate and optimized for the clinical problem presented.

4.2.3. Credentialed Providers. It is the responsibility of credentialed providers to be cognizant of all of their patient's radiation procedures (i.e., diagnostic imaging, nuclear medicine, and/or radiotherapy) and to maintain that information in the patient's medical record. Consideration of the patient's history regarding recent radiation procedures performed external to the facility is an important aspect of medical care, especially for females of child-bearing age. If a patient is pregnant or becomes pregnant prior to, or immediately after, a radiological procedure (e.g., x-ray of the pelvic region and/or nuclear medicine study), the provider shall immediately consult with a radiologist and/or regional consulting medical physics office regarding the calculation of the conceptus dose from radiation exposure. In addition, potential high-dose exposure, from multiple-computed tomography or high-level fluoroscopy procedures, should be reviewed/evaluated by the radiologist and/or regional consulting medical physics office.

4.2.4. Medical Physicists. Qualified organizational medical physicists as defined in **paragraph 4.3.3**, shall be directly responsible to the organization to which they are assigned,

and support those MTFs within their host organization's region of responsibility (reference **Attachment 6**).

4.2.4.1. Chief, Medical Physicist shall:

4.2.4.1.1. Oversee quality control (QC) programs and provide guidance on quality assurance (QA) practices for the MTF to which the medical physicist is assigned, and provide consultation on and review of QC programs and QA practices to clinics utilizing radiation for imaging within the consulting medical physics office's region of responsibility.

4.2.4.1.1.1. Provide medical physics education and training.

4.2.4.1.1.1.1. Oversee the training of medical physics trainees to a fully-qualified status for diagnostic imaging and/or radiation therapy medical physics. Training curricula shall follow the guidance provided by the American Association of Physicists in Medicine (AAPM) and the American Board of Radiology (ABR).

4.2.4.1.1.2. Provide medical physics lectures to support the Radiology Residency Training Programs.

4.2.4.1.1.3. Provide radiation safety and dose/image quality overview training to all fluoroscopy x-ray users, including non-radiology users. The training shall also include the guidance found in applicable AF/SG NOTAMs (e.g., SGNOTAM 10-002).

4.2.4.1.1.4. Support other AF training programs (e.g., technologists training programs in diagnostic imaging and nuclear medicine) related to the medical use of radiation as requested and upon availability.

4.2.4.1.1.5. Oversee technologists who perform nuclear medicine and diagnostic radiology QC and QA tasks. This is focused on periodic review of the data collection and associated analysis, and the appropriateness of the QC program.

4.2.4.1.2. Provide or supplement diagnostic medical physics contract services in support of the requirements in the Mammography Quality Standards Act, 21 CFR Parts 16 and 900. **Note**: For mammography support, diagnostic medical physicists must meet all qualifications specified in 21 CFR Part 900.

4.2.4.1.3. Provide or supplement acceptance testing services for all new diagnostic imaging and radiation therapy medical systems.

4.2.4.1.4. Consult on patient radiation protection requirements and dosimetry review programs, including those involving exposure to the conceptus and from advanced imaging modalities (i.e., computed tomography, interventional radiography, and nuclear medicine studies). This includes supporting regional sites in addition to the primarily assigned facility.

4.2.4.1.5. Create and maintain a repository of QC reports for each diagnostic medical x-ray system at each MTF within their region of responsibility as deemed appropriate by the regional consulting medical physics office. This is to include annual entrance skin exposure (ESE) calculations.

4.2.4.1.6. If a designated regional consulting medical physics office is unable to support a requirement for an MTF within their region of responsibility, the designated office shall first coordinate with other AF consulting medical physics offices to determine if support can be provided to the requesting MTF. If support can't be provided from within the AF or it is more cost effective to utilize a local contract medical physicist for the requested services, it is the responsibility of the requesting MTF to locate qualified support external to the AF (**paragraph 4.3.3.4**). All MTFs that utilize contracted medical physics services shall send a copy of the final report(s) to their designated regional medical physics office for review which must be completed within 30 duty days from receipt of report. The testing physicist can approve (full/partial) clinical use of the device based on testing results prior to final report review by the regional consulting medical physics office.

4.2.4.1.7. Consultative medical physics support for expeditionary and sustainment locations will be coordinated with the AF/SG Medical Physics Consultant and provided by a designated regional medical physics office.

4.2.4.1.7.1. A copy of all testing reports, regardless of which military service provides the medical physics support, shall be forwarded to the AF/SG Medical Physics Consultant NLT 45-duty days post system test by the supporting bioenvironmental engineer.

4.2.4.1.7.2. For sustainment locations, the equipment should be checked in the AOR by a qualified individual prior to clinical use. Requirements found in **paragraph 4.7.1** are applicable at sustainment locations where the AF is the lead service.

4.2.4.1.7.3. A copy of all testing reports, regardless of which military service provides the medical physics support, shall be forwarded to the AF/SG Medical Physics Consultant for review.

4.2.4.1.8. Provide semi-annual reports to the AF/SG Medical Physics Consultant regarding status of the regional support mission. Details of the report will be established by the AF Medical Physics Working Group (AF-MPWG) and used by the AF-RSC and the decision making processes for the Bioenvironmental Engineering Corporate Board (BCB).

4.2.4.1.8.1. Each MTF shall directly report any medical ionizing radiation or MRI related injury, IAW the FDA (i.e., 21 CFR) and/or accrediting body requirements (i.e., The Joint Commission or the Accreditation Association for Ambulatory Health Care), through their local MTF/SGH, and provide a copy of the report to AFMSA/SG3PB.

4.2.4.1.8.2. Urgent matters, such as equipment defects that may exist at multiple locations or situations that could results in patient injury, shall be reported to AFMSA/SG3PB and the AF/SG Consultant for Medical Physics.

4.2.5. Medical Equipment Repair Centers (MERC). Regional MERCs will support the collection of beam exposure data for ESE calculations IAW AFI 41-201, *Managing Clinical Engineering Programs,* and perform diagnostic imaging equipment acceptance testing as specified in **paragraph 4.7.1**.

4.2.5.1. Beam exposure data shall be collected annually for each radiographic and fluoroscopic unit including new equipment under warranty or service contract. The data shall be provided to the regional consulting medical physics office for review and the ESE calculations will be compared to the most recent National Evaluation of X-Ray Trends (NEXT).

4.2.5.2. Reports. The final ESE report will be maintained at the regional consulting medical physics office. A copy of the final ESE report shall be maintained by the regional MERC or local Biomedical Equipment Technician (BMET) office together with the appropriate survey report. An additional copy of the final ESE report will be forwarded to the radiology clinic's NCOIC. Radiology shall maintain this report, together with determined technique charts, for each diagnostic imaging unit and make such available during AFIA/SGO inspections.

4.2.6. Coordination and Funding Requirements for Medical Physics Services

4.2.6.1. Regional Medical Physics Offices shall have access to the Medical Equipment Web application managed by Medical Logistics Division Air Force Medical Operations Agency/Clinical Engineering Branch. The web application will provide information on any AFDIRB approved equipment procurement packages for diagnostic and nuclear medicine imaging equipment and radiation therapy equipment. This site can be used for regional support planning purposes by the regional consulting medical physics offices.

4.2.6.2. The equipment gaining MTF shall notify the designated regional consulting medical physics office of the equipment installation schedule for diagnostic or nuclear medicine imaging systems, radiation therapy equipment, and/or related construction (for room shielding calculations and/or evaluations) in a timely manner (3 months minimum in advance of equipment installation), and provide any updates regarding the schedule. This is for procurement and other types of equipment purchase packages.

4.2.6.2.1. The equipment gaining MTF shall notify the regional consulting medical physics office of any major repairs or upgrades (e.g., automatic exposure control device, tube head, and software upgrade) of equipment requiring evaluation. No diagnostic or nuclear medicine imaging system and no radiation therapy equipment shall be used clinically until acceptance testing has been completed by or confirmation is obtained from the regional consulting medical physics office that any repairs/upgrades on the device/system do not require testing.

4.2.6.2.2. Physics testing of clinical imaging devices and systems by a qualified diagnostic imaging medical physicist includes evaluation of radiation dose and/or image quality per Federal and AAPM guidelines and/or ACR accreditation requirements. The types of acceptance testing performed by manufacturers and the MERC or local BMET for the indicated systems may include some equivalent tests but overall evaluation, specifically for dose and image quality, is not equivalent to clinical medical physics testing requirements. MERC and local BMET acceptance testing is used for clinical qualification whereas medical physics testing is used for clinical use. For assistance in determining requirements, contact the designated regional consulting medical physics office found in **Attachment 6**.

4.2.6.3. It is the responsibility of the equipment gaining MTF to fund consulting medical physics services to include acceptance testing and associated annual surveys.

4.2.6.3.1. The specific equipment that requires medical physics acceptance testing and annual surveys by a qualified medical physicist, specified in **paragraph 4.3.3**, includes the following: mammography, computed tomography, computed radiography, digital radiography, teleradiology systems, magnetic resonance imaging, high level fluoroscopy systems (including interventional radiography), single photon computed tomography (SPECT) systems, and positron emission tomography (PET) or PET/CT systems.

4.2.6.3.2. If a qualified AF medical physicist is not available, a contract physicist should be hired based upon the qualification requirements stated in **paragraph 4.3.3**. Final reports of testing performed by non AF medical physicists must be forwarded to the designated regional consulting medical physics office for review (reference **Attachment 6**).

4.2.6.4. A list of the average and typical range of fees for specific contracted medical physics services (i.e., equipment testing only) is available in the AAPM annual salary survey report; a copy of the current list can be obtained for planning purposes from the regional consulting medical physics offices or from the AF/SG Medical Physics Consultant.

4.2.6.4.1. AF medical physicists may be able to provide additional services to the regional sites during site visits such as training (radiation safety, quality control program, etc.), testing of other diagnostic and/or nuclear medicine imaging systems, and audits of programs (i.e., radioactive material permit program compliance, modality specific quality control programs, and radiation safety program review) in addition to the equipment testing services. These medical physics services are typically not provided by contracted services, and if so, would be at an additional fee.

4.3. Qualifications.

4.3.1. Qualifications to Authorize the Use of Radiation in Medicine.

4.3.1.1. The use of machine produced radiation for the purposes of diagnosis or treatment of disease or injury shall only be prescribed by Doctors of Medicine or Osteopathy who are properly licensed by the United States or one of its territories/possessions or are certified by an appropriate certifying agency.

4.3.1.2. The Chief of Radiology may recommend to the Credentials Committee that other healthcare providers (physical therapy, occupational therapy, orthopedics, etc.) be granted privileges for ordering radiographic exams, consistent with their training and abilities. Doctors of Dental Surgery or Dental Medicine may request appropriate examinations of the head, neck and chest, although such requests are normally confined to the oral region. Podiatrists and Chiropractors may request x-ray examinations appropriate to their specialty.

4.3.1.3. All users of RAM for medical permitted activities must meet qualifications for that use as defined in 10 CFR 35. Additionally, they must be approved as an authorized user for the specific use identified on the RAM permit, and comply with associated

permit conditions and AFI 40-201, *Managing Radioactive Materials in the U.S. Air Force*.

4.3.1.4. Any request for specialized radiography and fluoroscopy, such as angiography, computerized tomography or other complex studies should be made by providers credentialed by the MTF.

4.3.1.5. Variances to the above qualification requirements may be authorized by the MTF's SGH in consultation with the Chief of Radiology for contingency, emergency or life-threatening situations.

4.3.2. Operator Qualifications.

4.3.2.1. Physician Performance. Eligible physicians include radiologists and other physicians (e.g., cardiologists, urologists, dentists) granted radiology privileges based on the needs of patients serviced by the MTF.

4.3.2.2. Technologist Performance. The application of radiation for diagnosis or treatment shall only be performed by technologists working under the supervision of an eligible physician or technologist (Diagnostic Imaging 4RXXX) who are trained and who have demonstrated proficiency in the use of the specific radiation source, or in the administration and use of the prescribed RAM. Such proficiency shall be IAW the American Society of Radiologic Technologists (ASRT) Position Statements and Scopes of Practice for technologists, assessed through national-performance-oriented evaluation procedures developed by the American Registry of Radiologic Technologists (ARRT), or by didactic training and practical experience equivalent to training programs and examination requirements of recognized credentialing organizations (ARRT or equivalent). Generally, technologists are the largest population responsible for implementing and following the requirements of this chapter. Technologists shall strive to ensure patient doses are ALARA while ensuring diagnostic and therapeutic objectives are achieved.

4.3.2.3. The above consideration for operators of radiation producing equipment should be implemented by the Chief Radiologist or NCOIC/Superintendent, Diagnostic Imaging through the responsible authority (MTF commander) in an instruction which details:

4.3.2.3.1. Who may operate diagnostic x-ray equipment and the supervision required;

4.3.2.3.2. The education and training and/or proficiency requirements for imaging technologists;

4.3.2.3.3. Requirements for continuing education and demonstration of proficiency; and

4.3.2.3.4. This policy should be reviewed periodically and revised as appropriate.

4.3.2.4. All individuals who operate medical devices that produce ionizing radiation must undergo initial and annual ALARA training when the potential to exceed 1 mSv (100 mrem) per year exists. . This training can be provided by radiology or the regional consulting medical physics office contingent upon the training curriculum being pre-approved by the regional consulting medical physics office.

4.3.3. Qualifications for Medical Physicists. Medical Physicists hereafter refers to any active duty (i.e., AFSC shreds: 43E3M or select 43E3G/43EA) or civilian (i.e., civil service and contract) personnel that are determined to be qualified as diagnostic imaging and/or radiation therapy medical physicists per the following standards:

4.3.3.1. Board Certified Diagnostic Imaging Medical Physicist. A diagnostic imaging medical physicist shall be considered qualified to provide diagnostic imaging physics services for nuclear medicine and diagnostic imaging modalities, except mammography, if the individual is board certified in diagnostic radiologic physics by the American Board of Radiology (ABR) or the American Board of Medical Physics (ABMP).

4.3.3.1.1. Board certified nuclear medical physicists are deemed qualified for handling RAM in nuclear medicine for equipment testing purposes. An individual's diagnostic medical physics training should have included the techniques and practices of handling nuclear medicine RAM; if not, then the individual shall perform a practicum approved by a qualified medical physicist, prior to performing tests.

4.3.3.1.2. Active duty, board certified, diagnostic imaging medical physicists that have not performed clinical medical physics tasks for 36 consecutive months due to a career broadening assignment should undergo a peer review period with a diagnostic board certified medical physicist to become reacquainted with all modalities within six (6) months of a clinical assignment.

4.3.3.1.3. Civilian or contract, board certified, diagnostic imaging medical physicists shall have in their duty requirements and/or contract a requirement to train or support the training program of active duty medical physics trainees toward ABR certification. The Chief of Medical Physics, if board-certified in the appropriate subfield, shall manage the training program.

4.3.3.2. Non Board Certified Diagnostic Medical Physicist. A non board certified, government employed, medical physicist or equivalently trained individual (e.g., not a contractor) may be considered qualified to perform surveys in specific diagnostic modalities, except mammography, if they have obtained a preceptor statement from a board certified diagnostic imaging medical physicist confirming that the individual has been trained and is qualified to perform surveys independently in the specified modalities. A copy of this preceptor statement shall be forwarded to and maintained in a file by the designated regional consulting medical physics office.

4.3.3.2.1. The minimum requirements for preceptorship completion shall include the performance of three (3) surveys for the particular modality (e.g., x-ray, type of fluoroscopy system, computed radiography system, digital radiography system, computed tomography system, magnetic resonance imaging system) within the past 36 months under the supervision of a qualified diagnostic medical physicist as specified in **paragraph 4.3.3.1**.

4.3.3.2.2. Government employed individuals obtaining qualifications via the preceptorship path shall complete either of the following within 36 months from the date of the preceptor statement: 1) achieve ABR board certification in diagnostic radiologic physics or 2) provide documentation of maintenance of qualifications to the designated regional consulting medical physics office; the minimum requirements

consist of three surveys for each modality that was approved on the preceptor statement and 50 hours or more of CME in radiologic physics. The AF/SG Medical Physics Consultant may require the individual to be requalified by a board-certified diagnostic medical physicist if board certification is not achieved after the first maintenance of qualifications cycle (i.e., 36 months). **Note:** All nuclear medical physics trainees shall complete a practicum, approved by a qualified medical physicist, on the methods and techniques of handling RAM prior to equipment testing. Individuals with previous nuclear medicine technology training satisfy this requirement.

4.3.3.3. Mammography Qualified Medical Physicist. For mammography surveys, individuals are considered to be qualified if they satisfy the standards as specified in the *Mammography Quality Standards Act* (MQSA) and have forwarded a copy of their State and/or FDA approved documentation to the designated regional consulting medical physics office.

4.3.3.4. Contracted Diagnostic Imaging Medical Physics Services. For contracted diagnostic and nuclear medicine physics services, a board certified diagnostic imaging medical physicist is preferred and will be deemed qualified.

4.3.3.4.1. MTFs should consult with the designated regional consulting medical physics office and/or the AF/SG Consultant for Medical Physics regarding the availability of active duty medical physicists prior to pursuing contracted medical physics services.

4.3.3.4.2. Contracted medical physics services for mammography surveys must be performed by a MQSA qualified individual (reference **paragraph 4.3.3.3**).

4.3.3.4.3. Contract diagnostic imaging medical physicists that are not board certified will not be deemed qualified; however, they can perform surveys under the supervision of a board certified diagnostic medical physicist and the final report cosigned by the certified physicist. The final survey report must be reviewed by the designated regional consulting medical physics office.

4.3.3.5. Radiation Therapy Medical Physicist. A qualified radiation therapy medical physicist shall be an individual board certified by the ABR or ABMP in therapeutic radiologic physics. This policy is mandatory for active duty, civil service, and contract personnel.

4.3.3.5.1. Individuals proposed to work in AF MTFs as authorized medical physicists as defined in 10 CFR 35.51 must be approved and listed on the RAM permit issued by the RICS prior to providing clinical services.

4.3.3.5.2. All primary civil service and/or contract radiation therapy medical physics positions at AF MTFs shall be filled by board certified medical physicists.

4.3.3.5.2.1. For secondary therapy medical physics positions, board eligible civil service and contract physicists must work under the supervision of a qualified radiation therapy physicist. **Note**: board eligible is defined as a person who has passed the ABR part 2 exam in therapeutic radiological physics and is eligible, in good standing, for part 3, oral exam).

4.3.3.5.2.2. Non board eligible individuals, i.e., reference board-eligible in **paragraph 4.3.3.5.2.1,** are not considered radiation therapy medical physicists and cannot function in that capacity. These individuals are considered medical physics assistants or junior physicists. They can provide radiation therapy support only under the direction of an on-site qualified radiation therapy medical physicist. Employment and/or training of these individuals shall not supersede the training and sustainment of available active duty medical physics trainees. MTFs shall consult with the AF/SG Consultant for Medical Physics regarding the availability of active duty medical physicists prior to pursuing civilian and/or contract medical physicists.

4.3.3.5.2.3. Supervisory and oversight responsibilities for non-military, including secondary and non-board eligible, medical physicists must be stated in a Memorandum of Understanding/Agreement. For clinics that employ government and contract medical physicists, a clear scope of responsibility and chain of command must be established by the radiation oncology flight commander in coordination with the MTF's chief medical physicist provided that individual is qualified in radiation therapy physics. The lead radiation therapy medical physicist shall be board certified.

4.3.3.5.3. Active duty radiation therapy medical physics trainees shall perform clinical physics duties under the direction of a qualified radiation therapy physicist.

4.3.3.5.4. Active duty board-certified radiation therapy medical physicists that have not performed clinical medical physics tasks for 36 consecutive months due to career-broadening assignments should undergo a peer-review period with a radiation therapy board-certified medical physicist to become re-acquainted with all modalities within 6-months of a clinical assignment.

4.3.3.5.5. Civilian or contract, board certified, radiation therapy medical physicists shall have in their duty requirements and/or contract a requirement to train or support the training program of active duty medical physics trainees toward ABR certification. The Chief of Medical Physics, if board-certified in the appropriate subfield, shall manage the training program.

4.4. General.

4.4.1. Nuclear Medicine and Radiation Therapy Clinics using Byproduct Material. All AF MTFs offering nuclear medicine or radiation therapy clinical services involving byproduct materials shall adhere to the applicable requirements of 10 CFR Parts 19, 20, 35 and conditions of their USAF RAM permit as specified in AFI 40-201.

4.4.2. Medical X-ray and Dental X-ray Clinics. All medical and dental x-ray systems used shall adhere to the most current requirements of 21 CFR, Parts 1020.30 through 33. A self-inspection of the radiation safety program shall be conducted annually.

4.4.3. Dosimetry Systems.

4.4.3.1. Dosimetry systems used in diagnostic radiology or nuclear medicine shall be calibrated annually using National Institute for Standards and Technology (NIST) traceable sources. Records of calibration shall be maintained for 36 months.

4.4.3.2. Dosimetry systems used in external beam radiation therapy shall have been calibrated by the NIST or by an AAPM Accredited Dosimetry Calibration Laboratory (ADCL). The calibration shall have been performed within the previous 24 months and after any servicing that may have affected calibration. A record of each dosimetry system calibration, intercomparison, and comparison shall be maintained for 36 months.

4.4.3.3. Each facility shall maintain two (2) independent dosimetry systems or alternatively have a means of obtaining an independent determination of beam calibration on an annual basis using an ADCL calibrated electrometer and ionization chamber(s) not normally used by the facility for routine calibration.

4.5. General Requirements in the Clinical Use of X-Ray, Electron Beam and Gamma-Ray Radiation. As a general principle, the dose to the patient shall be kept to a minimum and consistent with clinical objectives. Each MTF shall have and implement written operating and safety procedures. These procedures, including any restrictions of the operating technique required for the safe operation of the particular system, shall be made available to each individual operating a radiation producing machine. These procedures shall be reviewed annually and approved by the appropriate clinic chief, MTF radiation safety office, or regional consulting medical physics office. Specific requirements shall include, but are not limited to:

4.5.1. The useful beam shall be limited to the smallest area practical and consistent with the objectives of the radiological examination or treatment.

4.5.2. The tube potential, filtration and source-to-skin distance (SSD) employed in medical diagnostic procedures should be as large as practical, consistent with study objectives.

4.5.2.1. Special care should be taken to ensure that adequate and proper filtration is used for all diagnostic and therapeutic procedures.

4.5.2.2. The operator should use the maximum SSD consistent with medical requirements of the procedure. For diagnostic procedures, other than dental procedures, distances less than 30 cm should not be used. **Note**: For new systems, one must be aware of variable source to image distance options which could result in excessive wear on the x-ray tube.

4.5.3. Protection of the conceptus during radiological examination or treatment of women known to be pregnant shall be given special consideration, as described in **paragraph 4.6**.

4.5.4. Exposure techniques shall be appropriately modified for pediatric patients.

4.5.5. Sensitive body organs (e.g., lens of eye, gonads) should be shielded whenever they are likely to be exposed to a useful beam provided such shielding does not interfere with clinical diagnostic information or proper treatment. In addition:

4.5.5.1. Gonadal shielding using at least 0.5 mm lead equivalence shall be used whenever potentially procreative individuals are likely to receive direct gonadal radiation in an exam or treatment or the primary beam is within 2.5 cm of the gonadal area.

4.5.5.2. Shielding of breasts for scoliosis radiographs should be used due to the sensitivity of breast tissue in young women and the number of films likely to be taken in a lifetime. The shield must be placed between the source of x-rays and the patient. If there is no shield available, a posterior-anterior (PA) view should be used.

4.5.5.3. The lens of the eye should be shielded with a least 2 mm lead equivalence during tomographic procedures that include the eye in the useful beam (Not required for PA projections).

4.5.6. Fluoroscopy, with the exception of microampere systems used in orthopedics, shall not be used as a substitute for radiography, but should be reserved for the study of dynamics, spatial relationships, and guidance in spot film recording of critical detail and simulation in radiation therapy. Last image hold should be utilized whenever possible. System users shall understand the various dose mode options in multiple dose mode systems, and review those features annually in reference to exposure output data provided by the regional consulting medical physics office.

4.5.7. Intensifying screens, computed radiography imaging plates, digital imaging detectors, and other image recording devices shall be tested annually for uniform response and artifact evaluation. Defects identified shall be reviewed by the interpreting radiologist or medical physicist for corrective action. Imaging screens/plates/detectors shall be replaced as deemed necessary by the interpreting radiologist or medical physicist.

4.5.8. No person should routinely hold patients during diagnostic examinations.

4.5.8.1. If a patient must be held in position, then mechanical supports or immobilization aids must be used when feasible.

4.5.8.2. If mechanical devices cannot be used; the individual selected to hold the patient should be a willing adult (21 years of age or older) relative who is not an occupational radiation worker. If no willing adult family member is available, the next best option is to have a properly shielded non radiation safety worker position the patient. Only if this is not feasible, a monitored occupational radiation worker may hold the patient. Diagnostic Imaging (4RXXX) technologists may hold the patient when consistent with ALARA objectives as determined by the RSO.

4.5.8.3. Pregnant women or persons under 18 years of age should not be permitted to hold patients.

4.5.8.4. If a patient must be held by someone, the individual holding the patient shall be protected with appropriate shielding devices such as gloves and aprons. Positioning should be arranged so that no part of the holder's torso is struck by the useful beam, and that the holder's body is as far as possible from the useful beam.

4.5.9. Only individuals whose presence is necessary shall be in diagnostic x-ray, fluoroscopy or low-energy (< 150 kV) radiotherapy rooms during exposures. These personnel shall be protected with appropriate shielding for the specific situation (e.g., leaded aprons, leaded gloves and/or portable shields).

4.5.10. Individuals, other than the patient, shall not be in linear accelerator vaults or high dose rate brachytherapy vaults during machine use.

4.5.11. Protective devices, including lead aprons, gloves, and shields (including those items at dental, cardio-lab, angiography and operating rooms) shall be stored properly and checked initially and annually for defects such as holes, cracks, and tears. These checks may be performed by the flight/element personnel by visual means (obvious rips or tears) for dental and medical clinics. Additional checks of defects may be performed using x-ray imaging

(fluoroscopy recommended). Each MTF must develop proper procedures for disposal of rejected lead protective material.

4.5.12. The machine operator shall stand behind a barrier, if provided, and shall observe the patient during diagnostic or therapeutic procedures. The operator shall be able to see, speak to, and hear the patient.

4.5.13. Radiographic film shall not be used beyond its expiration date. Unprocessed film shall be protected by appropriate shielding and should not be exposed to more than 0.002 mGy (0.2 mrad) prior to use.

4.6. Exposure of Fertile Women to Radiation for Treatment or Diagnosis. Due to the increased radiation exposure risks to the conceptus (including all embryonic and fetal development stages), the possibility of pregnancy shall be addressed for any woman of reproductive capacity when considering any nuclear medicine procedure and/or any radiographic examination involving the lower abdomen.

4.6.1. Before any medical procedure involving radiation is performed, the patient shall be asked if she is pregnant; if the patient is pregnant or there is a possibility that she could be pregnant, the attending physician shall be immediately informed. Before any therapeutic procedure, a pregnancy test shall be given to all fertile women no sooner than 48 hours preceding the treatment. Any inconclusive pregnancy test or statement made by the patient as to the potential of being pregnant should delay treatment until a conclusive negative pregnancy test can be obtained. **Note:** A pregnancy test cannot detect the presence of an embryo until several days post implantation.

4.6.2. Ideally elective abdominal/pelvic examination of a woman of childbearing age should be performed during the first 14 days following the onset of menses to minimize the possibility of irradiation during pregnancy. In practice, the timeliness of medical needs should be the primary consideration in deciding the timing of an examination.

4.6.3. Radiographic and nuclear medicine examination and/or waiting rooms, other than those used exclusively for dentistry, shall be posted with appropriate signs alerting patients that if they may be pregnant to notify the physician or technologist before the examination.

4.6.4. In the event of a conceptus exposure (i.e., the fetus or embryo is exposed to radiation) the responsible clinic shall:

4.6.4.1. Have a qualified medical physicist (**Attachment 6**) determine by measurement or modeling the estimated dose to the conceptus. These calculations shall be reported to the patient's physician who will interpret them to the patient.

4.6.4.2. Adhere to the requirements of **paragraph 4.14**.

4.7. Quality Control (QC). A QC program provides adequate confidence that a diagnostic x-ray or nuclear medicine clinic will produce consistently high quality images with minimum dose to the patients and medical staff. Similarly, a radiation therapy QC program assists in assuring treatments are performed with the optimum dose to tumor tissues and minimal dose to healthy tissue of the patient with the least possible exposure to medical staff.

4.7.1. Diagnostic Imaging Clinics and Other Clinics Utilizing Radiation Sources. All clinics, utilizing radiation sources (i.e., non-radioactive materials) for the purposes of medical or dental imaging shall implement a QC program, approved by the regional consulting medical

physics office, with the objective to minimize patient and staff dose while obtaining optimal clinical objectives. As a minimum, the program shall consist of the following elements:

4.7.1.1. Acceptance Tests. Medical physics testing shall be based on national standards and guidance as appropriate (i.e., ACR credentialing standards, AAPM task group reports, manufacturer QC instructions, NEXT data, abide all pertinent federal regulations (e.g., 21 CFR 1020), AF Instructions (e.g., AFI 41-201 and AFI 48-148), and manufacturer QC instructions).

4.7.1.1.1. A qualified diagnostic imaging medical physicist, defined in **paragraph 4.3.3**, shall perform acceptance testing (including image quality and dose assessment, as appropriate), prior to clinical use or any major component change, on the following equipment: mammography imaging devices, computed tomography systems, high-level fluoroscopy systems, magnetic resonance imaging systems, computed radiography devices, digital radiography systems, nuclear medicine imaging systems, interventional radiology systems, and cardiac catheterization systems. **Note:** Acceptance testing for picture archiving systems (PACS), telemedicine systems (for radiation related applications such as teleradiology and teledental systems), and diagnostic review workstations are performed by Air Force Clinical Engineering (AFMOA/SGALE). Annual compliance testing should be performed by medical physicist or qualified MERC/local BMET.

4.7.1.1.2. A qualified MERC technician or local BMET, in coordination with the regional consulting medical physics office, may perform acceptance testing of computed tomography, computed radiography, and digital radiography systems located at regionally supported sites, sustainment sites, and expeditionary locations. Final reports must be forwarded to the designated regional consulting medical physics office for review.

4.7.1.1.3. All other diagnostic x-ray systems shall receive acceptance testing, prior to clinical use, by a qualified MERC technician or local BMET as defined in AFI 41-201. Acceptance testing for clinical use authorization shall not be performed by the vendor or their representative. Systems having undergone a significant component change (e.g., a new x-ray tube) shall be retested as appropriate prior to clinical use.

4.7.1.1.4. Final acceptance testing reports shall be placed in the Equipment Data File (EDF) at the owning MTF for the duration of system use.

4.7.1.1.5. Radiology Flight Commanders or the Radiology Services NCOIC shall ensure technique charts are provided to the regional medical physics office for preparation of ESE charts. They shall also ensure compliance with the QC programs and equipment testing requirements.

4.7.1.2. Compliance Inspections.

4.7.1.2.1. Qualified medical physicist shall perform an annual evaluation of the equipment specified in **paragraph 4.7.1.1.1**. These inspections should have a focus on dose and image quality.

4.7.1.2.2. Radiological compliance inspections shall be conducted at least annually on all diagnostic and dental x-ray systems.

4.7.1.2.2.1. Inspection results shall be maintained IAW AFI 41-201. A copy of each inspection report shall be forwarded to the equipment custodian. It is the responsibility of the equipment custodian to ensure the report is forwarded along with appropriate technique factors to the regional consulting medical physics office for computation of the ESE data as appropriate and IAW with **paragraph 4.2.5**.

4.7.1.3. Preventive Maintenance. Maintenance shall be performed in accordance with requirements of AFI 41-201. It is recommended that MERC/local BMET, the manufacturer, or a qualified third party contractor perform preventive maintenance on all diagnostic imaging devices to include those listed in **paragraph 4.7.1.1.1**. Any service contract should provide for additional site visits when problems occur.

4.7.1.3.1. The medical physicist should be informed of any major work/upgrades performed on the imaging devices listed in **paragraph 4.7.1.1.1**.

4.7.1.4. Contracted Mobile Imaging Services. Contracted mobile imaging services used on AF installations shall provide the MTF with recent QC data associated with the specific scanner prior to clinical use. This data shall be reviewed and acceptable to the MTF radiologist and/or regional consulting medical physics office. At minimum, patient and staff dose evaluation, image quality tests, and public exposure evaluation about the mobile unit (i.e., exposure for x-ray systems and Gauss line for magnetic resonance imaging systems) must be evaluated by a qualified medical physicist or qualified MERC/local BMET prior to clinical use.

4.7.1.5. Quality Control Program. Each clinic, including deployed AF locations, AF lead Joint Basing locations, and contracted mobile imaging services used on AF installations, shall establish a QC program for each diagnostic imaging modality in coordination with, and approved, by the MTF radiologist and/or regional consulting medical physics office. These programs shall be based on national standards and guidance as appropriate (e.g., ACR credentialing standards, AAPM task group reports, NEXT data, manufacturer's QC instructions) and abide all pertinent Federal regulations (e.g., 21 CFR 1020).

4.7.2. Nuclear Medicine Clinics. All nuclear medicine clinics shall implement a QC program approved by the regional consulting medical physics office with the objective to minimize patient and staff dose while obtaining optimal clinical objectives. As a minimum, the program shall consist of the following elements:

4.7.2.1. Acceptance Tests. Testing shall be based on national standards and guidance as appropriate (e.g., AAPM task group reports, ACR credentialing standards, manufacturer's QC instructions) and abide all pertinent Federal regulations (e.g., 10 CFR 35). Final acceptance testing reports shall be placed in the local BMET EDF at the owning MTF for the duration of system use.

4.7.2.1.1. A qualified diagnostic or nuclear medical physicist shall perform acceptance testing of all nuclear medicine imaging modalities (e.g., gamma cameras, positron emission tomography (PET), PET/CT imaging systems) prior to clinical use.

4.7.2.2. Radioactive Materials. Radioactive materials must be acquired and used IAW 40-201.

4.7.2.3. Compliance Inspections. Annual evaluations of imaging devices should be performed by a qualified diagnostic imaging or nuclear medicine medical physicist as specified in **paragraph 4.3.3**.

4.7.2.4. Preventive Maintenance. Maintenance shall be performed in accordance with requirements of AFI 41-201. It is recommended that the manufacturer or a qualified third party contractor perform preventive maintenance on gamma cameras and PET systems twice a year. The service contract should provide for additional site visits when problems occur. The medical physicist should be informed of any work performed on these imaging devices.

4.7.2.5. Contracted Mobile Imaging Services. Contracted mobile imaging services (e.g., gamma cameras and PET imaging systems) used on AF installations shall provide the MTF with recent QC data associated with the specific scanner prior to clinical use. This data shall be reviewed and approved by the MTF radiologist and/or regional consulting medical physics office. At minimum, image quality tests and public exposure evaluation about the mobile unit must be performed by a qualified nuclear medicine technologist and approved by the MTF radiologist or regional medical physics consulting office prior to clinical use.

4.7.2.5.1. The MTF's RAM permit must be amended by the permittee for authorized use location and possibly authorized RAM use prior to clinically using the mobile system. These requests must be submitted to and approved by the USAF RICS IAW AFI 40-201 prior to radioactive material procurement.

4.7.2.6. Quality Control Program. The clinic shall establish QC programs for each nuclear medicine imaging device and therapy modality in coordination with, and approved by, the MTF Nuclear Medicine Flight Commander and regional consulting medical physics office. These programs shall be based on national standards and guidance as appropriate (e.g., AAPM task group reports, ACR credentialing standards, manufacturer's QC instructions) and abide all pertinent Federal regulations (e.g., 10 CFR 19, 20, 35).

4.7.3. Radiation Therapy Clinics. All radiation therapy clinics shall implement a QC program approved by a qualified radiation therapy medical physicist with the objective to minimize staff dose while obtaining optimal clinical objectives. As a minimum, the program shall consist of the following elements:

4.7.3.1. Acceptance and Commissioning Tests. Testing shall be based on national standards and guidance as appropriate (e.g., AAPM task group reports, manufacturer's QC instructions) and abide all pertinent Federal regulations (e.g., 10 CFR 35, 21 CFR 1020). Specific testing requirements for external beam radiation therapy systems can be found in **paragraphs 4.11 and 4.12**. Final acceptance testing and commissioning reports shall be maintained by the therapy medical physics office for the duration of system use.

4.7.3.1.1. A qualified radiation therapy medical physicist shall perform acceptance testing and commissioning of all radiotherapy systems (e.g., linear accelerators and associated on-board imaging devices, brachytherapy systems, simulators, treatment planning systems, radiation oncology information management systems, diode or TLD dosimetry systems).

4.7.3.2. Radioactive Materials. RAM (i.e., brachytherapy sources and teletherapy units) must be authorized by the USAF RIC/S IAW AFI 40-201.

4.7.3.3. Compliance Inspections. Annual evaluations of radiation therapy devices/systems should be performed by a qualified radiation therapy medical physicist IAW **paragraphs 4.11 and 4.12** and should follow guidance provided by the AAPM.

4.7.3.4. Preventive Maintenance. Maintenance shall be performed in accordance with requirements of AFI 41-201. It is recommended that the manufacturer (preferred) or a qualified third party contractor perform preventive maintenance on radiation therapy devices and systems at least annually and when requested by the radiation therapy physicists. All work performed on these devices/systems must be reported to the radiation therapy medical physicist prior to clinical use.

4.7.3.5. Contracted Mobile Imaging Services. Contracted mobile services that are used or support AF radiation therapy clinics must comply with **paragraphs 4.7.1.4 and/or paragraph 4.7.2.5**.

4.7.3.6. Quality Control Program. The clinic shall establish QC programs for each modality in the therapy treatment chain. These programs shall be based on national standards and guidance as appropriate (e.g., AAPM task group reports) and abide all pertinent Federal regulations (e.g., 10 CFR 35). Specific requirements for external beam radiation therapy systems can be found in **paragraph 4.7.3.1; 4.7.3.4; 4.11; and 4.12**.

4.8. Specific Radiation Safety Requirements for Fluoroscopy. In addition to the general requirements specified under paragraph 4.5, the following requirements shall be applied to the use of fluoroscopic systems:

4.8.1. Specific Radiation Safety Requirements for General Fluoroscopy.

4.8.1.1. Fluoroscopic tabletop radiation exposure rates and spot films/cine exposures shall be measured annually. The regional and/or MTF medical physicist shall use this data to provide patient ESE calculations to each MTF or clinic. Clinicians shall be aware of the radiation exposure rates for the specific fluoroscopic system and for each mode of operation used during the clinical protocol.

4.8.1.2. The radiation exposure rate used in fluoroscopy should be as low as is consistent with fluoroscopic requirements and not exceed those listed in 10 CFR 1020.32 at the point where the beam enters the patient.

4.8.1.3. The total exposure time, dose area product (if available), cumulative absorbed dose (if available), physician, procedure type, and specific unit should be recorded in the patient's medical record and a log maintained for each unit for general fluoroscopy, angiography, and cardiac catheterization lab. This data should be monitored by a radiologist or qualified user, medical physicist, or technologist on a monthly or quarterly basis to identify equipment, procedural, or personnel issues resulting in higher than average exposures (i.e., average exposure values are listed in AAPM Report No. 58, *Managing the Use of Fluoroscopy in Medical Institutions*).

4.8.1.4. Tight collimation, short irradiation times, and minimal use of magnification shall be employed and consistent with clinical objectives. Always position the image intensifier as close to the patient as possible while maximizing the distance between the

x-ray tube and the patient. **Note**: for newer systems, a variable source-to-image distance option may be available. Review the operator's guide and/or consult with the regional medical physicist.

4.8.1.5. Medical fluoroscopy shall be performed only by or under the immediate supervision of a physician granted x-ray user privileges. The physician shall understand the proper use and limitations of the device to avoid needless exposure of the patient and other persons in the vicinity during use.

4.8.1.6. Leaded protective aprons shall be worn in the fluoroscopy room by all medical and support staff during all fluoroscopy procedures. **Note**: Radiation PPE normally ranges from 0.25-1 mm lead equivalent; a qualified expert (e.g., medical physicist or bioenvironmental engineer, as defined in the Glossary) shall specify and recommend all protective shielding for the specific modality.

4.8.1.7. Leaded gloves of at least 0.25 mm lead equivalence should be worn by physicians during near beam work. The physician shall not place hands in the direct beam, even with lead glove protection. If not using gloves, a ring dosimeter must be worn. It is recommended that ring dosimeters be worn when using lead gloves (i.e., ring dosimeter worn on finger inside of leaded glove).

4.8.1.8. Eye protection in the form of leaded glass shields or leaded (prescription) eye glasses should be used for clinicians with high fluoroscopic workloads, such as those frequently performing interventional angiography and cardiac catheterization procedures.

4.8.2. Specific Radiation Safety Requirements for Portable Fluoroscopic and Radiographic Procedures. In addition to the general requirements specified under **paragraph 4.5**, the following additional requirements are applicable to portable fluoroscopic and radiographic procedures, with the exception of microampere orthopedic units.

4.8.2.1. All unessential personnel, if possible, shall stand at least 2 m (approximately 6 ft) from the patient, the x-ray tube and the useful beam during procedures.

4.8.2.2. Portable equipment should be used only for examinations when it is not practical or of necessary expediency to transfer patients to locations of fixed radiographic or fluoroscopic units.

4.8.2.3. The operator and any individual within 2 m (approximately 6 ft) of the patient or the x-ray tube shall wear protective devices (i.e., lead aprons, thyroid shields) during any exposure.

4.8.3. Specific Radiation Safety Requirements for Cardiac and Fluoroscopically Guided Procedures. In addition to the general requirements specified under **paragraph 4.5**., the following additional requirements are applicable during cardiac and interventional radiography procedures and pain management procedures:

4.8.3.1. During serial (cine) radiography, the number of frames per second and the duration of the procedure should be kept to a minimum, consistent with clinical objectives. The total number of cine frames should be recorded in the patient's medical record and a maintained log for each unit.

4.8.3.2. Cardiac fluoroscopy shall be performed only by or under the immediate supervision of a physician properly trained in cardiac fluoroscopic procedures.

4.8.3.3. Avoidance of serious x-ray-induced skin injuries to patients during fluoroscopically-guided procedures. (reference Burlington B.D., *FDA Public Health Advisory: Avoidance of Serious X-Ray-Induced Skin Injuries to Patients During Fluoroscopically-Guided Procedures*). To avoid skin doses which may result in acute injury, clinics performing fluoroscopically guided procedures shall:

4.8.3.3.1. Establish and document standard operating procedures and clinical protocols for each specific type of interventional procedure performed. The protocols should address all aspects of the procedure, such as patient selection, normal conduct of the procedure, actions in response to complications and consideration of limits on fluoroscopy exposure time. Protocols should also include a mechanism to report estimated patient exposure to the patient's primary care physician for follow-up of lengthy procedures that might result in delayed effects.

4.8.3.3.2. Assess the impact of each procedure's protocol for the potential of radiation injury to the patient.

4.8.3.3.3. Modify the protocol, as appropriate; to minimize the cumulative absorbed dose to any irradiated area of the skin to accomplish the clinical tasks, and to avoid cumulative doses that would induce unacceptable adverse effects. Use equipment that aids in minimizing absorbed dose.

4.8.3.3.4. Obtain assistance from a qualified diagnostic medical physicist as needed, in implementing these protocols to ensure the clinical image quality/dose objectives of the procedure are not adversely affected.

4.8.3.3.5. Report, IAW AFI 44-119, *Medical Quality Operations*; the Safe Medical Devices Act of 1990 (SMDA); and 21 CFR Parts 803 to 807a, any fluoroscopically induced injuries, following the procedures established by the facility for such mandatory reporting. Practitioners who become aware of any medical device related adverse event or malfunction should report to their Medical Device User Facility Reporting individual. **Note:** Radiation induced injuries from fluoroscopy are not immediately apparent. Other than the mildest symptoms, such as transient erythema, the effects of the radiation may not appear until weeks following exposure. Providers performing these procedures may not be in direct contact with the patients following the procedure and may not observe the symptoms when they occur. Cumulative dose information therefore should be recorded in the patient's medical record and on the appropriate form in the AF Form 2100A series. Patients should be informed via consent form SF 522, *Request for Administration of Anesthesia and for Performance of Operations and Other Procedures*, of potential injuries and advised to report signs and/or symptoms of radiation induced injury to their attending physician.

4.9. Specific Radiation Safety Requirements for Mammography Procedures. In addition to the general requirements specified under paragraph 4.5, the following requirements are applicable during mammography procedures:

4.9.1. All mammography services offered by AF clinics shall adhere to the requirements of the *Mammography Quality Standards Act*, Title 21 CFR Parts 16 and 900.

4.9.2. Mammography imaging technologists shall meet the Federal standards of the MQSA (Public Law 102-539), and follow the American Society Radiation Technologists' Position Statements and Active Practice Standards for technologists.

4.10. Specific Radiation Safety Requirements for Computed Tomography Procedures. In addition to the general requirements specified under paragraph 4.5, the following requirements are applicable to computed tomography procedures.

4.10.1. The slice thickness and collimator pitch should be as great as practical and the number of slices in the study should be as small as possible.

4.10.2. Contrast studies should be performed only when necessary to provide critical diagnostic information.

4.10.3. The user shall be familiar with the relationship between patient dose (i.e., both the maximum value and its distribution) and the exposure scanning technique factors (i.e., kVp, mAs per slice, slice thickness and collimator pitch). Special attention shall be given to pediatric patients; patient size shall be matched to appropriate dose-reducing exposure settings, and inappropriate provider referrals should be prohibited.

4.10.4. A qualified diagnostic medical physicist shall measure the radiation output of computed tomography systems on an annual basis, and compute the dose for all applicable clinically used exposure scanning techniques, or when a major change (i.e., tube replacement) occurs in the system or in system operation. The computed tomography dose index (CTDI) or the multiple scan average dose (MSAD) shall be determined for common system parameters using the current procedures prescribed by the AAPM (reference AAPM Monograph 20, *1991 AAPM Annual Summer School Proceedings held at University of California, Santa Cruz*). Qualified MERC/local BMET personnel approved by the regional consulting medical physics office may perform the radiation output measurements and provide the data to the consulting physicist for dose calculation.

4.11. Specific Requirements for External Beam Radiation Therapy Systems Using Energies Less Than 500 keV. In addition to the general requirements specified under paragraph 4.5, the following requirements are applicable to radiation therapy systems operated at a potential less than 500 keV:

4.11.1. Operating Procedures.

4.11.1.1. When a patient must be held in position for radiation therapy, mechanical supporting or immobilization aids shall be used.

4.11.1.2. The tube housing assembly shall not be held by an individual during operation unless the assembly is designed to require such holding and the peak tube potential of the system does not exceed 50 kVp. In such cases, the holder shall wear protective gloves and lead apron of not less than 0.5 millimeters lead equivalency at 100 kV.

4.11.1.3. A copy of the current operating and emergency procedures shall be maintained at the therapeutic radiation machine control console.

4.11.2. Record Keeping. The clinic shall maintain a record of each calibration for five years. The record shall include: the date of the calibration; the manufacturer's name, model number, and serial number for both the therapeutic radiation machine and the x-ray tube; the model numbers and serial numbers of the instruments used to calibrate the therapeutic radiation

machine; and the signature of the radiation therapy medical physicist responsible for performing the calibration.

4.11.3. Periodic QC Checks.

4.11.3.1. Periodic QC checks shall be performed on therapeutic radiation machines IAW written procedures established by a qualified radiation therapy medical physicist with the procedures specifying the frequency at which tests or measurements are to be performed, and the acceptable tolerance of the check. Checks shall include monthly assessment of:

4.11.3.1.1. Electrical interlocks at each external beam radiation therapy room entrance;

4.11.3.1.2. The "BEAM ON" and termination switches;

4.11.3.1.3. Beam condition indicator lights on the access door(s), control console, and in the radiation therapy room;

4.11.3.1.4. Viewing systems; and

4.11.3.1.5. If applicable, electrically operated treatment room doors from inside and outside the treatment room.

4.11.3.2. The cause for a parameter exceeding a tolerance limit set by the radiation therapy medical physicist shall be investigated and corrected before the system is used for patient irradiation.

4.11.3.3. Whenever a QC check indicates a significant change in the operating characteristics of a system, as specified in the radiation therapy medical physicist's QC check procedures, the system shall be recalibrated as required.

4.11.3.4. The clinic shall have the radiation therapy medical physicist review and sign the results of each radiation output QC check within 1 month of the date that the QC was performed.

4.11.3.5. The clinic shall maintain a record of each QC check for five (5) years. The record shall include: the date of the QC check; the manufacturer's name, model number, and serial number of the therapeutic radiation machine; the manufacturer's name; model number and serial number for the instrument(s) used to measure the radiation output of the therapeutic radiation machine; and the signature of the individual who performed the periodic QC check.

4.12. Specific Requirements for External Beam Radiation Therapy Systems using Energies Greater Than 500 keV. In addition to the general requirements specified under paragraph 4.5, the following requirements are applicable to radiation therapy systems operated at a potential energy of 500 keV and above, and for electron or proton therapy systems:

4.12.1. **Radiation Therapy Medical Physicists:** The services of a qualified radiation therapy medical physicist shall be required in facilities having therapeutic radiation machines with energies of 500 kVp and above. The radiation therapy medical physicist shall be responsible for:

4.12.1.1. Full calibration(s) and protection surveys;

4.12.1.2. Supervision and review of patient dosimetry;

4.12.1.3. Beam data acquisition and transfer for computerized dosimetry, and supervision of its use;

4.12.1.4. Quality assurance, including QC check review;

4.12.1.5. Consultation with the radiation oncologist in treatment planning, as needed; and

4.12.1.6. Performance of calculations/assessments regarding medical events. If the radiation therapy medical physicist is not a full time employee of the MTF, the operating procedures required by **paragraph 4.12.2**. shall also specifically address how the radiation therapy medical physicist is to be contacted for problems or emergencies, as well as the specific actions, if any, to be taken until the radiation therapy medical physicist can be contacted.

4.12.2. Operating Procedures.

4.12.2.1. No individual, other than the patient, shall be in the treatment room during treatment or during any irradiation for testing or calibration purposes.

4.12.2.2. Therapeutic radiation machines, when not in operation, shall be secured to prevent unauthorized use.

4.12.2.3. When adjustable beam-limiting devices are used, the position and shape of the radiation field shall be indicated by a light field.

4.12.2.4. If a patient must be held in position during treatment, mechanical supporting or immobilization devices shall be used.

4.12.2.5. A copy of the current operating and emergency procedures shall be maintained at the therapeutic radiation machine control console.

4.12.3. Acceptance Testing, Commissioning and Full Calibration Measurements.

4.12.3.1. Acceptance testing, commissioning and full calibration of a therapeutic radiation machine subject to **paragraph 4.12** shall be performed by a qualified radiation therapy medical physicist meeting the requirements in **paragraph 4.3.3.5**.

4.12.3.2. Acceptance testing shall be performed IAW manufacturer's recommendations and any additional contractual requirements following installation or reinstallation of the therapeutic radiation machine and prior to clinical use. Commissioning should be accomplished using, *AAPM Code of Practice for Radiotherapy Accelerators: Report of AAPM Radiation Therapy Task Group 45,* and any updated/new and applicable AAPM reports following installation or reinstallation of the therapeutic radiation machine, and shall be conducted prior to clinical use of that particular beam or modality.

4.12.3.3. Full calibration should include measurement of all parameters specified in Table II of *Comprehensive QA for Radiation Oncology: Report of AAPM Radiation Therapy Committee Task Group 40* and should be performed IAW *AAPM Code of Practice for Radiotherapy Accelerators*: *Report of AAPM Radiation Therapy Task Group 45*. Although it shall not be necessary to complete all elements of a full calibration at the same time, all parameters (for all energies) shall be completed at intervals not exceeding 12 calendar months. Any new applicable AAPM reports associated with equipment testing that is associated with the radiation therapy process should be considered as well.

4.12.3.4. The radiation therapy medical physicist shall perform all elements of a full calibration necessary to determine that all parameters are within acceptable limits:

4.12.3.4.1. Whenever QC check measurements indicate that the radiation output differs by more than 2% from the value obtained at the last full calibration and the difference cannot be reconciled; therapeutic radiation machines with multi-energy and/or multi-mode capabilities shall only require measurements for those modes and/or energies that are not within their acceptable range; and

4.12.3.4.2. Following any component replacement, major repair, or modification of components that could significantly affect the characteristics of the radiation beam. If the repair, replacement or modification does not affect all modes and/or energies, measurements shall be performed on the effected mode/energy that is in most frequent clinical use at the department.

4.12.3.5. The clinic shall maintain a record of each calibration in an auditable form for five years. The record shall include: the date of the calibration; the manufacturer's name, model number and serial number for the therapeutic radiation machine; the model numbers and serial numbers of the instruments used to calibrate the therapeutic radiation machine; and the signature of the radiation therapy medical physicist responsible for performing the calibration.

4.12.4. Periodic QC Checks.

4.12.4.1. Periodic QC checks shall be performed on all therapeutic radiation machines subject to **paragraph 4.12** at intervals not to exceed those specified in "*Comprehensive QA for Radiation Oncology*: Report of AAPM Radiation Therapy Committee Task Group 40" or more current report

4.12.4.2. QC checks shall include determination of central axis radiation output and a representative sampling of periodic QC checks contained in, *Comprehensive QA for Radiation Oncology: Report of AAPM Radiation Therapy Committee Task Group 40*. Representative sampling shall include all referenced periodic QC checks in an interval not to exceed 12 consecutive calendar months.

4.12.4.3. The clinic shall perform periodic QC checks IAW procedures established by the radiation therapy medical physicist.

4.12.4.4. The clinic shall review the results of each periodic radiation output check according to the following procedures:

4.12.4.4.1. The radiation oncologist or radiation therapy medical physicist shall be immediately notified if any parameter is not within its acceptable tolerance. The therapeutic radiation machine shall not be made available for subsequent clinical use until the radiation therapy medical physicist has determined that all parameters are within their acceptable tolerances.

4.12.4.4.2. If all QC check parameters appear to be within their acceptable range, the QC check shall be reviewed and signed by either the radiation oncologist or radiation therapy medical physicist.

4.12.4.4.3. The clinic shall have the radiation therapy medical physicist review and sign the results of each radiation output QC check within one (1) month of the date that the QC was performed.

4.12.4.5. Therapeutic radiation machines subject to 4.12. shall have safety QC checks listed in *Comprehensive QA for Radiation Oncology: Report of AAPM Radiation Therapy Committee Task Group 40* performed at intervals not to exceed one (1) week; these checks shall ensure proper operation of:

4.12.4.5.1. Electrical interlocks at each external beam radiation therapy room entrance;

4.12.4.5.2. Proper operation of the "BEAM-ON", interrupt and termination switches;

4.12.4.5.3. Beam condition indicator lights on the access doors, control console, and in the radiation therapy room;

4.12.4.5.4. Viewing systems; and

4.12.4.5.5. Electrically operated treatment room door(s) from inside and outside the treatment room.

4.12.4.6. Emergency power cutoff switch. If more than one emergency power cutoff switch is installed and not all switches are tested at once, each switch shall be tested on a rotating basis (monthly or by manufacturer recommendation). Safety QC checks of the emergency power cutoff switches may be conducted at the end of the treatment day in order to minimize possible stability problems with the therapeutic radiation machine.

4.12.4.7. The clinic shall promptly repair any system identified above that is not operating properly.

4.12.4.8. The clinic shall maintain a record of each QC check for five (5) years. The record shall include: the date of the QC check; the manufacturer's name, model number, and serial number of the therapeutic radiation machine; the manufacturer's name; model number and serial number for the instrument(s) used to measure the radiation output of the therapeutic radiation machine; and the identity of the individual who performed the periodic QC check.

4.12.5. The following information or records shall be maintained by the MTF in auditable form:

4.12.5.1. Reports of acceptance testing;

4.12.5.2. Records of all surveys, calibrations, and periodic QC checks of the therapeutic radiation machine required by **paragraphs 4.11 and 4.12**, as well as the name(s) of person(s) who performed such activities;

4.12.5.3. Records of maintenance and/or modifications performed on the therapeutic radiation machine after the date of this instruction, as well as the name(s) of person(s) who performed such services; and

4.12.5.4. Signature of person authorizing the return of therapeutic radiation machine to clinical use after service, repair, or upgrade.

4.13. Report and Notification of a Medical Event.

4.13.1. An AF radiation therapy clinic shall report all radiation therapy medical events as specified in **paragraphs 4.13.3** and **4.13.4**. A radiation therapy medical event is defined as an adverse event which places the patient at risk of injury (except for an event that results from patient intervention) in which the administration of machine produced radiation results in:

4.13.1.1. A dose that exceeds 0.05 Sv (5 rem) effective dose equivalent, 0.5 Sv (50 rem) to an organ or tissue, or 0.5 Sv (50 rem) shallow dose equivalent to the skin, and any of the following:

4.13.1.1.1. The total dose delivered differs from the prescribed dose by 10% or more;

4.13.1.1.2. The fractionated dose delivered exceeds the prescribed dose, for a single fraction, by 50% or more;

4.13.1.1.3. An administration of a dose or dosage to the wrong individual;

4.13.1.1.4. An administration of a dose or dosage delivered by the wrong mode of treatment; or

4.13.1.1.5. A geographic miss causing a dose to an organ or tissue other than the treatment site that exceeds 50% or more of the prescribed fraction size.

4.13.2. The clinic shall report any event resulting from intervention of a patient or human research subject in which the administration of machine produced radiation results or will result in an unintended permanent functional damage to an organ or a physiological system, as determined by a physician.

4.13.3. The clinic shall notify the responsible MTF's SGH, or their designee, no later than the next calendar day after discovery of the medical event. The SGH will then determine, based upon consultation with the responsible radiation oncologist, radiation therapy physicist, and referring physician or clinic, whether or not the event involves risk of permanent injury to the patient or subject. If such risk is determined to be present, the event becomes a reportable medical event. The clinic will then telephonically report the medical event to the AFMSA/SG3PB (Radiation Program) within seven (7) calendar days.

4.13.4. The clinic shall submit a written report to AFMSA/SG3PB (Radiation Health) through the responsible MAJCOM/SGP within 15 calendar days after discovery of the medical event.

4.13.4.1. The written report must include:

4.13.4.1.1. The clinic's organization and base;

4.13.4.1.2. The name and unit of the prescribing physician, Chief of Radiation Oncology, and radiation therapy medical physicist;

4.13.4.1.3. A brief description of the event and its cause;

4.13.4.1.4. The effect, if any, on the individual(s) that received the radiation therapy;

4.13.4.1.5. The actions that have been taken or are planned to prevent recurrence;

4.13.4.1.6. The status of notification to the individual (or the individual's responsible relative or guardian) and any information that was provided; and

4.13.4.1.7. The report may not contain personal identifiable information.

4.13.5. Notification.

4.13.5.1. The clinic shall notify the referring physician, or, if unavailable, the responsible referring clinic chief within 24 hours.

4.13.5.2. The individual affected by the medical event shall also be notified no later than 24 hours after its discovery, unless the referring physician or clinic chief elects to personally inform the individual or that, based on medical judgment, telling the individual would be harmful.

4.13.5.3. The clinic is not required to notify the individual without first consulting the referring physician. However, if the referring physician or the affected individual cannot be reached within 24 hours, the clinic shall notify the individual as soon as possible thereafter.

4.13.5.4. The clinic may not delay any appropriate medical care for the individual, including any necessary remedial care as a result of the medical event, because of a delay in notification. Therefore to meet the requirements of this section, notification may be made that individual's responsible relative or guardian.

4.13.6. If the individual was notified under **paragraph 4.13.5** of this section, the clinic shall also provide a written report to the individual within 15 days after discovery of the medical event. A clinic may send either:

4.13.6.1. A copy of the report that was submitted to the MAJCOM/SGP; or

4.13.6.2. A brief description of both the event and the consequences that may affect the individual.

4.13.7. The clinic shall retain a record of a medical event for 50 years per the Air Force records Disposition Schedule (AFRIMS). The record shall contain the information specified in **paragraph 4.13.4.1**

4.14. Report and Notification of a Dose to a Conceptus (Embryo/Fetus) or Nursing Child.

4.14.1. A medical or dental clinic shall report any cumulative absorbed dose to a conceptus (i.e., embryo/fetus) or nursing child that is greater than 50 mSv (5 rem) dose equivalent that is a result of the administration of machine produced radiation and/or RAM to a pregnant individual or nursing mother from studies/treatments where the resultant dose was not specifically approved, in advance, by the requesting physician.

4.14.2. The clinic shall notify the responsible MTF's Chief of Hospital Services (SGH), or their designee, and the IRSO within 24 hours, and notify by telephone AFMSA/SG3PB (Radiation Health) no later than seven (7) calendar days after discovery of a dose to the conceptus that requires a report in **paragraph 4.14.1**.

4.14.3. The clinic shall submit a written report to AFMSA/SG3PB (Radiation Health) through the IRSO and appropriate MAJCOM/SGP no later than 15 days after discovery of a dose to the conceptus or nursing child that requires a report in **paragraph 4.14.1**

4.14.3.1. The written report must include:

4.14.3.1.1. The organization and base;

4.14.3.1.2. The name and organization of the prescribing physician, chief of the clinic, and responsible medical physicist or RSO;

4.14.3.1.3. A brief description of the event;

4.14.3.1.4. Why the event occurred;

4.14.3.1.5. The effect, if any, on the conceptus or nursing child; and

4.14.3.1.6. What actions, if any, have been taken or are planned to prevent recurrence.

4.14.4. Notification.

4.14.4.1. The clinic shall notify the referring physician, or if unavailable, the referring physician's clinic chief and the pregnant individual , hereafter referred to as the mother, no later than 24 hours after discovery of an event that would require reporting under **paragraph 4.14.1** , unless the referring physician or chief of clinic personally informs the clinic either that he or she will inform the mother or that, based on medical judgment, telling the mother would be harmful.

4.14.4.2. To meet the requirements of this section, the notification may be made to the mother's or child's responsible relative or guardian instead of the mother, when appropriate.

4.14.4.3. The clinic is not required to notify the mother without first consulting with the referring physician. If the referring physician or mother cannot be reached within 24 hours, the clinic shall make the appropriate notifications as soon as possible thereafter.

4.14.4.4. The clinic may not delay any appropriate medical care for the embryo/fetus or for the nursing child, including any necessary remedial care as a result of the event, because of any delay in notification.

4.14.5. If notification was made under **paragraphs 4.14.3 or 4.14.4** , the clinic shall also furnish a written report to the mother or responsible relative or guardian within 15 days after discovery of the event. The clinic may send either of the following:

4.14.5.1. A copy of the report that was submitted to AFMSA/SG3PB (Radiation Health)

4.14.5.2. A brief description of both the event and the consequences as they may affect the conceptus or nursing child.

4.14.6. A clinic shall retain a record of a dose to an embryo/fetus or a nursing child reported in accordance with this section for 50 years per the Air Force Records Information management System. The record shall contain the information specified in **paragraph 4.14.3.1**.

Chapter 5

RADIATION PROTECTION OF THE PUBLIC

5.1. General. Public dose is the dose received by a member of the public from exposure to radiation or radioactive material used in AF practices and radon. Public dose does not include occupational dose or doses from other background radiation sources or any medical administration the individual has received.

5.2. Control of Public Exposures from AF Practices. Administrative and engineering controls, policies, or procedures are implemented to control public dose from AF practices. The following requirements are specified for all AF practices that may result in public exposure:

5.2.1. All practices shall be conducted to minimize public dose considering the state of technology, economics of improvements in relation to the state of technology, the economics of improvements in relation to benefits to the public health and safety, and other societal and socioeconomic considerations.

5.2.2. Members of the public shall not occupy vehicles during cargo or vehicle inspections that use scanning systems employing radiation.

5.2.3. All practices shall be conducted in such a manner that a member of the public will not exceed the applicable dose limits specified in **Table A5.1**. For non-medical practices, the IRSO shall be responsible for the determination, either by measurement or calculation that this dose is not exceeded, and maintain this record permanently. For medical practices, the PRSO, responsible medical physicist, health physicist or BE, as appropriate, shall be responsible for this determination.

5.2.4. Unrestricted areas. The dose in any unrestricted area resulting from AF controlled radiation sources will not exceed 0.02 mSv (2 mrem) in any one hour, or 1 mSv (100 mrem) in a year, occupancy and use factors being taken into account. This requirement does not apply to those excepted by 10 CFR 35.75. The IRSO shall be responsible for the assessment, either by measurement or calculation, that these dose rates are not exceeded and maintain this record permanently.

5.2.4. **(MOODYAFB)** Although the area may be considered an unrestricted area, doses to the general public and occupational workers must remain ALARA, regardless of the dose limits and dose rate limits.

5.2.5. Control of Visitors.

5.2.5.1. Visitors to any restricted area must be accompanied by persons knowledgeable about the protection and safety measures in the area.

5.2.5.2. Visitors must be provided adequate information and instruction before they enter a restricted area to ensure appropriate protection of the visitors and of other personnel in the area.

5.2.5.3. Visitors entering defined "Radiation Areas" or "High Radiation Areas" or that could incur a deep dose equivalent in excess of 0.10 mSv (10 mrem) shall be provided personal monitoring devices. A log of all such monitored individuals shall be maintained and the results communicated to the individual.

5.2.6. Public Overexposures. If an individual member of the public may have received a dose in excess of the applicable limit in **Table A5.1**, that dose and practice shall be immediately investigated by the IRSO. If appropriate, the unit, organization or IRSO should implement, with commander's approval, protective actions to mitigate further exposures.

5.2.6.1. Notification: When a member of the public may have received a dose above the applicable limits in **Table A5.1**, the IRSO shall be contacted immediately. The IRSO shall contact USAFSAM/OEH to validate the assessment of dose. If the dose is validated, the IRSO should then notify, within three (3) hours, the MAJCOM BE and AFMSA/SG3PB.

5.2.6.2. Reporting: The IRSO, with the assistance of USAFSAM/OEH, shall investigate suspected exposures above the limits specified in **Table A5.1**, with a written report of the investigation submitted through the MAJCOM BE to AFMSA/SG3PB within seven (7) days of notification. The written report must include:

5.2.6.2.1. The organization's name and office symbol where the exposure occurred;

5.2.6.2.2. A brief description of the event;

5.2.6.2.3. A description of the person(s) exposed and their estimated dose equivalent;

5.2.6.2.4. Why the event occurred; and

5.2.6.2.5. What actions, if any, have been taken or are planned to prevent recurrence.

5.3. Protection of the Public from Avertable Doses. This section specifies requirements to protect members of the public from two specific sources of radiation: public exposures resulting from radon in AF controlled structures and public exposures that may occur from an incident or accident involving radioactive materials. For each case, interventions may be necessary to protect the public. The radon exposure at which an intervention is merited is referred to as a remedial action level. For incidents or accidents, a protective action or intervention is merited when dose can be avoided by conducting the intervention or implementing the protective action.

5.3.1. Radon Exposure: The following requirements are specified to address those facilities and operating locations not originally assessed in the AF Radon Assessment and Mitigation Program of 1987 (USAF OEHL Report 87-132RZ0111KRD, *The Bioenvironmental Engineer's Primer to the Air Force Radon Assessment and Mitigation Program (RAMP)*). These facilities or operating locations are medium or high risk if any of the screening measurements identified at least one area with radon concentration greater than 4 pCi/L (EPA medium risk guideline), or 20 pCi/L (EPA high risk guideline).

5.3.1.1. Newly constructed or acquired structures on installations designated as medium and/or high risk: Any new enclosed structure for installations that underwent a detailed assessment as part of the 1987 RAMP shall be assessed by the IRSO for exposure to radon progeny, with the exception of: hangars, maintenance bays, dedicated storage facilities, structures occupied less than four hours per day, temporary facilities, and elevated structures with unobstructed air flows underneath.

5.3.1.1.1. New structures should not be tested for one year after construction to allow for foundation settling. Contact USAFSAM for designing an appropriate sampling plan.

5.3.1.1.2. Monitoring shall be performed using a long term monitor (e.g., alpha track detector or long-term electret) deployed in the lowest occupied location of the structure.

5.3.1.1.3. Evaluation.

5.3.1.1.3.1. No further action is required for structures identified with annual average radon concentrations of less than 4 pCi/L.

5.3.1.1.3.2. Structures that exceed 4 pCi/L shall be remediated by CE to levels ALARA, but shall result in exposures less than 4 pCi/L. The remediation schedule should adhere to the following schedule: > 20 pCi/L within one year, 4 - 20 pCi/L, within 1-3 years.

5.3.1.1.3.3. **(Added-MOODYAFB)** Moody Air Force Base was classified as low risk by the RAMP (i.e. all samples from buildings built during or before 1987 had radon concentrations less than 4 pCi/L.

5.3.1.1.4. Remediated structures shall be reassessed by the IRSO for ambient radon concentrations no less than two weeks and no greater than six months post remediation to validate the efficacy of the remedial action. A combination of short-term and long-term monitoring should be conducted to assure acceptable radon levels have been achieved. If remediation is unsuccessful, maintain exposure limits at ALARA until further remediation actions are implemented.

5.3.1.2. New installations or installations never previously assessed: For all new, permanent operating locations, a random sampling of the structures shall be assessed for radon. USAFSAM shall be consulted on designing an appropriate sampling program. Any installation found to have a single structure with concentrations greater than 4 pCi/L shall undergo a detailed assessment.

5.3.1.3. CE should design and construct new structures on medium and high risk bases with radon resistant features.

5.3.2. Uncontrolled Exposures Resulting from Radiological Incidents and Accidents. Members of the public that are under the protection of an installation commander shall be protected during emergency situations from exposure to radiation to the greatest extent possible. Specific protective measures that can be implemented are listed in **Attachment 7**. The decision to invoke a protective action, i.e., sheltering or evacuation, should be based on whether the action will allow the population to avert (or avoid) the recommended threshold dose provided with a given action or intervention.

Chapter 6

RADIATION PROTECTION OF AF PERSONNEL DURING INTERVENTIONS

6.1. General. This chapter provides guidance for protecting AF personnel conducting actions in uncontrolled radiation environments. These environments may include deployed locations where known or suspected nuclear or radiological hazards exist and radiological environments created by hostile action or industrial, medical, nuclear incident or accident. Interventions are specific actions performed in these environments to mitigate the source(s) of exposure, to save life or limb, protect high value assets, or achieve higher objectives that may merit personnel incur risks greater than those permitted for practices. This section also applies to nuclear capable units and units with 91(b) material. The following summarizes the radiation protection considerations described in Joint Publication, *Operations in Chemical, Biological, Radiological, and Nuclear (CBRN) Environments*, Aug 2008. For specific guidance refer to JP 3-11; USACHPPM TG 244, The Medical CBRN Battlebook, Oct 2008; and/or AFRRI's Medical Management of Radiological Casualties, Apr 2003.

6.2. Radiation Protection Principles and Guidance for Interventions.

6.2.1. There are three basic principles of ionizing radiation protection used to manage the risks associated with ionizing radiation exposure.

6.2.1.1. Justification. No unnecessary exposure should be undertaken.

6.2.1.2. Optimization. The level of individual doses, number of people exposed and likelihood of incurring exposure should be kept ALARA.

6.2.1.3. Limitation. The dose to an individual shall be limited according to appropriate DOD guidance.

6.2.2. Applying radiation protection principles should not introduce a higher level or more severe risk to the unit or mission. Complete risk management requires:

6.2.2.1. Information. Personnel must assess risk using an all hazards approach using all available information such as measurements, visual observations, and modeling.

6.2.2.1.1. **(Added-MOODYAFB)** During the all-hazards response, the IRSO is responsible for conducting a health risk assessment IAW AFMAN 48-153.

6.2.2.1.2. **(Added-MOODYAFB)** The health risk must be delivered within the operational context, especially if military operations take place (i.e. airlift into a contaminated area, recovery of classified material). Recommendations to control hazards, limit and reduce contamination, and maintain exposures ALARA will be analyzed and communicated.

6.2.2.2. Justification. Consideration of threshold and random health consequences of radiation exposure needs to be considered. Unnecessary risks to health should not be accepted.

6.2.2.3. Optimization. To minimize the potential effects of exposure, intervention planning should involve efforts to reduce the time in a radiation area, maintaining the

maximum distance possible from radiation sources, and using shields between exposed personnel and radiation sources to keep radiation exposures ALARA.

6.2.3. Dose guidance should be developed for a given intervention. Dose limits do not apply for interventions. Instead, the operational exposure guidance (OEG) is applicable. The commander's decision to allow this exposure should be based on the Radiation Exposure Status (RES) category found in JP 3-11 and made in the context of the situation and balance the anticipated benefit with both short and long-term health risks the exposure may cause. The exposure will vary depending on whether the mission is critical, priority, or routine and the severity of the radiological threat (catastrophic, critical, marginal, or negligible). An OEG should be set for each mission with potential for exposure, for decontamination of personnel or equipment, or for immediate or operational decontamination.

6.2.3.1. Dose due to the ingestion or inhalation of RAM cannot be accurately measured in the field but can be estimated for operational purposes. Depending on the type of RAM and its dispersed form, the internal dose may be much larger than the external dose for a given operation.

6.2.3.2. Medical authorities should recommend personal protective equipment (PPE), including respiratory protection. Commanders will dictate PPE use based on operational risk analysis. Operational risk analysis should include the detriment posed by reduced vision, hearing, mobility, and tactile sensation; increased heat and psychological stress; increased time to complete tasks, and decreased task accuracy when wearing PPE. These in turn can lead to increases in external radiation exposures and the magnitude of other mission risks.

6.2.4. Voluntary Participation. Where practical, interventions predicted to result in significant exposures (those greater than practice limits) should make use of volunteers to the greatest extent possible.

6.2.4.1. Experienced volunteers who do not plan on having more children may be preferred. The concern over temporary/permanent sterility is reduced. Also, experienced workers typically can complete an operation in a more timley manner, reducing their dose.

6.2.5. Excluded Populations. Operational personnel that are, or may be pregnant, or that are less than 18 years of age shall not conduct interventions.

6.3. Allowable Contamination Levels.

6.3.1. Recommended contamination levels for clothing and equipment or skin should be managed to levels that are ALARA using methods found in AFMAN 44-161(l), AFJMAN 44-151, and NUREG 1575 (most current version). In some operational environments where thorough decontamination options may be unavailable, skin contamination should not exceed the levels specified in NATO Standardization Agreement 2473.

6.3.2. Operational implementation of these contamination standards should follow procedures in Allied Engineering Publication-49, *NATO Handbook for Sampling and Identification of Radiological Agents (SIRA), Volume 1 (Operational) and 2 (Forensic)*, or guidance provided by the assigned radiation protection personnel.

6.4. Training and Risk Communication. Personnel participating in an intervention shall be informed of the potential health risks their radiation exposure may result in, and trained in the necessary principles and procedures to minimize their exposure consistent with **paragraph 3.3** of this instruction. Depending on the nature of the intervention, training should be provided by medical radiation protection personnel.

6.5. Monitoring During Interventions. Implementing operational risk management and supporting operational dose guidance requires the ability to estimate or measure individual doses. This is optimally done through the use of radiation survey instruments, environmental sampling, personnel dosimeters, and/or bioassays.

6.5.1. Personnel deployed in known or potential radiation environments shall be provided individual dosimeters if the potential to exceed 1 mSv (100 mrem) exists.

6.5.2. Personnel entering contaminated environments should be afforded in-vitro or in-vivo bioassays to determine the extent of internal exposures if internal contamination is suspected.

6.5.3. If individual dosimetry or prompt bioassay measurements are not available, efforts will be made to estimate individual doses through group dosimetry, radiation survey, environmental monitoring data, modeling, dose reconstructions, or other methods consistent with currently accepted scientific practice.

6.5.4. Systematic, individual dose records for external and internal exposures shall be maintained indefinitely, even if the dosimeter or bioassay result is zero. Results shall be maintained in the AF MRER, and the individual's medical record.

6.5.4.1. For units for which group dosimetry is used, doses as measured shall be averaged and applied to the entire group for the purposes of applying the Commanders OEG. The RSO supporting the unit shall track unit doses, determine the acceptability of the unit dose to reflect individual doses, and forward this dose information for each unit member for inclusion in the MRER and the individuals medical records.

6.5.4.2. Classification of exposure data may prevent its inclusion in the MRER. Medical authorities for the command conducting classified operations shall maintain and be able to access all classified exposure data, ensuring all releasable data is available for maintenance in the MRER and to the monitored individual.

6.5.5. Reports of dosimetry or bioassay results shall be given promptly to the potentially exposed individual and their commander or delegated representative.

6.5.6. The risks of stochastic health effects (e.g., cancer) are considered directly proportional to the total dose received by an individual. Commanders need to be aware of individual dose histories when planning future operations where radiation threats exist.

6.6. Medical Diagnosis and Treatment. Personnel receiving unknown doses of radiation or doses exceeding the limits established for practices should receive care IAW AF Manual 44-161 (l), *Treatment of Nuclear and Radiological Casualties*, NCRP Report 161, *Management of Persons Contaminated with Radionuclides*; and NATO Manual AMedP-6(B), *NATO Handbook on the Medical Aspects of NBC Defensive Operations, Part I - Nuclear*, or consistent with currently accepted practice.

6.7. Medical Surveillance.

6.7.1. Doses greater than 50 mSv (5 rem): On completion of a military operation or operations involving radiation exposure, long term, periodic health monitoring is required for individuals receiving cumulative effective doses in excess of 50 mSv (5 rem). Such follow-up shall be formed through USAFSAM, and may include, but not be limited to:

6.7.1.1. Creation of a registry for the impacted population;

6.7.1.2. Submission of biological or bioassay samples to determine absorbed dose and residual burdens of RAM, respectively; and/or

6.7.1.3. Annual or biannual medical examination, particularly following the latent periods of known radiogenic cancers. Known radiogenic cancers include leukemia, multiple myeloma, lymphomas, thyroid, breast, lung, esophageal, stomach, urinary tract, skin, and colon.

6.7.2. Doses less than 50 mSv (5 rem): For those personnel who have received doses less than current occupational dose limits, testing and monitoring should be limited to those testing and monitoring programs included in guidelines for the general population (e.g., routine mammography, pap smears, and prostate specific antigen testing).

6.7.2.1. Health monitoring may include submission of bioassay samples to document intakes of radionuclides, and screening for cancers consistent with current medical practices.

CHARLES B. GREEN
Lieutenant General, USAF, MC, SFS
Surgeon General

(MOODYAFB)

BILLY D. THOMPSON, Colonel, USAF
Commander

Attachment 1

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Abbreviations and Acronyms

AFRAT—Air Force Radiological Assessment Team

AAPM—American Association of Physicists in Medicine

ABC—Automatic Brightness Control

ABR—American Board of Radiology

ABH—American Board of Health Physics

ABMP—American Board of Medical Physics

ACR—American College of Radiology

ADCL—Accredited Dosimetry Calibration Laboratory

AFDIRB—Air Force Diagnostic Imaging and Radiotherapy Board

AFI—Air Force Instruction

AFIA—Air Force Inspection Agency

AF-RSC—Air Force Radiation Safety Committee

AF-RSCES—Air Force Radiation Safety Committee Executive Secretariat

AFMAN—Air Force Manual

AFMPWG—Air Force Medical Physics Working Group

AFMSA—Air Force Medical Support Agency

AFOSH—Air Force Occupational Safety and Health

AFPD—Air Force Policy Directive

AFRRAD—Air Force Radioactive Recycling And Disposal

AFSC—Air Force Specialty Code

ALARA—As Low As Reasonably Achievable

ALI—Annual Limits of Intake

AOR—Area of Responsibility

ARRT—American Registry of Radiologic Technologists

ASRT—American Society of Radiologic Technologists

BCB—Bioenvironmental Engineering (BE) Corporate Board

BE—Bioenvironmental Engineering

BEE—Bioenvironmental Engineer

BMET—Biomedical Equipment Technician

CE—Civil Engineering

CEDE—Committed Effective Dose Equivalent

CFR—Code of Federal Regulations

CBRNE—Chemical, Biological, Radiological, Nuclear and high-yield Explosive

COCOM—Combatant Command

CONUS—Continental United States

CME—Continuing Medical Education

CRCPD—Conference of Radiation Control Program Directors

CT—Computed Tomography

CTDI—Computed Tomography Dose Index

DHS—Department of Homeland Security

DMC—Deployed Medical Commander

EDF—Equipment Data File

EM—Emergency Management

EPA—Environmental Protection Agency

ESE—Entrance Skin Exposure

ESEG—Entrance Skin Exposure Guide

ESOH—Environmental, Safety, and Occupational Health

eV—Electron Volt

FDA—Food and Drug Administration

GSU—Geographically Separated Unit

IAW—In Accordance With

ICRP—International Commission on Radiological Protection

HLC—High Level Control

JTF—Joint Task Force

IAEA—International Atomic Energy Agency

INRAD—Intrinsic Radiation

IRSO—Installation Radiation Safety Officer

MAJCOM—Major Command

MERC—Medical Equipment Repair Center

MRER—Master Radiation Exposure Registry

MRI—Magnetic Resonance Imaging

MSAD—Multiple Scan Average Dose

MQSA—Mammography Quality Services Act

MTF—Military Treatment Facility

NARM—Naturally Occurring or Accelerator Produced Material

NATO—North Atlantic Treaty Organization

NCOIC—Non-Commissioned Officer In Charge

NCRP—National Council on Radiation Protection

NEXT—Nationwide Evaluation of X-ray Trends

NIST—National Institutes of Standards and Technology

NRC—Nuclear Regulatory Commission

OCONUS—Outside the Continental United States

ODG—Operational Dose Guidance

OEG—Operational Exposure Guidance

OPR—Office of Primary Responsibility

ORM—Operational Risk Management

PCi/L—picoCurie/Liter

PCM—Primary Care Manager

PET—Positron Emission Tomography

PPE—Personal Protective Equipment

PRSO—Permit Radiation Safety Officer

QA—Quality Assurance

QAP—Quality Assurance Program

QC—Quality Control

RADIAC—Radiation Detection Instrumentation and Calculation

RAM—Radioactive Material

rem—Roentgen Equivalent Man

RES—Radiation Exposure Status

RIC—Radioisotope Committee, see USAF RIC, reference AFI 40-201

RICS—Radioisotope Committee Secretariat, reference AFI 40-201

RSC—Radiation Safety Committee

RSO—Radiation Safety Officer

SMDA—Safe Medical Devices Act of 1990

SMO—Senior Medical Officer

SSN—Social Security Number

SSD—Source-to-Skin Distance

Sv—Sievert

SPECT—Single Photon Emission Computed Tomography

TDY—Temporary Duty

TEDE—Total Effective Dose Equivalent

TO—Technical Orders

URSO—Unit Radiation Safety Officer

USAF RIC—United States Air Force Radioisotope Committee

USAFSAM—United States Air Force School of Aerospace Medicine

WLM—Working Level Month

Terms

91(b) Material—RAM exempted from NRC licensing controls under Section 91(b) of the AEA of 1954, as amended, in the interest of national defense, under the possession of the DOD. These include the RAM in nuclear weapons – e.g., uranium isotopes, plutonium isotopes, tritium, and other radioactive components of nuclear weapons. These also include the components of nuclear reactors that fall under the definition of 91(b).

Absorbed Dose—The energy imparted by radiation per unit mass of irradiated material. The units of absorbed dose are the rad and the gray (Gy); (1 rad = 0.01 Gy).

Activity—The rate of disintegration (transformation) or decay of radioactive material. The units of activity are the Curie (Ci) and the Becquerel (Bq).

Administrative Dose—An arbitrary value assigned in a dose report in cases where a dosimeter is not returned for processing at the end of the wear period, is damaged, or which cannot be evaluated due to other factors. Administratively assigned doses must be investigated by the installation RSO as "Abnormal Exposures" IAW AFMAN 48-125.

ALARA—Acronym for “as low as is reasonably achievable” means making every reasonable effort to maintain exposures to radiation as far below applicable dose limits as is practical, consistent with the purpose for which the activity is undertaken, taking into account the state of technology, the economics of improvements in relation to benefits to the public health and safety, and other societal and socioeconomic considerations and in relation to utilization of nuclear energy, RAM, and radiation in the public interest.

Annual Limit of Intake—The derived limit of RAM taken into the body of an adult worker by inhalation or ingestion in a year. The ALI is the smallest value of intake of a given radionuclide in a year that would result in a committed effective dose equivalent of 0.05 Sv (5 rem) or a committed dose of 0.5 Sv (50 rem) to any organ or tissue.

Background Radiation—Radiation from cosmic sources; naturally occurring RAM, including radon (except as a decay product of source or special nuclear material); and global fallout as it

exists in the environment from the testing of nuclear explosive devices or from past nuclear accidents such as Chernobyl that contribute to background radiation. "Background radiation" does not include radiation from source, byproduct, or special nuclear materials regulated by the NRC or from NARM that the AF regulates through AFI 40-201.

Becquerel (Bq)—The International System of Units (SI) unit of radioactivity is equivalent to one disintegration per second (dps, or Bq). One curie (Ci) is equivalent to 3.7 E 10^{10} (37 billion) Bq.

Bioassay (Radio-bioassay)—The determination of kinds, quantities or concentrations, and, in some cases, the locations of RAM in the human body, whether by direct measurement (in-vivo bioassay) or by analysis and evaluation of materials excreted or removed from the human body (in-vitro bioassay).

Byproduct Material—(1) Any RAM (except source material and special nuclear material) yielded in or made radioactive by exposure to the radiation incident to the process of producing or utilizing special nuclear material; (2) the tailings or wastes produced by the extraction or concentration of uranium or thorium from ore processed primarily for its source material content, including discrete surface wastes resulting from uranium solution extraction processes. Underground ore bodies depleted by these solution extraction operations do not constitute "byproduct material'' within this definition.

Committed Dose Equivalent—The dose equivalent to organs or tissues that will be received from the intake of RAM by an individual during the 50-year period following the intake.

Conceptus—The developing human organism from conception until time of birth. Referred to as the embryo/fetus.

Curie (Ci)—A unit of radioactivity equal to 37 billion Becquerels. See definition of Becquerel.

Declared Pregnant Woman— A woman who informed her employer, in writing, of her pregnancy and the estimated date of conception.

Derived Air Concentration (DAC)—The concentration of a given radionuclide in air that, if breathed for a working year of 2,000 hours under conditions of light work (inhalation rate of 1.2 cubic meters of air per hour), results in an inhalation of one ALI.

Deterministic Effect—Biological effects for which the severity of the effect in an exposed individual varies with the dose, and for which a threshold usually exists (e.g., skin erythema and cataracts).

Diagnostic Source—In the healing arts, a source of ionizing radiation used in the diagnosis of injury or disease. Includes: x-ray units (fixed, portable, fluoroscopy, computed tomography, etc.), RAM (capsules, liquids or gases) used in nuclear medicine, and RAM used in a laboratory setting to perform "in-vitro" studies (on blood, urine, cells, etc.).

Deep Dose Equivalent—The dose assigned to personnel from **external** whole-body exposure, it is the dose equivalent at a tissue depth of one cm (1000 mg/cm^2) which is expressed in units of rem or Sievert (Sv).

Dose Equivalent—A quantity used for radiation protection to indicate the biological effectiveness of different radiations to cause stochastic health effects (e.g., cancer). The product of the absorbed dose in tissue (***DT***) and the quality factor (***Q***), and all other necessary modifying

factors at the location of interest where ***HT = DT Q***. The units of dose equivalent are the rem and Sievert (Sv). (0.01 Sv = 1 rem). The dose equivalent in Sv is equal to the absorbed dose in gray multiplied by the ***Q***; 1 Sv = 100 rem. Its purpose is to have a single unit, regardless of the type of radiation, describing the biological effect due to exposure to radiation on man

Effective Dose—The sum over specified tissues of the equivalent dose or dose equivalent in a tissue multiplied by a weighting factor for that tissue, w_T. Expressed in unit of Sievert (Sv) or rem (100 rem = 1 Sv).

Electron Volt (eV)—A unit of energy equal to 1.6 □ 10^{-19} joule.

Equivalent Dose—A quantity used for radiation protection to indicate the biological effectiveness of different radiations to cause stochastic health effects (e.g., cancer). The equivalent dose equals the absorbed dose multiplied by a radiation weighting factor, w_R. Expressed in units of Sievert (Sv) or rem.

Exposure—In radiation protection, the act or occurrence of being exposed to radiation or RAM. In risk management, the frequency and length of time subjected to a hazard.

Gray (Gy)—The SI unit of any absorbed dose. One gray is equal to the absorption of one joule per kilogram of material (1 Gy =100 rad).

Hazard—Any real or potential condition that can cause injury, illness, death of personnel, damage to or loss of equipment or property, or mission degradation.

High Radiation Area—Any area with dose rates greater than 100 millirem (1 milliSievert) in one hour 30 centimeters from the source or from any surface through which the ionizing radiation penetrates. Areas at licensee facilities must be posted as "high radiation areas" and access into these areas is maintained under strict control.

Installation—A grouping of facilities located in the same vicinity, which support particular functions. Installations may be elements of a base. The term “installation” applies to real properties such as depots, arsenals, hospitals, terminals, and other special mission installations.

Intervention—An activity that is not part of a controlled practice and is intended to reduce or mitigate sources of existing exposure (e.g., radon in structures); actions that save life or limb or mitigate threats greater than that posed by radiation; or those that are done to achieve higher objectives, including those of national security.

Intrinsic Radiation—Radiation emitted through the weapon surface or directly from exposed components of nuclear weapons.

Investigation Level—1) A dose equivalent value or radionuclide intake activity set by the installation RSO that requires further investigation when exceeded. Levels are normally tailored to each using section’s historical dosimetry data in order to promptly identify and correct adverse trends; (2) The CEDE from radioactive material ingested, inhaled, or otherwise taken into the human body or dose equivalent from an external radiation source to which the worker is occupationally exposed which justifies further investigation. Such an investigation generally includes a review of the circumstances associated with the apparently abnormal internal or external personnel dose equivalent, assessment of the consequences and mitigation or prevention of such a personnel dose equivalent of similar magnitude in the future.

Ionizing Radiation—Any electromagnetic or particulate radiation capable of producing ions, directly or indirectly during its passage through matter. It includes gamma rays, x-rays, alpha particles, beta particles, neutrons, protons and other particles and electromagnetic waves capable of producing ions.

Lens Dose Equivalent (LDE)—The dose equivalent to the lens of the eye from external exposure of the lens of the eye to some radiation source. It is measured at an eye lens tissue depth of 0.3 cm (300 mg/cm^2).

Medical Event—An event that meets the criteria in **paragraph 4.13.1**.

Member of the Public—Any individual except when that individual is receiving an occupational dose.

Occupational Dose—The dose received by an individual in the course of employment in which the individual's assigned duties involve exposure to radiation or to RAM from regulated and unregulated sources of radiation, whether in the possession of the licensee or other person. Occupational dose does not include dose received from background radiation; from any medical administration the individual has received; from exposure to patients administered RAM and released IAW applicable regulations; from voluntary participation in medical research programs; or as a member of the public.

Practice—Routine, controlled operations that produce radiation exposures as an unavoidable and unintentional aspect of the activity. For the purpose of this Instruction, exposures received as result of high-flyer operations.

Prescribed Dose—(1) For brachytherapy, stereotactic radiosurgery, or other radiation therapy procedures involving radioactive material, the total dose as documented in the written directive; (2) For external beam radiotherapy, the total dose and dose per fraction as documented in the written directive;

Public—All persons who are not already considered occupationally exposed by a source or practice under consideration.

Qualified Expert—A person who, by virtue of training and experience, can provide competent authoritative guidance about certain aspects of radiation safety or a person having knowledge and training to provide advice regarding radiation protection principles, standards and measurements. In general, a BE or health physicist with training and experience appropriate to the radiation protection issues to be managed is considered a qualified expert. Ideally, persons should have certification from the American Board of Health Physics, the American Board of Medical Physics, the American Board of Radiology, or the American Board of Industrial Hygiene, to be considered a qualified expert in these respective fields.

Note: A biomedical equipment repair technician is considered a qualified expert to support QC, equipment repair, collection of data and calibration IAW AFI 41—201.

Note:—Nuclear medicine or diagnostic radiology: A medical or health physicist may be considered a qualified expert when they have had modality specific training and performed three surveys for a particular type of equipment under the supervision of a board certified medical physicist or a medical physicist who meets all the education, training and experience necessary to complete all aspects of ACR or ABMP board certification.

Note:—Radiation therapy: A medical physicist must be supervised by a board certified radiation therapy physicist until all education, training and experience necessary to complete board certification is met.

Note:—Mammography physics: A qualified expert must meet the education, training and experience in FDA's MQSA, 10 CFR Part 900.

Note:—A senior health physicist or BE with readiness experience (i.e., a current or former member of AFRAT or BE NBC unit type codes) can be considered a qualified expert for purposes of accident/incident response and/or consequence management involving radiological and nuclear materials.

Quality Factor—The modifying factor (listed in 10 CFR 20.1004, tables 1004(B).1 and 1004(B).2) that is used to derive dose equivalent (biological effectiveness) from absorbed dose.

Rad—A conventional unit for the measurement of absorbed dose. One rad is equal to an absorbed dose of 100 ergs per gram of material (1 rad = 0.01 Gy = 1 cGy).

Radiation—For the purposes of this regulation, unless otherwise specified, ionizing radiation and specific, medical uses of non-ionizing radiation, such as radio- or microwaves, or visible, infrared, or ultraviolet light.

Radiation Area—Any area with radiation levels greater than 5 millirem (0.05 milliSievert) in one hour at 30 centimeters from the source or from any surface through which the radiation penetrates.

Radiation Safety—For the purposes of this instruction, a scientific discipline whose objective is the protection of people and the environment from unnecessary exposure to radiation. Radiation safety is concerned with understanding, evaluating, and controlling the risks from radiation exposure relative to the benefits derived. Same as "health physics" and "radiation protection."

Radiation Safety Committee—An advisory committee for the commander to assess the adequacy of an organization's radiation safety program. Same as "radiation control committee" and "radiation protection committee."

Radiation Safety Officer—The person that the commander designates, in writing, as the person responsible for the installation, organization or unit radiation safety program. It is the same as a "radiation protection officer" or "health physics officer." (Reference **Attachment 3**)

Radiation Safety Program—A program to implement the objectives of radiation safety regulations directives and instructions.

Radiation Source—Any non-exempt quantity of RAM, equipment, or devices which spontaneously generate or are capable of generating ionizing radiation. Examples include nuclear reactors, medical and dental radiographic and fluoroscopic x-ray systems, particle generators and accelerators, certain electromagnetic generators operating at electrical potentials that result in the production of x-rays, x-ray diffraction, industrial radiographic and spectrographic equipment, electron microscopes, electron-beam welding, melting, and cutting equipment, nuclear moisture or density gauges, byproduct, source, and special nuclear materials, natural or accelerator-produced radioactive materials, materials containing induced or deposited radioactivity and radioactive commodities.

Radiation Worker—An individual who may be occupationally exposed in the course of their duties or designated by the IRSO. This includes flight crews that routinely fly average flight profiles that are equal to or greater than 40,000 ft.

Radio-bioassay—See bioassay

Reference Levels—A dose established at one-quarter the applicable annual dose limit, for monitoring performed on a quarterly basis, and one-tenth the applicable annual dose limit for monitoring performed on a monthly basis.

Rem—The conventional unit of any of the quantities expressed as dose equivalent. The dose equivalent in rem is equal to the absorbed dose in rad multiplied by a radiation quality factor, Q.

Repeats (or retakes)—are those images that had to be obtained a second time due to error (dark, light, motion, processor, etc.) thus resulting in a repeated "exposure" to the patient.

Restricted Area—An area, access to which is limited by the facility for the purpose of protecting individuals against undue risks from exposure to radiation sources and RAM. Restricted area does not include areas used as residential quarters, but separate rooms in a residential building may be set apart as a restricted area.

Risk—Chance of hazard or bad consequences; exposure of chance of injury or loss. Risk level is expressed in terms of hazard probability and severity.

Risk Assessment—The identification and assessment of hazards (first two steps of the risk management process).

Occupancy Factor—The fraction of time an area of interest is physically occupied by the same individual.

Operational Risk Management—A logical six step thought process, applicable to any situation or environment, for identifying and controlling hazards to protect the force.

Shallow or Skin Dose Equivalent—The external exposure of the skin or an extremity, taken as the dose equivalent at a tissue depth of 0.007 cm (7 mg/cm^2 – the average depth of the germinal cell layer) averaged over an area of 1 cm^2usually expressed in units of rem or Sievert (Sv).

Sievert (Sv)—The SI unit of any of the quantities expressed as equivalent dose. The equivalent dose in Sievert is equal to the absorbed dose in gray multiplied by appropriate radiation weighting factors, w_R. (1 Sv = 100 rem). One milliSievert (mSv) is 0.001 Sv [(0.1 rem) or (100 mrem)].

Stochastic Effects—Health effects that occur randomly and for which the probability of the effect occurring, rather than its severity, is assumed to be a linear function of dose without threshold. Hereditary effects and cancer incidence are examples of stochastic effects.

Surveys—An evaluation of the radiological conditions and potential hazards incident to the production, use, transfer, release, disposal, or presence of radioactive material or other sources of radiation. When appropriate, such an evaluation includes a physical survey of the location of radioactive material and measurements or calculations of levels of radiation, or concentrations or quantities of radioactive material present.

Therapeutic Source—In the healing arts, a source of radiation used in the treatment of disease, normally cancer. Includes high energy linear accelerators generating x-rays and/or electron

beams used in radiation therapy, RAM based therapy units (e.g., using Co-60), sealed radioactive sources (e.g., Cs-137, Ir-192, I-125) temporarily or permanently implanted within a patient, and unsealed radioactive drugs (e.g., I-131) used for patient treatment.

Total Effective Dose—The sum of the deep-dose equivalent (for external exposures) and the committed effective dose (for internal exposures).

Unrestricted Area—An area, access to which is neither limited nor controlled (for the purposes of radiation safety).

Use Factor—The fraction of time a particular device is utilized, or the fraction of time the primary beam of a device is directed towards a given area.

Very High Radiation Area—An area accessible to individuals, in which radiation levels exceed 500 rad (5 Gy) in one hour at 1 meter from the source or from any surface that the radiation penetrates.

[Note: For very high doses received at high dose rates, units of absorbed dose (e.g., rad and Gray) are appropriate, rather than units of dose equivalent (e.g., rem and Sievert)].

Weighting Factor (Radiation)—A factor that accounts for differences in biological effectiveness between different radiations.

Weighting Factor (Organ or Tissue)—For an organ or tissue, the proportion of the risk of stochastic effects resulting from irradiation of that organ or tissue to the total risk of stochastic effects when the whole body is irradiated uniformly.

Working Level—The concentration of radon progeny which has a potential alpha energy release of 1.3×10^5 MeV per liter of air. One working level month is the exposure incurred from being exposed to a concentration of one working level for 170 hours.

Prefix Symbols for Units

10^{18} exa –E	10^{-3} milli - m
10^{15} peta – P	10^{-6} micro - μ
10^{12} tera – T	10^{-9} nano - n
10^{9} giga – G	10^{-12} pico - p
10^{6} mega – M	10^{-15} femto - f
10^{3} kilo -k	10^{-18} atto - a

Attachment 2

QUANTITIES AND TERMS USED IN RADIATION PROTECTION

A2.1. General. Ionization is the process by which atoms lose, or sometimes gain electrons and thus become electrically charged. Ionizing radiation are those forms of sub-atomic particles and electromagnetic waves that are capable of causing ionization in matter. Historically, the quantities used to measure the "amount" of radiation have been defined by the gross number of ionizing events in a given mass of material. The most commonly used quantity reported in early radiation protection was the Roentgen (R) equal to the quantity of x-ray or gamma radiation producing ions in air carrying one electrostatic unit of charge per cubic centimeter of air. At standard temperature and pressure, this is equivalent to 2.58×10^{-4} Coulomb per kilogram of air. Although this quantity is still in use, the more appropriate units for radiation protection are based on the ionizations that occur in given mass of human tissue, and the subsequent risk of biological affect that that "dose" may cause.

A2.2. Absorbed Dose. The fundamental dose quantity in radiation protection is the *absorbed dose, D*. This is the energy absorbed per unit mass and is in units of joule per kilogram, which is given the special name gray (Gy). One Gy is equal to 100 rad, the conventional quantity of absorbed dose equal to 100 ergs/gm.

A2.3. Quality Factors and Dose Equivalent. The probability of stochastic health effects (like cancer) is dependent not only on the absorbed dose, but also on the type of radiation causing the dose. This has been taken into account by weighting the absorbed dose at a point in tissue with a quality factor, Q, for a given radiation. The quality factor in turn is based on the density of ionization along a track of the radiation as it traverses a tissue, referred to as its linear energy transfer or LET. The weighted absorbed dose under this system is called the *dose equivalent*, H, and expressed in units of rem. The dose equivalent is the dosimetric quantity used in Title 10, Code of Federal Regulations, Part 20, for the purposes of radiation protection from RAM. For the purposes of individual monitoring using personnel dosimetry, three specific quantities of dose equivalent are used:

A2.3.1. Deep Dose Equivalent, H_d, is the dose equivalent measured at a tissue depth of 1 cm (1000 mg/cm^2) beneath the outer surface of the skin. It is the primary dose reported in the AF personnel dosimetry program. The deep dose equivalent is derived from the more general Individual Dose Equivalent, Penetrating, $H_p(d)$. This is defined as the dose equivalent in soft tissue at a depth "d" in the body that is appropriate for penetrating radiations.

A2.3.2. Shallow Dose Equivalent, H_s, is the dose equivalent measured at a tissue depth of 0.007 cm (7 mg/cm^2, the average depth of the germinal cell layer) averaged over an area of 1 cm^2. This is otherwise referred to as the shallow or skin dose in the AF personnel dosimetry program. The shallow dose equivalent is derived from the Individual Dose Equivalent, Superficial, $H_s(d)$. This is defined as the dose equivalent in soft tissue at a depth "d" in the body that is appropriate both weakly and strongly penetrating radiations.

A2.3.3. Lens Dose Equivalent, H_E, is the dose equivalent to the lens of the eye from external irradiation. It is measured at an eye lens tissue depth of 0.3 cm (300 mg/cm^2).

A2.4. Radiation Quality Factors and Equivalent Dose. The most recent paradigm in radiation protection emphasizes the absorbed dose averaged over a tissue or organ (as opposed to

a point) and weighted for the radiation quality. The weighting factor for this purpose is currently called the radiation quality factor (Q) and is selected for the type and energy of the radiation incident on the body or, in the cases of sources within the body, emitted by the source. The absorbed dose in a tissue, multiplied by the radiation weighting factors is called the *equivalent dose,* H_T. This can be expressed as:

$$H_T = \sum_R Q_R \cdot D_{T,R}$$

where $D_{T,R}$ is the absorbed dose averaged over the tissue or organ T, due to radiation R. The unit of equivalent dose is the joule per kilogram, with the special name Sievert (Sv).

The radiation weighting factor, w_R, for a given type and energy of radiation is representative of the *relative biological effectiveness* (RBE) of that radiation to inducing stochastic health effects at low doses. The values of Q, and are summarized in **Table A2.1**.

Table A2.1. Quality Factors.

Type and energy range	Quality Factor[1], Q
Photons, all energies	1
Electrons, muons, all energies	1
Neutrons < 10 keV	2
10 keV to 100 keV	5
> 100 keV to 2 MeV	10
> 2 MeV to 20 MeV	8
> 20 MeV	5
Protons, other than recoil protons	10
Alpha particles, fission fragments, heavy nuclei.	20

1. Title 10, Code of Federal Regulations, Part 20, *Standards for Protection Against Radiation*

A2.5. Tissue Weighting Factors and Effective Dose. The relation between the probability of stochastic effects and equivalent dose also depends on the organ or tissue irradiated. The effective dose is used to express the probability of occurrence of cancer and hereditary effects whether the dose is received by the whole body via uniform irradiation or by partial body or individual organ irradiation. The factor by which the equivalent dose in tissue or organ T is weighted is called the *tissue weighting factor*, w_T, and represents the relative contribution of that organ or tissue to the total detriment due to cancer and hereditary effects resulting from uniform irradiation of the whole body. The weighted equivalent dose is given the name effective dose equivalent, or more simply, the *effective dose*, E, and again has units of joule per kilogram with the special name Sievert (Sv). The effective dose is the sum of the weighted equivalent doses for all irradiated tissues or organs.

$$E = \sum_T w_T \cdot H_T$$

where H_T is the equivalent dose in tissue or organ T and w_T is the weighting factor for tissue T. Tissue weighting factors are given in **Table A2.2**. So that a uniform whole body equivalent dose

results in an effective dose that is numerically the same, the sum of the tissue weighting factors is one.

Table A2.2. Tissue Weighting Factors for Different Organs and Tissues.

Organ	10 CFR 20[1]
Gonads	0.25
Bone marrow	0.12
Colon	
Lung	0.12
Stomach	
Bladder	
Breast	0.15
Liver	
Esophagus	
Thyroid	0.03
Skin	
Bone Surfaces	
Remainder	0.3

1. Title 10, Code of Federal Regulations, Part 20, *Standards for Protection Against Radiation*

A2.6. Committed Equivalent Dose and Committed Effective Dose. Following an intake of radionuclides in the body, there is a period during which the material irradiates various organs and tissues. The committed equivalent dose, $H_T()$ is the integral of the equivalent dose rate in a specific tissue (T) following intake of a radionuclide in the body. For the purpose of radiation protection, the time of integration is taken as 50 years for occupational exposures and 70 years for members of the public. The committed effective dose, E(), for each internally deposited radionuclide is calculated by summing the products of the committed equivalent doses and the appropriate w_T values for all tissues irradiated. The expression for the occupational committed effective dose equivalent is given as:

$$E(50) = \sum_T w_T \cdot H_T(50)$$

A2.7. Total Effective Dose. This is the sum of the deep dose equivalent, H_d, and committed effective dose, E(50). Limits on occupational and public radiation dose apply to the sum of relevant doses from external exposure deep-dose equivalent in a year and the 50 year committed effective dose from intakes in the same year. The total effective dose can be expressed as:

$$E_t = E(50) + H_d$$

A2.8. Working Level Months. Exposure to radon and its progeny is not measured in the conventional dosimetric quantities described above. Instead exposures are generally indicated by the working-level, a unit originally developed to describe exposures of uranium miners to radon. One working level is defined as that concentration of radon daughters in air that has a potential alpha energy release of 1.3×10^5 MeV in a liter of air, equivalent to 2×10^{-5} Joule per cubic meter of air (J/m^3). One working level month (WLM, or 3.5×10^{-3} Jh/m^3) would be the exposure received by being present in that concentration for one working month, equivalent to 170 hours.

Exposure to one WLM results in an estimated absorbed dose to the lung of 4 - 13 mGy (0.4 to 1.3 rad). Based on data gather regarding occupational exposures uranium miners, lung cancer probabilities are estimated in the broad range of 1 to 4 x 10^{-4} per WLM. One WLM is approximately equal to an annual exposure of 4 pCi per liter of radon, if the radon decay products are in 50% equilibrium with the radon.

A2.9. Conversion Between SI and English Units. *Rule of Thumb*: For radiological protection purposes, and only for x-ray/gamma radiation environments: 1 R 1 rad. (1 rad = 0.01 Gy and 1 rem = 0.01 Sv)

A2.10. Annual Average Effective Dose Equivalent in the US Population.

Table A2.3 **provides a list of common exposure sources and the annual average dose an individual in the US receives from each source. The rounded annual dose for non-smokers per year is 6.2 mSv, or 620 mrem. It provides a basis for comparison to the limits and dose guidance specified in this instruction.**

Below is a comparison showing the increase in the annual average exposure based on results from National Council on Radiation Protection and Measurements, Report 160 -- *Ionizing Radiation Exposure of the Population of the United States* and Report 93-- *Ionizing Radiation Exposure of the Population of the United States* showing recent data indicating an increase of effective dose. This is namely due to medical exposures.

Table A2.3. Common Exposure Sources.

Exposure Category	**Effective Dose per Individual in the U.S. Population (mSv)**	
	2006	**early 1980s**
Ubiquitous Background	3.11	3.00
Medical	3	0.53
Consumer	0.13	0.05-0.13
Industrial, Security, medical, educational, and research	0.003	0.001
Occupational	0.005	0.009
Total	**6.2**	3.6

Attachment 3

TRAINING AND QUALIFICATIONS

A3.1. General. This attachment provides the training requirements and qualifications required for all AF radiation workers and RSOs. In general, the AF has three types of RSOs – IRSOs, URSOs and PRSOs. In some instances, RSOs require additional qualifications that are outlined in other AF publications. Training requirements are outlined in **Table A3.1**.

A3.2. IRSO Qualifications.

A3.2.1. Be a BE (4E3EX) who meets at least one of the following:

A3.2.1.1. Attended the BE Officer Course prior to 1999 or after 2007; or

A3.2.1.2. Awarded the G shred (i.e., 43E3G); or

A3.2.1.3. Is a certified health physicist; or

A3.2.1.4. Has received 40 hours of RSO training (reference **Table A3.1**) via a formal course.

A3.2.2. In the absence of a BE officer or if a unit is geographically separate from one, the IRSO should be a 7-level BE Craftsman (4BO71). In the absence of a BE officer or 7-level BE Craftsman, an AF civilian employee may be the IRSO provided one of the training and experience criteria in **paragraph A3.2.1** are met.

A3.2.3. Contractors cannot be the IRSO. Permittees listed on a permit issued by the AF Radioisotope Committee Secretariat cannot be an IRSO unless no other individual meets the above qualifications.

A3.2.4. In addition to the qualifications listed in this instruction, IRSOs must meet the qualifications listed in AFI 40-201 or AFI 91-108 if they are applicable to the installation.

A3.2.5. Know the hazards of the sources of radiation on the installation (including INRAD from nuclear weapons and radioactive waste sites).

A3.3. URSO Qualifications.

A3.3.1. URSO qualifications are based upon the RAM or radiation sources in the unit. **Table A3.1** lists the training required for URSOs.

A3.3.2. URSO for units conducting non-destructive inspections using radiation sources. In addition to A3.3.1, must meet the requirements in TO 33B-1-1.

A3.4. PRSOs Qualifications.

A3.4.1. PRSOs must also meet the training and qualification requirements listed in AFI 40-201 or AFI 91-108 as applicable.
A3.4.2. There are three types of permits – non-template, template, and medical permits. Additional training requirements for the RSOs for permits are delineated in AFI 40-201.

A3.4.3. Know the hazards of the sources of radiation relevant to the permit.

A3.5. Radiation Workers. All radiation workers will have the training listed in **Table A3.1**. Initial and annual training can often be provided by the IRSO, URSO, or PRSO depending on scope of use.

Table A3.1. Training guidance for radiation safety officers and radiation workers.

Topic/Audience	IRSO[1]	URSO Only Machine Produced	URSO NDI Machine Produced	Radiation Worker	PRSO[2]
Radiation vs. contamination	X	X	X	X	X
Internal vs. external exposure and dose	X			X	X
Biological effects of radiation	X	X	X	X	X
Types and hazards associated with RAM or devices possessed	X	X	X	X	X
ALARA concept	X	X	X	X	X
Training in the principles of time, distance, and shielding to minimize exposure	X	X	X	X	X
Radiation detection and measurement	X	X	X	If required by position	X
Personnel dosimetry	X	X	X	X	X
Applicable regulations	X	X	X	X	X
License/Permit conditions, amendments, renewals	X				X
Locations of use and storage of RAM	X				X
Material control and accountability	X				X
Annual audit of radiation safety program	X	If required by position	If required by position	If required by position	X
Transfer and disposal	X				X

Topic/Audience	IRSO[1]	URSO Only Machine Produced	URSO NDI Machine Produced	Radiation Worker	PRSO[2]
Record keeping	X	X	X	X	X
Prior events involving permitted material	X				X
Managing incidents/mishaps	X	X	X	X	X
Recognition and assurance of radiation warning signs; visibility and legibility	X	X	X	X	X
Inspection by regulatory agencies	X	X	X		X
Requirement for complete and accurate information	X	X	X		X
Employee protection	X	X	X	X	X
Deliberate misconduct	X	X	X	X	X
Emergency response procedures	X	X	X	X	X
Protective action guides	X				
Air monitoring procedures	X				
Bioassay techniques	X				X
Special Requirements	X		TO 33B-1-1		AFI 40-201

[1]These are the minimum topics for a 40 hr course to train IRSOs.

[2]Specific training requirements for PRSOs are contained in AFI 40-201.

Attachment 4

MEDICAL CONSEQUENCES OF RADIATION EXPOSURE

A4.1. General. Exposure of living cells by radiation can result in changes to their atoms and molecules and thus may result in damage to the cells. If cellular damage does occur, and is not adequately repaired, it may prevent the cell from surviving or reproducing, or it may result in a viable but modified cell. These two outcomes have very different implications for the organism as a whole.

A4.2. Risk of Deterministic Effects. Most organs and tissues of the body are unaffected by the loss of large number of cells following irradiation, but if the number lost is substantial, there will be observable harm resulting in a loss of tissue function. Such effects require a "threshold" radiation dose to be exceeded before an effect is observed, and generally the severity of the effect once this threshold is exceeded increases with dose. These "deterministic" effects include skin erythema, sterility, opacities of the lens of the eye (cataracts) and depression of the blood forming process. The effects may occur early, within hours or days following exposure; or late, occurring months or years after exposure.

Table A4.1. Estimated Threshold Doses for Acute Radiation Effects [1, 2.]

Effect	Organ	Dose
Temporary sterility	Testis	150 mGy (15 rad)
Depression of blood cell forming process	Bone Marrow	500 mGy (50 rad)
Reversible skin effects (e.g., erythema)	Skin	1000-2000 mGy (100 – 200 rad)
Permanent sterility	Ovaries	2500-6000 mGy (250 – 200 rad)
Temporary hair loss	Skin	3000-5000 mGy (300 – 500 rad)
Permanent sterility	Testis	3500 – 6000 mGy (350 – 600 rad)
Cataracts	Lens of the eye	2000 – 10,000 mGy (200 – 1000 rad)

1. Institute of Medicine, *Potential Radiation Exposure in Military Operations, National Academy Press*, Washington DC, 1999.
2. Report 60, *1990 Recommendations of the International Commission on Radiological Protection. ICRP Publication 60*. Annals of the ICRP 21(1-3), 1991.

A4.3. Risk of Neoplasia and Hereditary Effects. For cells that have been altered or modified and can still reproduce, the potential outcomes are very different. Despite highly effective defense mechanisms, the reproduction of radiation-modified cells may result, after a prolonged and variable delay called the latency period, in the manifestation of a malignant condition or cancer. The probability of cancer usually increases with increments in dose, probably with no threshold, and in a way that is roughly proportional to dose, at least for doses well below the thresholds for deterministic effects.

A4.3.1. For the purposes of radiation protection, the risk of stochastic effects is assumed proportional to dose without threshold, this is also referred to as the linear-no threshold hypothesis. The severity of cancer is not affected by dose. This kind of effect is called "stochastic" in that it is random or of a statistical nature. If the damage occurs in a cell whose function is to transmit genetic information to later generations (i.e., sperm or ovum), any resulting effects, which may be of many different kinds and severity, are expressed in the progeny of the exposed individual. This type of stochastic effect is called "hereditary".

A4.3.2. The nominal risks given in **Table A4.2** are for radiation induced fatal cancers for the general population. In addition, non-fatal cancers occur. Therefore, typical ratios for total cancers (fatal plus non-fatal) are given in **Table A4.2**. **Note:** These values should not be used to interpret individual risks, which are dependent on numerous factors such as age, sex, heredity and environment.

Table A4.2. Nominal Risks[1] at Low Doses and Low Dose Rates for Low-LET[2] Radiation Expressed as Severe Hereditary Disorders and Fatal Cancers (Lifetime)[3].

Effect	Nominal risk per milliSievert (100 mrem)	Ratio: total cancer to fatal cancers
Hereditary	$10x10^{-6}$ (all generations)[5]	-
Cancer	*Fatal probability*	
Leukemia (active marrow)	$5x10^{-6}$	1.01
Bone Surfaces	$0.5x10^{-6}$	1.4
Breast (females only)	$4x10^{-6}$	2.0
Lung	$8.5x10^{-6}$	1.05
Thyroid	$0.8x10^{-6}$	10
Colon	$8.5x10^{-6}$	1.8
Esophagus	$3x10^{-6}$	1.05
Skin	$0.2x10^{-6}$	500
Stomach	$11x10^{-6}$	1.1
Liver	1.5×10^{-6}	1.05
Bladder	$3x10^{-6}$	2.0
Ovaries	$2x10^{-6}$	1.4
Other (combined remaining)	$5x10^{-6}$	1.8
Sum of fatal cancer risk for whole body irradiation[4]	$50x10^{-6}$ (1 in 20,000) per milliSievert	
Baseline cancer mortality	0.15 (1 in 6.7) to 0.25 (1 in 4)	

1. The nominal risks are average values for a population comprised equally of males and females.
2. LET is the *linear energy transfer*; low-LET radiation refers to sparsely radiations such as gamma rays, x-rays and beta particles.
3. Report 62, Radiological Protection in Biomedical Research, (Also includes Addendum 1 to ICRP Publication 53, *Radiation Dose to Patients from Radiopharmaceuticals*, and a *Summary of the Current ICRP Principles for Protection of the Patient in Diagnostic Radiology*), May, 1993.
4. For infants and children, the nominal risk is likely 2-3 times higher than $50x10^{-6}$. For adults over 50 at the time of exposure, the risk is 5-10 times less.
5. A doubling dose of 1 Gy (100 rad) is assumed.

A4.4. Effects Following Irradiation In-Utero. The potential effects on the conceptus following irradiation depend on the time of irradiation relative to conception. When the number of cells in the conceptus is small, and their nature is not yet specialized, the effect of damage to these cells is likely a failure to implant or the undetectable death of the conceptus. Irradiation in the first three weeks following conception is not likely to result in deterministic or stochastic effects in the live-born child, despite the heart and central nervous system development in the third week. During the rest of the period of organogenesis, malformations may be caused in the organ(s) under development at the time of irradiation. These effects are deterministic in nature, with a threshold estimated to be 0.1 Gy (10 rad). Two additional effects of irradiation on the developing fetus include severe mental retardation and cancers that may develop in childhood or in adult life. The periods of sensitivity after conception for these described effects are summarized in **Table A4.3**.

Table A4.3. Types of Effects Following Irradiation In-Utero[1.]

Time After Conception	Effect	Normal Incidence in live-born
First three weeks	No deterministic or stochastic effects in live-born children	-
3rd through 8th weeks	Potential for malformations of organs[2]	0.06 (1 in 17)
8th through 25th weeks	Potential for mental retardation, probability: 1 to $4x10^{-4}$/mSv[3]	$5x10^{-3}$ *(1 in 200)*
4th week throughout pregnancy	Cancer in childhood or adult life, probability: 28 to $130x10^{-6}$/mSv[4]	$1x10^{-3}$ *(1 in 1000)*

1. Report 60, *1990 Recommendations of the International Commission on Radiological Protection.* ICRP Publication 60. Annals of the ICRP 21(1-3), 1991.

2. Malformations of organs appear to be deterministic effect, with a threshold dose in man, estimated from animal experiments to be 0.1 Gy.

3. The risk of severe mental retardation is associated with an observed shift in IQ of 30 IQ units per Sv to the brain during the 8th through 15th week after conception, with lesser shifts from the 16th through 25th week). At absorbed doses on the order of 0.1 Sv, no effect would be detectable in the general distribution of IQ in an irradiated group.

4. The risk of fatal cancers expressed in childhood or in adult life for individuals irradiated in utero may be similar to the risk of individuals irradiated in the first decade of life, which is somewhat larger than that for the population as a whole.

Attachment 5

DOSE AND CONTAMINATION LIMITS FOR PRACTICES

Table A5.1. Annual Dose Limits for Practices[1, 3.]

Application	Occupational	Declared Pregnant Females	Minors (16 - 18 years)[4]	Public
Total Effective Dose Equivalent[2] Deep-dose Equivalent + Committed Dose Equivalent	50 mSv (5 rem) in a single year, and 500 mSv (50 rem) to any tissue, except lens of the eye	5 mSv (500 mrem) for remainder of pregnancy to the embryo/fetus (no more than 50 mrem/month is recommended)	5 mSv (500 mrem) per year 50 mSv (5 rem) to any tissue, except lens of the eye	1 mSv (100 mrem) in a year[5]
Annual Dose Equivalent				
The lens of eye[6]	150 mSv (15 rem)		15 mSv (1.5 rem)	
The skin[6]	500 mSv (50 rem)		50 mSv (5 rem)	
The hands and feet	500 mSv (50 rem)		50 mSv (5 rem)	

Notes:

1. Based on the requirements of Title 10, CFR, Part 20.
2. The limits apply to the sum of relevant doses from external exposure in a period of 1 calendar year and the 50 year committed dose from intakes in the same period.
3. The mSv is the preferred unit of dose for radiation protection purposes. Current AF instrumentation uses the Gy or R as their basic unit of measure, and the MRER reports doses in rem. For low LET penetrating radiations (x-rays, gamma rays), the following conversions can be applied: 10 mSv = 1 cSv ≈ 1 cGy =10 mGy = 1 rad ≈ 1 R.
4. Conditions for Minors: No person under the age of 16 years shall be subjected to occupational exposure, and no person under the age of 18 shall be allowed to work in a restricted area unless supervised, and then only for the purposes of training
5. In special circumstances, an effective dose of up to 5 mSv in a single year, provided the average over five years does not exceed 1 mSv per year. AFMSA/SG3PB shall be contacted to obtain this variance. Also, general public shall not be exposed to more than 0.02 mSv (2 mrem) in any one hour.
6. Averaged over 1 cm^2, regardless of the area exposed.

Table A5.2. Acceptable Surface Contamination Levels[1] (Bq/cm^2 and dpm/100 cm^2).

Nuclide	Removable[2,4]	Total (Fixed + Removable) [2,3]
U-nat, ^{235}U, ^{238}U, and associated decay products	[7] 0.17 Bq/cm^2 (1,000 dpm/100 cm^2)	[7] 0.83 Bq/cm^2 (5,000 dpm/100 cm^2)
Transuranics, ^{226}Ra, ^{228}Ra, ^{230}Th, ^{228}Th, ^{231}Pa, ^{227}Ac, ^{125}I, ^{129}I	0.0033 Bq/cm^2 (20 dpm/100 cm^2)	0.017 Bq/cm^2 (100 dpm/100 cm^2)
Th-nat, ^{232}Th, ^{90}Sr, ^{223}Ra, ^{224}Ra, ^{232}U, ^{126}I, ^{131}I, ^{133}I	0.033 Bq/cm^2 (200 dpm/100 cm^2)	0.17 Bq/cm^2 (1,000 dpm/100 cm^2)
Beta-gamma emitters (nuclides with decay modes other than alpha emission or spontaneous fission) except ^{90}Sr and others noted above[5]	0.17 Bq/cm^2 (1,000 dpm/100 cm^2)	0.83 Bq/cm^2 (5,000 dpm/100 cm^2)
Tritium and tritiated compounds[6]	1.7 Bq/cm^2 (10,000 dpm/100cm^2)	N/A

Note: This table is extracted from 10CFR835, Appendix D and NUREG-1575, Multi-Agency Radiation Survey and Assessment of Materials and Equipment Manual (MARSAME), Supp.1, Table E.1. In general, this table will not apply to contingency operations. For contingency operations follow the COCOM, or equivalent, directives.

1 The values in this appendix, with the exception noted in footnote 5, apply to radioactive contamination deposited on, but not incorporated into the interior or matrix of, the contaminated item. Where surface contamination by both alpha-and beta-gamma emitting nuclides exists, the limits established for alpha-and beta-gamma-emitting nuclides apply independently.

2 As used in this table, dpm (disintegrations per minute) means the rate of emission by radioactive material as determined by correcting the counts per minute observed by an appropriate detector for background, efficiency, and geometric factors associated with the instrumentation.

3 The levels may be averaged over one square meter provided the maximum surface activity in any area of 100 cm2 is less than three times the value specified. For purposes of averaging, any square meter of surface shall be considered to be above the surface contamination value if: (1) From measurements of a representative number of sections it is determined that the average contamination level exceeds the applicable value; or (2) it is determined that the sum of the activity of all isolated spots or particles in any 100 cm^2 area exceeds three times the applicable value.

4 The amount of removable radioactive material per 100 cm2 of surface area should be determined by swiping the area with dry filter or soft absorbent paper, applying moderate pressure, and then assessing the amount of radioactive material on the swipe with an appropriate instrument of known efficiency. (Note: The use of dry material may not be appropriate for tritium.) When removable contamination on objects of surface area less than 100 cm2 is determined, the activity per unit area shall be based on the actual area and the entire surface shall be wiped. It is not necessary to use swiping techniques to measure removable contamination levels if direct scan surveys indicate that the total residual surface contamination levels are within the limits for removable contamination.

5 This category of radionuclides includes mixed fission products, including the Sr-90 which is present in them. It does not apply to Sr-90 which has been separated from the other fission products or mixtures where the Sr-90 has been enriched.

6 Tritium contamination may diffuse into the volume or matrix of materials. Evaluation of surface contamination shall consider the extent to which such contamination may migrate to the surface in order to ensure the surface contamination value provided in this appendix is not exceeded. Once this contamination migrates to the surface, it may be removable, not fixed; therefore, a "Total" value does not apply.

7 (alpha)

Attachment 6

REGIONAL CONSULTING MEDICAL PHYSICS OFFICES



Note: The AF/SG Consultant for Medical Physics, or Associate Corps Chief for Bioenvironmental Engineering, will identify the medical physics office to support other regions, including OCONUS and deployed locations, as appropriate.

Attachment 7

PUBLIC INTERVENTION LEVELS

A7.1. REMEDIAL ACTION LEVELS.

Table A7.1. Remedial Action Levels from Natural Sources of Radiation.

Exposure to Radon Decay Products	> 1 WLM/y (3.5 x 10^{-3} Jhm^{-3}/year)
Exposure to Radon	> 4 pCi/L average radon concentration

A7.2. Department of Homeland Security (DHS) and EPA PROTECTIVE ACTION GUIDE (PAG).

Table A7.2. DHS and EPA Protective Action Guides.

Phase	Protective Action	Protective Action Guide
Early	Limit Emergency Worker Exposure	50 mSv (5 rem) (or greater under exceptional circumstances)[1]
	Sheltering of Public	1 to 50 mSv (5 rem) projected dose[2]
	Evacuation of Public	1 to 50 mSv (5 rem) projected dose[3]
	Administration of Prophylactic Drugs	For potassium iodide, FDA Guidance dose values [4 5]
Intermediate	Limit Worker Exposure	50 mSv/yr (5 rem/yr)
	Relocation of Public	20 mSv (2 rem), projected dose first year. Subsequent years: 5 mSv/yr (500 mrem/yr) projected dose
	Food Interdiction	5 mSv/yr (500 mrem/yr) projected dose
	Drinking Water Interdiction	5 mSv/yr (500 mrem/yr) dose
Late	Final Clean-up Actions	Late phase PAG based on optimization

1. In cases when radiation control options are not available or, due to the magnitude of the incident, are not sufficient, doses above 5 rem may be unavoidable.

2. Should normally begin at 1 rem; however, sheltering may begin at lower levels if advantageous.

3. Should normally begin at 1 rem.

4. Provides protection from radioactive iodine only.

5. For other information on medical prophylactics and treatment please refer to http://www.fda.gov/cder/drugprepare/default.htm or http://www.bt.cdc.gov/radiation/index/asp or http://www.orau.gov/reacts.

6. ''Potassium Iodide as a Thyroid Blocking Agent in Radiation Emergencies,'' December 2001, Center Drug Evaluation and Research, FDA, HHS (http://www.fda.gov/cder/guidance/5386fnl.htm).

7. ''Accidental Radioactive Contamination of Human Food and Animal Feeds: Recommendations for State and Local Agencies,'' August 13, 1998, Office of Health and Industry Programs, Center for Devices and Radiological Health, FDA, HHS (http://www.fda.gov/cdhr/dmqrp/84.html).

Attachment 8

OPERATIONAL DOSE GUIDANCE FOR INTERVENTIONS

Table A8.1. OPERATIONAL DOSE GUIDANCE (NATO Standardization Agreement 2473).

Total Cumulative Doses[1]	Radiation Exposure Status Category	Recommended Protection and Surveillance Actions[2]	Increased Risk of Long Term Fatal Cancer[5]
0 to 0.5 mSv (0 to 0.05 rad)	0	None	Negligible
0.5 to 5 mGy (0.05 to 0.5 rad)	1A	Record individual dose Initiate periodic environmental monitoring	1:4,000
5 to 50 mGy (0.5 to 5 rad)	1B	Record individual dose Continue monitoring Initiate radiation survey Prioritize tasks Establish dose control measures during operations	1:400
50 to 100 mGy (5 to 10 rad)	1C	Record individual dose Continue monitoring Update radiation survey Continue dose control measures Execute priority tasks only[3]	1:200
100 to 250 mGy (10 to 25 rad)	1D	Record individual dose Continue monitoring Update radiation survey Continue dose control measures Execute critical tasks only[4]	1:80
250 to 750 mGy [6] (25 to 75 rad)	1E	Record individual dose Continue monitoring Update radiation survey Continue dose control measures Execute critical tasks only[4]	1:30

Notes:

1. The use of the measurement milliSievert (mSv) is preferred in all cases. For low LET, whole body irradiation (x-rays, gamma rays): 1 cGy =10 mGy = 1 rad ≈ 10 mSv ≈ 1 R.
2. All doses should be kept ALARA. This will reduce individual risk as well as retain maximum operational flexibility for future employment of exposed personnel.
3. Priority missions are those missions that avert danger to people, prevent damage from spreading, or support the organization's mission essential task list (METL).
4. Critical missions are those missions that are essential to the overall success of a higher headquarters' operation, emergency lifesaving missions, or like missions.

5. This is in addition to the 1:5 and 1:4 incidence of fatal cancer among the general population. Increased risk is given for induction of fatal cancer. Total lifetime risk is assumed to be 4 – 7 percent per ~1,000 mSv (100 rad). It must be recognized that higher radiation dose rates produce proportionally more health risks than the same total dose given over longer periods of time.
6. NATO STANAG 2083, *Commander's Guide on Nuclear Radiation Exposure of Groups,* states 125 cGy (125 rad) as the commander's upper dose limit.
7. Reference AFMAN 10-2503, Chemical, Biological, Radiological, Nuclear, and High-Yield Explosive (CBRNE) Operations, for recommended operational exposure guidelines based on the commander's assessment of the mission criticality and acceptable risk level.

Table A8.2. Military Contamination Limits for 7-day Operations[1.]

Commander Dose Guidance	**Maximum Contamination Limits[2] 7 day mission duration**	
	Equipment and Protective Clothing [4]	
	High Toxicity Alpha Emitter [3]	**Beta and Low Toxicity Alpha Emitters [3]**
Category 1A 0.05 - 0.5 rad	5 Bq/cm^2 (30x10^3 dpm/100cm^2)	50 Bq/cm^2 (300x10^3 dpm/100cm^2)
Category 1B 0.5 - 5 rad	50 Bq/cm^2 (300x10^3 dpm/100cm^2)	500 Bq/cm^2 (3000x10^3 dpm/100cm^2)
Category 1C 5 - 10 rad	100 Bq/cm^2 (600x10^3 dpm/100cm^2)	1000 Bq/cm^2 (6000x10^3 dpm/100cm^2)
Category 1D 10 - 25 rad	250 Bq/cm^2 (1500x10^3 dpm/100cm^2)	2500 Bq/cm^2 (15000x10^3 dpm/100cm^2)
Category 1E[6] 25 – 75 rad	750 Bq/cm^2 (4500x10^3 dpm/100cm^2)	7500 Bq/cm^2 (45000x10^3 dpm/100cm^2)

Notes:
1. From NATO Standardization Agreement 2473, Ratification Draft 1. The specified contamination limits are those estimated that would limit the total cumulative dose from inhalation of re-suspended material and ingestion of contamination to less the category Commander Dose Guidance for the operations period.
2. Radioactive contamination is removable (assessed with swipes).
3. If the alpha emitter is undetermined, use high-toxicity emitter column. Low toxicity alpha emitters include natural uranium, ^{235}U, and ^{238}U. All other alpha emitters are considered to be high toxicity.
4. It is recommended that in Cat 1A, gloves and booties be worn. It is required that in Cat 1B and above, booties, coveralls, gloves and respiratory protection be worn.

Table A8.3. Military Contamination Limits for 3 Month Operations[1] .

Commander Dose Guidance	Maximum Contamination Limits [2] 3 month mission duration	
	Equipment and Protective Clothing [4]	
	High Toxicity Alpha Emitters [3]	Beta and Low Toxicity Alpha Emitters [3]
Category 1A 0.05 – 0.5 cGy	0.5 Bq/cm^2 ($3x10^3$ dpm/100 cm^2)	5 Bq/cm^2 ($30x10^3$ dpm/100 cm^2)
Category 1B 0.5 - 5 cGy	5 Bq/cm^2 ($30x10^3$ dpm/100 cm^2)	50 Bq/cm^2 ($300x10^3$ dpm/100 cm^2)
Category 1C 5 - 10 cGy	10 Bq/cm^2 ($60x10^3$ dpm/100 cm^2)	100 Bq/cm^2 ($600x10^3$ dpm/100 cm^2)
Category 1D 10 – 25 cGy	25 Bq/cm^2 ($150x10^3$ dpm/100 cm^2)	250 Bq/cm^2 ($1500x10^3$ dpm/100 cm^2)
Category 1E 25 – 75 cGy	75 Bq/cm^2 ($450x10^3$ dpm/100 cm^2)	750 Bq/cm^2 ($4500x10^3$ dpm/100 cm^2)

Notes:

1. From NATO Standardization Agreement 2473, Ratification Draft 1. The specified contamination limits are those estimated that would limit the total cumulative dose from inhalation of re-suspended material and ingestion of contamination to less the category Commander Dose Guidance for the operations period.
2. Radioactive contamination is removable (assessed with swipes).
3. If the alpha emitter is undetermined, use high-toxicity emitter column. Low toxicity alpha emitters include natural uranium, ^{235}U, and ^{238}U. All other alpha emitters are considered to be high toxicity.
4. It is recommended that in Cat 1A, gloves and booties be worn. It is required that in Cat 1B and above, booties, coveralls, gloves and respiratory protection be worn.

Attachment 9

AIR FORCE RADIATION SAFETY COMMITTEE (AF-RSC)

A9.1. Purpose. The AF-RSC is responsible for providing oversight of sources of radiation not covered by AFI 40-201 or AFI 91-108 to ensure they are operated in accordance with Federal, AF and host nation requirements. The AF-RSC is also responsible for the development and execution of the AF's radiation protection program for both practices and interventions as is outlined in this AFI.

A9.2. AF-RSC Members Organizations and Responsibilities.

A9.2.1. **Table A9 1** indicates the membership and general roles for each member of the AF-RSC. At the request of the AF-RSCES or the AF-RSC Chairman additional advisors can be invited.

A9.2.2. AF-RSC Executive Secretary (AF-RSCES): Organization, scheduling and planning of the AF-RSC meeting, and generation of meeting minutes.

A9.2.3. AF-RSC Chair: Open AF-RSC meetings, and preside over their proceedings. Assist in maintaining the discussion focus of the agenda items, and resolve disagreements. Adjourn meeting once completed.

A9.2.4. AF-RSC Members: Ensure either they or their alternates attend scheduled AF-RSC meetings and are prepared to address agenda items.

Table A9.1. AF-RSC Membership and Responsibilities.

Office	AF-RSC Membership			Responsibilities
	Chair	Voting	Alternate	
Chair, AF-RSC	1	1		Provides oversight/leadership of RSC
AF/SG Consultant for Health Physics (Chief, Radiation Health)		1	1	Can act in Chair's absence; develops RSC agenda; provides direction and guidance for Radiation Health; conducts daily ops and ensures completion of RSC action items
Bioenvironmental Engineering (AFMSA/SG3PB)		1	1	Provides career specific input relevant to the safe use of radiation sources
AF/SG Consultant for Medical Physics		1		Liaison for the Air Force Medical Physics Working Group (**paragraph 2.10.6**) and supports AF-RSC regarding issues specific to Chapter 4.
Medical Readiness (AFMSA/SGX)		1	1	Provides guidance on readiness issues with respect to the use of radiation.
Assistant Secretary of the Air Force (Installations, Environment and Logistics)		1	1	Provides guidance, direction, and oversight [for] all matters pertaining to the formulation, review, and execution of Environment, Safety, and Occupational Health (ESOH) policies, plans, programs, and budgets.
Assistant Secretary of the AF for Acquisition, SAF/AQ		1	1	Ensures the AF-RSC is informed regarding the acquisition of new systems and upgrades that utilize sources of radiation not covered by AFI 40-201 or 91-108 prior to fielding.
Deputy Chief of Staff for Logistics, Installations, and Mission Support, USAF/A4/7		1	1	Provides guidance on the management of items covered by this AFI that produce radiation or result in radiation exposure.
Surgeon General, USAF/SG				Establishes AF policy for controlling radiation hazards and sets limits for exposure to radiation and ensures compliance with relevant Federal policy and accepted scientific practice
Assistant Surgeon General, Health Care Operations, HQ USAF/SG3				Responsible for the execution of the AF radiation safety program covered by this AFI and for the AF-RSC. Appoints the Chair, RSC.
The Air Force Inspection Agency, Medical Operations Directorate, AFIA/SG		1		Conducts inspections to assess compliance with AF and Federal requirements and provides reports to the AF-RSC Executive Secretary.
The Civil Engineer (HQ USAF/A7C)		1	1	Provides policy guidance for major accidents/incidents, enemy attack, and terrorist use of weapons of mass destruction involving nuclear and radiological materials
Command Surgeons, Air Force Materiel Command (AFMC/SG), Air Force Global Strike Command (AFGSC/SG), Air Combat Command (ACC/SG), U.S. Air Forces in Europe (USAFE/SG), Pacific Air Forces (PACAF/SG);		1	1	Provides policy guidance for the acquisition and logistical management of sources of radiation that are not covered by AFI 40-201 or AFI 91-108

Office	AF-RSC Membership			Responsibilities
Air National Guard (ANG/SG); AF Reserve (AFRES/SG)				
Commander, 711th Human Performance Wing (711 HPW/CC)		1	1	Provides advice on research and development matters. Appoints a voting member and alternate to the RSC.
US Air Force School of Aerospace Medicine (USAFSAM)		1	1	Provides to the RIC quarterly and annual summaries of occupational radiation exposure from sources of radiation covered by this AFI.
				Provides technical and on-site health physics support to the AF-RSC and AFSC/SEW as required to prevent, investigate and mitigate human or environmental exposures to 91(b) and permitted RAM.

A9.3. Business Practices.

A9.3.1. The AF-RSC Chair and the AF-RSCES will establish the detailed procedures for AF-RSC meetings.

A9.3.1.1. A formal AF-RSC meeting will occur at least semi-annually, but more frequently if determined necessary by the Chair or the majority of the committee. AF-RSC meetings are scheduled by the AF-RSCES.

A9.3.1.2. Ad hoc meetings can be called by the Chair on an as needed basis. These meetings are called to address emergent issues that require timely action by the AF-RSC.

A9.3.1.3. Rapid staffing of an action can be approved by the Chair to address emergent issues for which an ad hoc meeting cannot be convened. For a rapid staffing the AF-RSCES will prepare a staffing package that addresses the issue and provides the voting members of the AF-RSC the ability to vote without meeting. All rapid staffing actions will be discussed at the next AF-RSC meeting.

A9.3.1.4. Quorum for a meeting is established by having a least one-half of the voting members present.

A9.3.2. Meetings will be conducted IAW all applicable policies and procedures. The Chair and the AF-RSCES will establish the specific procedures for the conduct of routine meetings, ad hoc meetings and rapid staffing actions.

A9.3.3. Motions and Voting.

A9.3.3.1. Only members and guests recognized by the Chair may speak.

A9.3.3.2. Generally, before any item can be discussed, there should be a motion made and seconded. Once a motion has been seconded, discussion will follow. After discussion, one of four things can happen:

A9.3.3.2.1. **(1)** There can be a vote on the motion. In the event of a tie, the Chair casts the deciding vote.

A9.3.3.2.2. **(2)** The motion can be amended (second required). Then there can be discussion on the amendment. The amendment can be voted. If the amendment passes, the motion automatically passes. If the amendment fails, the motion still stands and can be discussed until voted.

A9.3.3.2.3. **(3)** The motion can be tabled (second required). There can be no discussion on a motion to table; a vote must be taken immediately. If the vote is to table, no further discussion can take place on the motion.

A9.3.3.2.4. **(4)** There may be no action on the motion; therefore it becomes old business at a future meeting.

A9.3.3.3. Motions must be clear and concise. A motion to "improve practices" would be vague and discussions could meander. However, a motion to "implement x-ray radiation safety practices for a new x-ray device" is specific and could be effectively discussed and acted on.

A9.4. Disagreement Resolution. The Chair of the meeting is responsible for maintaining order. On procedural questions, the Chair's ruling will be determinative and final.

A9.5. Record Keeping.

A9.5.1. The AF-RSCES will ensure that appropriate files for each meeting are maintained IAW all applicable requirements. This will include as a minimum, the agenda, meeting minutes, copy of all reference materials, background information, memoranda, standing reports, and presentations applicable to that quarters meeting.

A9.5.2. Committee members are responsible for providing applicable materials to the AF-RSCES as requested.

A9.5.3. Records of AF-RSC meetings, to include agendas, presentations, and meeting minutes, shall be kept for the duration of existence of the AF-RSC.